

THE BRAIN DOESN'T FLOAT

THE EMBODIED MIND · THE SEVEN CENTERS · VOLUME I

The Brain Doesn't Float

Wisdom and Neuroscience on How to Return to the Body

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The clinical cases have been anonymized. They are composites drawn from the author's clinical experience and altered to protect the confidentiality of the people involved.

*To my patients, who taught me that the deepest kind of safety is
not thought. It is felt.*

CONTENTS

Prologue. When the Trees Stop Dancing.....	9
PART I THE DIAGNOSIS	17
1 Stories of the Segment.....	19
2 Psychology and Psychodynamics.....	37
3 Interrelations of the Segment	49
PART II FROM THE SCIENCES	65
4 Musculoskeletal Anatomy	67
5 Glands, Hormones, and Behavior	79
6 The Sacral Plexus: The Wiring of the Base.....	87
7 The Neuropsychology of Grounding	97
8 The Muladhara Chakra	107
PART III THE PRACTICAL TOOLS.....	117
9 Pranayama: The Breath That Holds Us Up.....	119
10 Asanas: The Geometry of Stability.....	125
11 Atmandance: When I Dance, My Mind Stops	131
12 Mindfulness and Meditation	139
PART IV THE ACTION PLAN	145
13 Eight Clinical Cases.....	147
14 Practical Exercises and Everyday Life.....	159
Epilogue. The Ground That Was Already There	165
Letter from Marcos. One Year Later.....	169
Glossary	173
Recommended Reading	177
References.....	179
About the Author	183
Where to Find the Author	184

PROLOGUE

When the Trees Stop Dancing

KEY CONCEPTS grounding · neuroception · lower segment · bodily safety · uprootedness

Stop for a moment. Yes, right now, before you read on. Not as one of those rhetorical tricks so common in the prologues of self-help books—the ones that ask you to breathe deeply with a solemnity that would make a Tibetan shaman blush—but as a small clinical experiment in real time. Feel your feet. Don't think about them. Don't imagine them. Don't take for granted that they're there just because they always have been. Feel them. Notice the weight of your body settling downward. The pressure against the floor. The temperature. The real contact with the surface that is holding you up at this precise instant.

Now notice something interesting: how long did it take you to get there? One second? Two? Five? Did you have to close your eyes for a moment to manage it? Or are you still not feeling much, like someone trying to get a signal up in the mountains while the phone stubbornly refuses to connect?

If the honest answer is that it took you more than two seconds—or that your feet still feel vaguely distant, like news from another country—you are not doing anything wrong. Quite the opposite: you have just detected, with considerable precision, one of the most widespread and least diagnosed phenomena of our time. Welcome to the club. It has a great many members, all of them rather busy and all of them fairly convinced that the problem lies somewhere else.

There is an image that has stayed with me through more than thirty years of clinical practice. A tree moving in the wind. The trunk sways, the branches bend with an elegance no choreographer could improve on, the leaves tremble, and the whole crown seems to surrender to the movement with an enviable freedom. Beautiful. But what truly matters is happening below, where almost nobody looks. The roots stay firm. Sunk into the earth. Silent. They don't appear in photographs. They don't get likes. They have no story to tell. And yet they are what makes everything above possible. The tree can dance because it is rooted. It can yield because it has a base. It can move because something in it does not move.

Without roots, there is no flexibility. There is collapse. The tree would not dance: it would be torn loose in the first storm, its elegance intact and its lovely crown spinning through the air toward parts unknown.

This image is not a decorative metaphor. To me, it is the most accurate description I know of what happens in my consulting room every week, with people of utterly different ages, professions, histories, and vocabularies. An advertising executive in Barcelona who cannot remember the last time she felt truly relaxed. A forty-year-old architect who analyzes his emotions with a sophistication that would put a Jungian therapist to shame, yet cannot sleep more than five hours at a stretch. A teenage girl who meditates every morning with an app, goes to the gym, and eats gluten-free, and still drags around a diffuse anxiety she cannot understand, cannot locate, and cannot shake off. A fifty-five-year-old man who has built a company, raised three children, and never complained about anything that mattered, but who, in our first session, when I ask him to close his eyes and bring his attention to his body, opens them again after ten seconds with an expression of genuine bewilderment: "There's nothing there."

That "there's nothing there" is perhaps the sentence I have heard most often in my career. And also, paradoxically, the one that carries the most information.

Oddly enough, almost nobody walks into my office saying, "I'm uprooted." The word sounds too poetic, too esoteric, too far removed from the language people use to talk about their real problems. What turns up in the

consulting room are other words. Anxiety. Exhaustion. Insomnia. A constant tension they can't pin down. A growing difficulty in feeling present. A vague sense that something doesn't quite add up, that life is going well on the surface but isn't quite holding together on the inside. A background hum, like the noise of a refrigerator you've stopped hearing because it has been there so long it has become part of the landscape. And the surprise is always the same: how had they not noticed before?

The answer is that they had noticed. But they had read that signal as something to be solved by thinking. By being more efficient. More positive. More mindful. Meditating more, sleeping better, taking more magnesium. And all of that helps—of course it helps. But when the root of the problem is not what you think but how you feel—or, more precisely, how much you have stopped feeling—all those strategies end up like painting the walls of a house whose structure is not properly anchored. It looks presentable. Until the first storm.

P.1. The Floating City

After so many years in practice, there is a way of looking at people on the subway that you simply can't help. It isn't that I want to be nosy. It's that the body speaks so loudly that, if you know how to listen, it is impossible not to hear it. And what I hear, again and again, in any subway car in any city in the world, is something remarkable: they float. This is not a metaphor. They are there, physically present, their bodies in the seat and their attention somewhere else, like buoys adrift on a sea of notifications. On some mornings, I catch myself counting them. The record is fourteen people in one car... suspended. Thirteen, if I discount the one who was asleep. At least the sleeper had an excuse.

They float in the most literal sense possible. Their shoulders have climbed up to their ears as if asking permission for takeoff. Their necks lunge forward toward their screens with an insistence that would have scandalized any nineteenth-century anatomist. They breathe little, and badly: short, rib-bound survival breaths. And their feet—their poor feet—perform a purely logistical function: they are there to get from one place to another.

Not to feel. As if the ground were a minor detail, a mere accessory to walking.

Let me be clear that there is no moral judgment in this: it is simply an observation. An observation about how the life we have built—remarkably rich in information, in virtual connection, and in stimulation—is also a life that has gradually emptied out the experience of one's own body. We have gained connectivity and lost rootedness. And the price of that loss, though hard to quantify, is paid in the currency of anxiety, chronic fatigue, and a vague sense of not being fully present in the life one is living.

P.2. An Epidemic with No Official Name

Psychiatric manuals have a category for almost everything. The *Diagnostic and Statistical Manual of Mental Disorders* (DSM), published by the American Psychiatric Association, orders mental disorders with the thoroughness of an obsessive librarian and catalogs some three hundred distinct conditions (American Psychiatric Association, 2022). But it has no name for what I am describing. For that vast category of people who meet the criteria for no specific disorder, who function with reasonable effectiveness, and who nonetheless live with a kind of background disconnection that makes them feel their life is happening to them without their ever quite inhabiting it.

Let us call it, provisionally, uprootedness. Not in the geographical sense—I am not speaking of emigrants or homeless people, though their experiences have points of contact—but in a more precise one: the loss of contact with the base. With the body that holds us up. With the ground beneath our feet, even when nobody feels it.

Uprootedness does not hurt sharply. It doesn't present like a toothache or a fracture. It presents as tiredness that rest doesn't cure; as anxiety with no clear object; as trouble concentrating that doesn't respond to changes in how one works; as that background irritability that makes us react more strongly than expected to situations that, seen in perspective, didn't warrant it. And it also presents as a relationship with one's own body that has become purely instrumental: the body is for getting to meetings, for carrying

the groceries, for reproducing if one wishes, for looking presentable in photos. But it is not felt.

We live in an extraordinary age in technological terms and—with all due respect—a rather clumsy one in bodily terms. We have never been so connected and so disconnected at the same time: connected to screens, to news, to other people's opinions, to productivity metrics; disconnected from our own physical sensations, from our biological rhythm, from that slower and deeper intelligence that lives not in the brain but in the whole body.

We spend hours with our breathing reduced to a minimum, our jaw clenched as if chewing stress were a competitive sport, our neck thrust forward as if the screen would improve if we got two inches closer, and our attention hijacked by stimuli that renew themselves before the nervous system has had time to process the last one. The result is not just postural fatigue, though it is that too. It is a progressive erosion of our capacity to feel safe, present, and regulated.

And here we arrive at one of the central themes of this book: personal safety is not merely an idea. It is not a mental decision or a conviction one adopts after reading the right book. It is, before anything else, a bodily experience. A signal the nervous system either receives or does not. A physiological state the organism registers long before the mind puts words to it, and one that determines—silently but decisively—how we respond to life, to other people, and to ourselves.

When that experience is present, the organism can do more than survive: it can explore, bond, rest, create, trust. When it is missing, life reorganizes itself around a far more basic and far less glamorous priority: staying safe. And the brain, that remarkable and somewhat hyperactive tool we carry on our shoulders, does what it does best: it makes up for the missing base with more thinking, more anticipation, more control. An intelligent solution—elegant, even. But a limited one. Like building a spectacular skyscraper on foundations that were never fully set: the structure holds as long as there is no wind. But sooner or later, the wind always comes.

P.3. Why This Book, and Why Now

When I trained as a psychologist in the 1980s, the dominant paradigm was strongly cognitive. It was taken for granted—with the calm certainty of someone who has never questioned something very obvious—that psychological problems resided mainly in the mind, that therapy consisted above all in changing thoughts, and that the body was, at best, a biological support to be kept in reasonable shape so the mind could function comfortably. A vehicle, not a territory. A taxi, not a home. You took the body to the doctor when something specific hurt. For everything else, there was the head.

My encounter with body therapies, with yoga, with somatic psychology, and with various contemplative traditions of the East changed that view radically. I discovered something that may seem obvious today but was for a long time ignored, or actively resisted, in academic circles: the body is not the mind's passive vehicle. It is the place where psychological experience happens. Where it becomes flesh. Where it is expressed. Where it defends itself and stiffens. And, in many cases, where it can also be transformed with a depth and speed that purely cognitive work cannot always reach.

Contemporary science took its time, but it got there. We now know that emotions are not purely mental phenomena but complex psychophysiological processes involving the autonomic nervous system, the endocrine system, the immune system, posture, breathing, and muscle tone.

We know that the capacity to perceive one's own body from within—what neuroscience calls interoception—is closely bound up with emotional regulation, decision-making, and overall mental health (Craig, 2002; Khalsa et al., 2018). We know that trauma that has not been processed in the body can stay “stuck” in the nervous system for decades, producing symptoms that no purely verbal intervention fully resolves. Bessel van der Kolk documented this with a clarity that made history in the field: the body keeps the score (van der Kolk, 2014). And it does so with a bookkeeping accuracy that would put any auditor to shame.

This is not a quick-fix self-help book. There are no seven steps, no promises of transformation in fifteen days, no photographs of people on

mountaintops wearing beatific expressions. It will not ask you to change your mental attitude or try to convince you that what you lack is a bigger dose of positive thinking. There has been enough positive thinking already, and anxiety, stubborn as ever, has not received the memo.

What this book proposes is a different journey. A descent, really. A trip downward, toward the region that is most ignored and that holds the most: the lower segment, the part of the body that runs from the feet to the pelvis, by way of the legs, the sacrum, the pelvic floor, and all the invisible territory that supports the spine and, with it, everything else. A territory neuroscientists write more about every year, which the yogic traditions have for millennia called *Muladhara*—the root, the first chakra, the foundation—and to which most well-informed people of the twenty-first century pay barely any attention, except when something specific hurts.

The journey will be scientific and poetic, rigorous and entertaining, clinical and philosophical, because human reality is all of these at once and it is unwise to sacrifice one dimension to satisfy the others. We will talk about anatomy without anyone having to dissect anything. About neuroscience without the book turning into an academic manual. About contemplative tradition without anyone having to give up their healthy, hard-won skepticism. And about real people—clinical histories carefully anonymized, yet entirely true in their essence—because concepts only come alive when they have a face, a name, and a particular Tuesday afternoon on which something broke.

So, before we move on to the first chapter, do me a small favor. Feel your feet again. Not forever, and without great ceremony.

Just for a moment. Notice the contact with the floor. The weight. The temperature. The quiet texture of that surface that holds you up and asks nothing in return.

That is the lower segment. It is not mysticism. It is not gymnastics. It is not an idea. It is the most concrete, the simplest, and the most forgotten experience there is: the feeling of being here, inside this body, with your feet on the ground.

Everything else can wait thirty seconds more.

Barcelona, 2026

THE PROLOGUE AT A GLANCE

- **Uprootedness**—the loss of felt contact with the bodily base—has no diagnostic label, yet it is among the most common presentations in clinical practice.
 - **Safety is first a bodily state**, registered by the nervous system long before the mind names it.
 - **When the base is missing, the mind compensates** with more thinking, anticipation, and control: an intelligent but costly strategy.
 - **The lower segment**—feet, legs, pelvis, sacrum, and pelvic floor—corresponds to what the yogic tradition calls Muladhara.
 - **Practice:** feel your feet for thirty seconds; notice weight, pressure, and temperature.
-

PART I

The Diagnosis

From disconnection to a diagnosis of the lower segment

Before working with the body, we need to understand why we abandon it.

This first part explores the lower segment through history, psychology, and the body's interrelations.

Here you will find the map of the territory: who studied it, how it works psychologically, and how it converses with the rest of the organism.

CHAPTER 1

Stories of the Segment

From the ignored body to the neurological base of the self. A two-hundred-year journey.

KEY CONCEPTS somatics · interoception · body neuroscience · somatic psychology · grounding

María was thirty-five and had a life that, seen from outside, looked reasonably well put together. Not the cinematic existence of golden light and elegant crises you find in streaming series, but one of those correct, functional urban biographies that prompt remarks like “she can’t complain.” She worked at an advertising agency in Barcelona, had a partner, and lived in an apartment with a cat named Einstein who seemed considerably more emotionally balanced than most of her colleagues. Her calendar filled up with meetings, deadlines, and that suspicious availability now elegantly disguised as “flexible working.” From outside, everything fit together with the satisfying precision of a finished jigsaw puzzle.

Inside, however, something never quite settled. What María carried around was not exactly sadness, nor fear in the classic sense, nor some dramatic void that could be described in motivational-postcard phrases. It was something more elusive, fainter, and, precisely for that reason, more unsettling: a kind of persistent unreality, as if her life were happening to her and, at the same time, to a version of herself slightly displaced from where she stood. She took part in conversations, answered emails, met deadlines, smiled in the right meetings, but she never fully inhabited her own days. It was like living inside a well-directed film in which you know the script per-

fectly but never feel the real weight of the actor's body. The character works. The actor, somewhere between scenes three and four, has gone off for a coffee.

She noticed it most in the mornings, in that fragile moment between sleep and waking when she opened her eyes and set her feet on the cold bedroom floor. Something appeared there that took her years to name: a diffuse anxiety, not like a roaring lion but like the hum of a refrigerator you no longer hear until someone suddenly switches it off and the silence is deafening. It wasn't panic with a racing heart and catastrophic images. It was a subtle inner hesitation, as if her body doubted for a second that the floor would really hold her. As if every morning it needed to check that reality was still solid and hadn't evaporated during the night.

Then she would get up, shower, make her coffee, and repeat her little rational incantation: "I'm a functional person. I have everything under control." The phrase sounded impeccable, almost professional. The trouble was that, over time, it began to sound hollow, like repeating "nothing's wrong here" in a house where the doors close by themselves.

For years she had managed to function like that, with that background tremor turned into familiar scenery. At the agency, moreover, this state was not the exception: it was the climate. Her colleagues—young, creative, and intelligent—went from meeting to meeting with laptops in tow, ate in front of their screens without quite registering what they were chewing, answered messages at eleven at night, and reached Sunday afternoon in a mixture of emptiness, irritation, and resignation that nobody questioned. María had accepted it as part of the toll for being an adult in a competitive city and an economy apparently designed to keep everyone slightly off balance. She thought that perhaps this was simply "being an adult": a fixed dose of unease and disconnection in exchange for a certain outward stability.

Until it stopped being bearable.

It didn't happen in some great dramatic moment worthy of a documentary. It happened on an ordinary Tuesday, in a meeting with one of those clients who confuse firmness with aggression and seem to enjoy dismantling other people's work as if training for a championship in verbal demolition.

María had spent weeks preparing the campaign. The client interrupted every thirty seconds, questioned decisions he had previously approved, and demanded mutually incompatible changes with the calm inconsistency of someone who never has to carry anything out. Outwardly, María kept speaking with professional courtesy, explaining, arguing her case. Inwardly, something quieter and deeper happened: a crack. It wasn't a collapse, and she didn't burst into tears. It was like the ice on a lake beginning to split while the surface still looks firm. What broke was not her patience. It was the illusion that she could go on living like this indefinitely.

That afternoon she left the meeting dazed. She was walking to the subway with her earphones in, though she wasn't really listening to the music—that electronic background noise that works as a social shield without ever quite being company—when she tripped on the usual curb, the same city curb that had been in the same spot for as long as she could remember. She didn't actually fall, but in that instant, as her body fought not to hit the ground, she felt something that stopped her dead in the middle of the sidewalk: she couldn't feel her legs.

Not literally—her legs were there; they had held her up for thirty-five years with commendable loyalty—but she didn't feel them as a living part of her experience. It was as if her consciousness lived exclusively in her head and the rest of her body ran as an automated peripheral system, a silent Uber carrying her brain from one point to another without anyone having had the courtesy to tell her the route.

She stepped out of the stream of pedestrians and stood still. When was the last time she had really felt her body? She found no answer. A small voice, considerably wiser than her rational discourse, whispered that there had to be another way to live.

María's story is not exceptional. Change the names, the jobs, the cities, and the scenery, and you will see the pattern repeat with a frequency that no longer surprises anyone working in mental health. People who produce, care for others, solve problems, keep up appearances, and who inside describe the same sensation: floating. Being there, but not quite. Inhabiting their life like someone subletting an apartment where they never finish unpacking.

What unites these stories is not a difficult childhood, an identifiable trauma, or a weak personality. What unites them is something more specific and more correctable: the progressive loss of contact with the lower segment.

1.1. What the Lower Segment Is and Why It Matters

The lower segment is not a poetic term invented to make book titles sound good. It is the name we use, within the system of work this book proposes, for a functional region of the body that includes everything from the feet to the pelvis: the feet and ankles, the legs, the pelvic girdle, the sacrum and coccyx, the pelvic floor muscles, the psoas, and the deep hip muscles, and all the nervous, hormonal, and structural systems that support them and connect them to the rest of the organism.

In the yogic tradition, this territory corresponds mainly to the first chakra, Muladhara, though it also includes elements of the second. In somatic psychology and the Western body traditions, it is the “base” par excellence: the first body segment in terms of energetic and functional organization.

But before getting into anatomy or tradition, we should answer the most basic question: why does it matter?

The short answer is that the lower segment is, literally, the structure that holds up everything else. Not metaphorically: completely literally. The feet are our permanent point of contact with the surface that supports us. The legs carry that contact upward. The pelvis organizes the relationship between the lower and upper body. The sacrum connects the spine to the pelvic anchorage. The psoas continuously regulates the relationship between the nervous system and posture. When this base works well, the rest of the organism can partly relax, trust its support, and distribute effort more efficiently. When it fails—or when we stop perceiving it—the whole system climbs upward in search of stability, and the result is exactly what María described: a diffuse background tension, a head working overtime to make up for what the body has stopped providing, and a general sense of never being entirely safe, even when nothing in particular is happening.

The longer—and more interesting—answer has to do with something neuroscience has documented over recent decades with a precision that doctors in the 1980s would have found implausible: the body, and this lower region of it in particular, plays an active and decisive role in emotional regulation, in generating a basic sense of safety, and in organizing the inner states that determine how we respond to life. It is not just that emotions are “reflected” in the body. It is that emotions, to a very significant degree, happen in the body. The distinction is not semantic. It has enormous practical consequences.

1.2. A Story That Begins Two Hundred Years Ago

To understand how we came to ignore something as fundamental as the lower half of our own body so systematically, a very brief historical detour is useful. Not because history is itself the subject of this chapter, but because understanding where we come from helps us understand why we are where we are—and that is always useful before deciding where we want to go.

The philosophical split between mind and body is not a recent invention. In 1637, René Descartes formulated it with a clarity that proved very influential and, in the long run, very problematic: “I think, therefore I am” (Descartes, 1637/2006). A brilliant philosophical starting point, but with the slight drawback of suggesting that what matters is thought, and that existence is verified through it. In this scheme, the body holds a subordinate position. It is the machine that must be kept running so the mind can think. An engine with philosophical aspirations, but an engine all the same.

This Cartesian view deeply permeated Western medicine, academic psychology, and, by extension, popular culture. Medicine specialized in parts of the body treated in relative isolation from one another. Psychology, when it emerged as a discipline in the nineteenth century, did so largely with the mind as its central object of study and the body as peripheral. And contemporary culture inherited, without fully knowing it, the same premise: the important self, the self that makes decisions and is defined by its achieve-

ments, lives in the head. The body is what you take to the gym three times a week and in for an annual check-up.

Throughout the twentieth century, however, a radically different view gradually emerged from several directions at once. Somatic psychology, pioneered by Wilhelm Reich in the 1930s (Reich, 1933/1949), planted the seed of an idea that now seems almost obvious but was disruptive at the time: unresolved emotions do not disappear. They are stored in the body as chronic muscle tension, compensatory postural patterns, and restricted breathing. Reich was not exactly an easy character—he mixed genuinely brilliant ideas with some far more questionable ones—but his central insight about muscular armor, that shell of tension we build to protect ourselves emotionally and which ends up imprisoning us, anticipated by decades what contemporary neuroscience would confirm with data.

Alexander Lowen, Reich's student and the founder of bioenergetics, introduced in the 1950s and 1960s a specific emphasis on connection with the ground—grounding—as a central element of therapeutic work (Lowen, 1975). For Lowen, contact with the earth was not a metaphorical exercise but a concrete practice: literally standing on your own feet, feeling the weight of the body settle downward, developing the capacity to rest on your own base. We will return to both pioneers later in this chapter, because their clinical observations anticipated much of what follows.

Meanwhile, in the East, the contemplative traditions had for millennia been working with a map of the body completely different from the Western one, yet surprisingly consistent with these observations. The chakra system, in its various versions, organizes bodily experience into energy centers that correspond to specific regions of the organism and are associated with particular psychological and spiritual functions. The first chakra, Muladhara—which in Sanskrit literally means “root” or “foundation”—is located at the base of the spine, in the region of the perineum, and is associated with the themes at the heart of this book: safety, survival, rootedness, basic trust in existence.

The agreement between these ancient elaborations and contemporary clinical observation is no accident. Systematic observation, whether scientific or contemplative, tends to converge.

1.3. The Great Synthesis: When Neuroscience Talked to the Body

It was in the last three decades, however, that the most powerful synthesis took place—the one that definitively changed the rules of the scientific game. Several developments came together almost simultaneously.

In the 1990s, the neuroscientist Antonio Damasio published *Descartes' Error* (Damasio, 1994), a book whose title made clear from the outset which side it was on. Damasio's central thesis, based on decades of research with patients who had specific brain lesions, was that emotions are not irrational interference in the process of logical thought. On the contrary, they are indispensable cognitive instruments. Patients who lost the capacity to feel emotions through damage to certain brain regions did not become more rational or more efficient: they became unable to make adaptive decisions, even the simplest ones. Reason and emotion are not opposing systems. They are inseparable. And both, as Damasio meticulously documented, have deep bodily roots.

At almost the same time, Stephen Porges put forward Polyvagal Theory (Porges, 1995, 2011), a proposal that reorganized our understanding of the autonomic nervous system in a way that proved decisive for somatic psychology and trauma work. Porges described how the vagus nerve—that remarkably long nerve connecting the brain with almost every internal organ, including the heart, the lungs, and the abdominal viscera—actually operates through several distinct “circuits,” each associated with a different state of activation or deactivation of the system. The most evolutionarily recent of these, the ventral vagal circuit, switches on when the nervous system registers safety, and allows states of social connection, active calm, and exploration. The more primitive ones switch on in response to perceived threat and produce fight, flight, or collapse. Several of the theory's evolutionary and anatomical claims remain debated in the psychophysiological literature

(Grossman & Taylor, 2007); its usefulness as a clinical model of safety and threat, however, is widely recognized.

What matters here is that the activation of these circuits is not just a response to what we think about the situation. It is a response to signals that the body—including the lower body, posture, muscle tone, breathing, contact with the ground—constantly sends to the central nervous system.

Bessel van der Kolk, with his monumental work on trauma culminating in *The Body Keeps the Score* (van der Kolk, 2014), added what was perhaps the missing piece. He documented exhaustively that traumatic experiences are not stored mainly as narrative memories—stories we tell ourselves about what happened—but as somatic states. Patterns of muscle tension, altered breathing, defensive posture, and nervous-system regulation that were imprinted on the body and are triggered by stimuli the nervous system associates, consciously or not, with the original situation. And he established something with enormous therapeutic consequences: for those states to change, talking about them is not enough. You have to work with the body that holds them.

In retrospect, this convergence—of Western somatic psychology, Eastern contemplative traditions, and contemporary neuroscience—is not surprising. They were three different ways of observing the same thing: a human organism in which mind and body are not two separate entities that occasionally communicate, but an integrated system in which what happens in each region of the body affects mental state, and vice versa. A system in which the base—the lower territory of the body, contact with the ground, the sense of rootedness—plays a role that was underestimated for centuries and which, with considerable evidence behind it, can no longer be ignored.

1.4. Why We Keep Ignoring It Anyway

But if science has reached these conclusions, if the contemplative traditions have pointed the same way for millennia, and if any body therapist can confirm these patterns clinically every week, why do we go on living from the

neck up? Why is the lower segment still the great forgotten territory of contemporary self-awareness?

The first reason is cultural, and goes back to what Descartes had already suggested: we live in a civilization that has historically bet on mind over body, thought over sensation, the abstract over the concrete. That bet has produced remarkable achievements—science, technology, medicine, philosophy, art. But it comes at a price, the very price María and people like her are paying: a progressive disconnection from direct bodily experience that ends up producing that sense of floating, of having no base, of being in one's own life without quite inhabiting it.

The second reason is technological and more recent. The digital revolution has created living environments in which it is possible to spend hours, whole days, functioning almost exclusively from the head and the hands. Office work, screen work, digital entertainment, device-mediated communication: none of these activities requires the lower body except as passive support. And as any biological system knows, what is not used tends to fade into the background. The lower body doesn't usually hurt. It doesn't fail dramatically. It simply grows quieter, more distant, less present on the person's inner map. Until someone trips on a curb and discovers they can't feel their legs.

The third reason is perhaps the most paradoxical: the most common response to anxiety, exhaustion, or insecurity is to look for the solution where we spend most of our time—in the head. More thinking. More analysis. More cognitive strategies. More meditations on apps that time our minutes of "mental presence." All of that has value, no doubt. But when the source of the problem is precisely disconnection from the bodily base, responding with more mental activity is like trying to put out a fire with an electric fan. Lots of movement; debatable effect.

1.5. A Map for the Journey Ahead

This book is not an anatomy treatise, a yoga manual, or a neuroscience text, though there will be elements of all three along the way. It is, above all, a practical and conceptual guide to the territory María rediscovered in the

humblest and most effective way possible: by paying attention to her feet for one minute, three times a day.

The path we propose is systematic. We begin with the history and psychology of the lower segment—what it is, how it works, what happens when it doesn't—and from there we move into anatomy and neurophysiology, not because you need to know the names of all the bones, but because understanding how the base is built helps you relate to it differently. Next we explore the hormonal and nervous systems that regulate this region of the body and have a direct impact on states of anxiety, safety, and emotional regulation. And finally we reach practical territory: the concrete tools—postural, breathing, movement, and attention practices—that make it possible to restore or strengthen contact with the lower segment in a progressive and sustainable way.

There is also a chapter devoted to the chakra system, specifically to Muladhara—not as an invitation to adopt any particular belief, but in recognition that these traditions have been observing the same territory for far longer than contemporary neuroscience, and that their maps, read with intellectual honesty, offer perspectives that scientific language has not yet fully translated.

María's story ends, for now, rather undramatically. Six months after that first session with the body therapist, she was still working at the same agency, living in the same apartment with Einstein, and leading, in essence, the same outward life. What had changed was harder to measure, but entirely real: she no longer lived evicted from her own body. When she got up in the morning and set her feet on the floor, she felt them. She noticed the contact, the temperature of the wood, the distribution of her weight. And with that sensation came something she described as a quiet certainty: the simple experience of being here.

The anxiety did not disappear completely—that would be promising you too much, and neither of us wants that. But it changed in quality. It stopped being an all-pervading background and became something more localized, more recognizable, more manageable.

“For years I thought I had to fix my anxiety by thinking better,” she told a friend a year later. “And the problem was that I was looking for it in the one place it couldn’t be: my head.”

Her friend looked at her with the same mix of skepticism and curiosity María herself had felt at the start.

María smiled before answering. “In my feet. And in everything between my feet and my waist. In the lower segment. In that part of the body that has spent your whole life trying to tell you something, while you spent your whole life not listening because you were too busy with what you thought about what you felt.”

Her friend considered this for a moment. “Sounds too simple.”

“It is,” said María. “And that is exactly the trap.”

1.6. A Doctor Who Doesn’t Care About the Ground

There is a scene that repeats itself, with minimal variations, in countless doctors’ offices across the Western world. The person arrives with symptoms that don’t quite fit any clear diagnosis: tiredness that rest doesn’t cure, diffuse muscle tension, difficulty concentrating, a kind of anxiety that comes and goes for no identifiable reason, aches that move around. The doctor, who has ten minutes per patient and forty more in the waiting room, asks the standard questions, orders some blood tests, and, when the results come back normal, concludes with the most repeated sentence in contemporary medicine: “There’s nothing organic. It could be stress.” And writes a prescription for anxiolytics, sometimes with the reassuring conviction of someone who has done their job.

The problem is not the doctor, who is usually perfectly competent within the limits of the system. The problem is the system, which inherited from Descartes the conviction that body and mind are separate entities that occasionally bother each other, and that one can therefore treat the body without asking too much about the mind, and vice versa. The result is that thousands of people with perfectly real symptoms—perfectly explicable from the standpoint of somatic psychology—end up trapped in a no man’s land: too

functional for the psychiatrist, too vague for the internist, too much “something” to be left without help.

1.7. The Difference Between Functioning and Inhabiting

Functioning means the operating systems are running. The work gets done, the bills get paid, relationships are sustained at a reasonable level of operation, and anyone looking from outside sees someone who “has their life in order.”

Inhabiting is something else. Inhabiting means being present in your own experience, feeling the weight of your own hands, tasting your own life, noticing when something is right and when something isn't before the body expresses it as a crisis.

The difference between the two states is exactly the difference between an actor who knows the script and an actor who is on stage. The first says the right lines. The second lives the character. The first can keep the performance going indefinitely as long as nothing changes too much. The second has access to something the first does not: the capacity to improvise, to respond to what is actually happening rather than to what was supposed to happen.

And the difference between the two, at least in clinical experience, usually has a great deal to do with the lower segment. With whether the feet are felt or not. With whether the pelvis has weight or floats. With whether the breath reaches the abdomen or stays perpetually in the chest, like a tenant who never quite moves in.

1.8. The Body María Had Forgotten

When María arrived at the office of the body therapist who would change her relationship with her body, she brought something no blood test can show: a history of progressive disconnection. It had not happened all at once. It was the accumulated result of years of priorities, of a culture that rewards productivity over presence, of a university education that teaches you to think and argue but not to feel and rest, and of a working environment

that regards being available at eleven at night as a sign of commitment rather than of self-abandonment.

Her body had slipped into the background so gradually that she herself hadn't noticed. First it was the weekend sport, which grew less frequent until it disappeared. Then the quiet, screen-free dinners, replaced by late-night scrolling that promised rest and produced exactly the opposite. Then the ability to sit and do nothing, which mutated into an inability to tolerate emptiness, as if silence had become threatening. And finally, almost imperceptibly, the sense of the body itself: that basic register of being inside it, perceiving its weight, feeling its temperature, hearing its signals.

The therapist had a way of working that bewildered María at first. She didn't ask about childhood. She didn't ask for relational patterns to be described. She didn't interpret. She simply asked for attention. Attention to the body, in the present moment, with a curiosity that sought not results but information.

The first session consisted almost entirely of sitting. Adjusting position, observing what happened when the feet were placed one way or another, noticing the difference between a pelvis that tips backward and one that settles onto the sitting bones. Breathing, with no special technique, simply observing where the air went and where it stopped.

At the end of that first session, María asked a question the therapist had heard hundreds of times: "And is this actually good for anything?"

The answer took weeks to become obvious. And when it did, it didn't arrive the way María expected.

1.9. Small Changes That Don't Sound Like Change

The first sign was that María began to sleep better. Not dramatically—not the restorative sleep of mattress commercials, where someone wakes up smiling with perfect hair—but better in the real sense of the word: she woke less often at three in the morning, took less time to fall back asleep, and her sleep had a different quality, denser, more real.

This was no accident. Work with the lower body, done consistently, changes the baseline state of the autonomic nervous system. It reduces

chronic sympathetic tone. It encourages activation of the parasympathetic system. And those physiological changes have direct consequences for sleep, digestion, resting heart rate, and the quality of overnight recovery. Physiology is not poetry; it is very literal.

The second change was that she began to notice when she was tense before reaching saturation point. That may not sound like much. But for someone who had lived without access to that information—who went from zero to a hundred with no gauge in between—it was revolutionary. She began to be able to say “this is costing me a lot” before the cost was so high it overwhelmed her. She began to be able to choose, instead of simply reacting.

The third, which María described with a mixture of embarrassment and wonder in one of the last sessions, was that she had started to feel pleasure in things that had long since stopped giving her any. A slow meal. A walk with no destination. A conversation with no agenda. The taste of her morning coffee, which for years had been mere fuel and was now, every so often, real coffee.

Small changes. Nothing epic. Nothing to justify a documentary about personal transformation. But changes that, taken together, traced something the therapist named precisely and without drama: María was learning to inhabit her life.

1.10. The Lower Segment in the History of Psychology

Now comes the complicated part, but don't be alarmed. Before we go on with María—who will be waiting patiently—we need to understand why this kind of work has so few years of formal history behind it, despite thousands of years of intuitive practice.

Western psychology was born at the end of the nineteenth century as a science of the mind. Not of the body. Wundt and Freud, and the schools that followed them, were preoccupied above all with the mind—its contents, its structures, its conflicts. The body appeared as a stage on which the mind projected its dramas—Freud's hysterical conversions—or as a relatively inert biological support that psychology could ignore with some peace of

mind. There was a notable exception. In 1884, William James proposed that we do not tremble because we are afraid, but are afraid because we tremble: the perception of bodily change *is* the emotion (James, 1884). It was a radically embodied idea, and the discipline set it aside for the better part of a century, until Damasio and others rescued it.

It's curious: human beings have been dancing, breathing, and touching one another to feel better for thousands of years. But until very recently, that wasn't "science." It was... what? Folklore? Mysticism? Gymnastics? I sometimes think psychology took a hundred years to discover the body simply because it was embarrassed to.

This orientation had its logic: psychology was trying to legitimize itself as a science in an academic setting that valued the abstract and mental over the concrete and bodily. The body was medicine's territory. The mind was psychology's territory. And the borders were respected with the rigidity of someone defending a guild.

The first break came from within psychoanalysis itself. Wilhelm Reich, a student of Freud whom Freud eventually distanced himself from for reasons both theoretical and personal, noticed something psychoanalytic orthodoxy preferred to ignore: patients didn't bring their psychological conflicts only in the form of words and free associations. They brought them in the body, too. In posture. In breathing. In muscle tone. In the way they sat, crossed their arms, tightened their chest, or hardened their jaw.

Reich developed the concept of character armor (Reich, 1933/1949): the shell of chronic muscle tension the organism builds as a defense against experiences it cannot process any other way, and which over time becomes so automatic and invisible that its wearer no longer perceives it as a defense but simply as "the way I am." Tense shoulders are not an emotional state: they are a posture. But that posture has emotions encoded in it with a fidelity no diary could match.

Reich's work was disruptive and flawed in equal measure. His personal life and his later theories became increasingly erratic, and the scientific community took decades to separate the genuinely brilliant insights from the rest. But the seed he planted—the body as psychological territory, not mere-

ly biological support—germinated into a tradition that grew slowly but steadily.

1.11. Alexander Lowen, Bioenergetics, and the Ground Beneath Our Feet

It was Alexander Lowen, Reich's student and later dissenter, who took these ideas into more systematic and more accessible territory. As the founder of bioenergetics, Lowen placed explicit emphasis on what he called grounding: contact with the ground as a central therapeutic element. Not as a metaphor, but as a concrete practice. Literally standing up and feeling the weight of the body. Bending the knees slightly and noticing the pressure in the soles of the feet. Breathing into the abdomen as the body settles onto its base (Lowen, 1975).

Lowen worked systematically with the legs and the pelvis. He observed that the patients with the greatest difficulty regulating their emotions were also, almost invariably, those with the least functional contact with the lower body. Tense or disconnected legs. A pelvis locked or collapsed. Feet that didn't clearly feel the floor. And he also observed that when that contact improved—when the lower body regained tone, mobility, and presence—emotional regulation improved along with it. Not because the intrapsychic conflict had been resolved, but because the nervous system now had a base from which to operate differently.

This observation, which Lowen grounded in bioenergetics using a vocabulary that blended psychoanalysis and bodywork, anticipated by decades what contemporary neuroscience is now documenting with data: the sense of rootedness—of having a base, of feeling the ground—has concrete neurophysiological correlates that change autonomic nervous system activation, the tone of the amygdala, and the quality of emotional regulation.

1.12. What Happened in María's Body While She Read This

At some point in this chapter—probably at several—María might have looked up from the book and brought her attention to her feet. Not be-

cause anyone had explicitly asked her to, but because the text was touching something in her that resonated with a particular familiarity: the description of her own experience, handed back to her in words.

And in that small gesture—lowering her gaze, bringing her attention downward, noticing the contact of her foot with the floor or the shoe—something would have been happening that this chapter is trying to describe: the lower segment re-entering the map of conscious experience. Not as a concept, but as a sensation.

There is no more effective way to learn about the first center than to feel it. Not as a formal exercise, not as a spiritual practice with optional incense, but as a small act of attention that can be done at any moment, anywhere, without anyone noticing and at no cost at all. Three breaths with your attention on your feet. The weight of your body, noticed. The temperature of the floor under your soles.

That gesture doesn't solve anything on its own. It doesn't reorganize years of disconnection in thirty seconds. But it does something important: it reminds you that something is there. That the lower body exists. That it has weight and temperature and contact with the world. And that every time you remember it, the distance between the mind and the ground gets a little shorter.

CHAPTER AT A GLANCE

- **Mind–body dualism**, formalized by Descartes in 1637, shaped Western medicine and psychology.
 - **Early embodied alternatives**—William James, Wilhelm Reich, and Alexander Lowen—were long marginalized; Lowen made grounding a clinical practice.
 - **Since the 1990s**, the work of Damasio, Porges, and van der Kolk has converged on one conclusion: emotion and safety are bodily processes.
 - **Cultural, technological, and cognitive habits** keep the lower body out of awareness.
 - **Functioning is not inhabiting**; the difference often lies in the lower segment.
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CHAPTER 2

Psychology and Psychodynamics

The invisible architecture of the self. How the inner base is built—or isn't.

KEY CONCEPTS basic safety · attachment · trauma · neuroplasticity · autonomic nervous system

Imagine for a moment that emotional balance worked like Wi-Fi. We switch on the router, take two deep breaths, repeat a well-meaning affirmation in front of the mirror, and, if things get demanding, add a playlist of Tibetan singing bowls. That should be enough to give us a stable connection to life, high-speed emotional coverage, and reasonable serenity even at rush hour. Unfortunately, the human nervous system has a design that is far older, far more complex, and—it must be said—far less obedient than a piece of telecommunications equipment. We are not a router; we are more like a tree. And trees don't reboot at the push of a button.

The question that organizes this chapter is simple to state and disconcerting in its implications: what happens psychologically when the lower segment is not active enough? What inner dynamics are set in motion when someone lives, like María, from the neck up? And how does that situation come about—not all at once, but woven over years with an almost artisanal patience?

2.1. Safety as an Experience, Not a Concept

The first thing to understand is that personal safety—in the deep sense of the term, not in the sense of “having a plan B” or “three months’ savings in

the bank”—is not a concept one adopts. As the prologue suggested, it is a bodily experience, registered by the nervous system long before the mind puts words to it.

When that experience is present—even partially, approximately, good enough—the organism can do something that goes beyond survival: it can explore, bond, rest, create, trust. It can, in a word, live. When it is missing, life reorganizes itself around a more basic and far less elegant priority: staying safe. And that reorganization has consequences that branch out into every aspect of existence.

This is where the psychodynamics of the lower segment come in. The brain—that remarkable and somewhat hyperactive tool installed at the top of the body—tries to make up for the missing base with more thinking, more anticipation, more mental control. It is an intelligent solution. You have to give it that. But it is also a costly one: it keeps the organism in a state of vigilance that, as we will see in Part II, carries a measurable physiological price—what the neuroendocrinologist Bruce McEwen called *allostatic load* (McEwen, 1998).

2.2. The Chakra System as a Psychological Map

Before turning to the developmental psychology that explains how this inner base is formed—or fails to form—it is worth briefly clarifying the frame of reference this book uses. When we speak of the lower segment, we are speaking of what the yogic tradition calls the first chakra, *Muladhara*, with influences from the second, *Svadhithana*. And something should be said about the chakras that is often left out in the contexts where they are used: they are maps, not dogmas.

The chakra system is rooted in the spiritual traditions of the Indian subcontinent, with antecedents in early Indian literature and a far fuller elaboration in the tantric texts over the centuries. The seven-chakra scheme that circulates in Western popular culture today spread mainly through the work of Sir John Woodroffe, writing as Arthur Avalon, in the early twentieth century (Woodroffe, 1919).

This is worth pointing out because it inoculates us against a very common problem: spiritual dogmatism with an exotic aesthetic. The chakras are not anatomical organs waiting to be discovered by a particularly sharp-eyed radiologist. They are symbolic maps of experience, developed over centuries through systematic observation of body and mind. And, as with any serious map, their value does not depend on reproducing the territory literally, but on helping us find our way through it.

This book proposes reading the energy centers in an integrative way: not as mystical entities floating on some invisible plane, nor as mere poetic metaphors with no bodily correspondence, but as functional zones where anatomical, physiological, neurological, and emotional systems converge. When we speak of a “blocked” center, we need not imagine some esoteric energy jammed in place. We can speak, far more precisely, of chronic muscle tension, restricted movement, nervous-system dysregulation, hormonal imbalances, or emotional patterns that repeat themselves and become embodied in a person’s posture, breathing, and quality of presence.

With that said, let us return to psychology.

2.3. How the Base Is Built: Early Development

The psychological base is not decided. It is not chosen. It does not appear one fine day as the result of a firm resolution to “work on myself more.” It is built—or not—through thousands of early experiences that the nervous system records long before we have language to describe them.

The human baby arrives in the world in a condition unique among mammals: completely dependent, with no fur to protect it, no teeth to defend itself, and not the faintest idea that it is not the center of the universe. A wolf cub can hunt within a few months. A human, at a few months, can barely hold up its head. In our case, evolution seems to have bet on brain development at the expense of everything else. As if someone had said: “Let’s make them brilliant, but useless for years. It’ll pay off in the end.”

John Bowlby (1969), the founder of attachment theory, showed rigorously that these repeated experiences are not anecdotal details of development: they are the materials from which the nervous system builds what he

called internal working models—implicit, nonverbal, prelinguistic schemas about how the world works, what we can expect from others, and whether or not we are the kind of person whose needs have a place in reality. These models are not built with words. They are built with skin, temperature, rhythm, closeness, and response time. With the body of another body.

When caregivers are sensitive and consistent enough—and the “enough” deserves emphasis, because Winnicott made it clear that what is needed is not the perfect mother but the good-enough mother (Winnicott, 1971)—the baby’s nervous system draws an implicit but fundamental conclusion: the world responds. My needs have a place. I am not alone in this. The ground holds. This conclusion is not thought. It is felt in the body. It settles in as a baseline state of relative safety from which the organism can operate. It is the root of the tree.

When care is inconsistent, unpredictable, neglectful, or threatening—and it need not be dramatically abusive to leave a mark—the conclusion changes. The world is not reliable. Needs may or may not be met. The ground can vanish without warning. And faced with that uncertainty, the organism does the only thing it knows how to do: it adapts. It develops strategies. It learns to survive with whatever resources are available. And those survival strategies, so intelligent at the time, are exactly what will still be running decades later, even when there is no longer any real danger to justify them.

2.4. The Silent Wound

There is an idea worth repeating because it is hard to accept: it doesn’t always take major trauma to disturb the base. The word trauma evokes intense, dramatic, identifiable scenes. And yes, trauma in that sense exists and leaves deep marks. But in clinical practice, many of the people who find it hardest to ground themselves have not lived through anything particularly catastrophic. What they lived through is more mundane and, in many ways, harder to recognize precisely because it has no clear name.

A loving but chronically anxious mother who, without realizing it, conveyed to her baby that the world was a dangerous place. A father who was

physically present but emotionally absent—that kind of distance that makes no noise but that a child’s nervous system records with great fidelity. A home with no real emotional stability, where the climate shifted unpredictably according to the adults’ moods. Nothing cinematic. Nothing you could report. But enough for a developing child’s nervous system to learn that the ground is not entirely reliable and that it pays to stay alert.

This silent wound—that of someone who had no identifiable trauma but never had enough base either—is perhaps the most prevalent and least diagnosed of the difficulties that reach the consulting room. Because the people who carry it usually function well. Sometimes they function very well. They have developed highly sophisticated cognitive strategies to compensate for the missing bodily base: they are more intelligent, more analytical, more foresighted, more controlling. They have built an elaborate mental architecture precisely because the bodily architecture was not solid enough.

The problem appears when that mental architecture, however impressive, comes up against something it cannot solve with more thinking. A loss. An illness. A failure. The accumulated exhaustion of years spent living from the neck up. And then, as in María’s case, something cracks. Not dramatically. Just enough for the crack to show.

2.5. Survival Strategies That Become Prisons

When the base is not sufficiently available, the organism develops a series of compensating strategies. In their origin, these strategies are intelligent, adaptive solutions. The problem arises when they become chronic and automatic—when they keep running even though external conditions changed decades ago.

The most common, and the one most visible in contemporary culture, is hypermentalization: living from the head. Thinking more, anticipating more, analyzing more, controlling more. If the body does not provide a sufficient sense of base, the mind tries to supply it through intellectual understanding. The implicit logic is: if I understand what’s happening, I’ll be able to handle it. If I anticipate every possible scenario, nothing will surprise me.

If I have everything under mental control, I don't need to trust something I can't control, like the bodily sense of safety.

It is a strategy that works rather well in many contexts. Academic life, many professional careers, environments where cognitive performance is highly valued: all of these reward people who have developed a very active mind. The price, which usually comes due between the ages of thirty and forty-five, is chronic mental fatigue, difficulty truly resting, and that vague sensation María described: being in one's own life without quite inhabiting it.

Another frequent strategy is postural and behavioral rigidity. If the inner ground is unreliable, firmness is sought outside: rigid posture, a tightly structured schedule, repetitive rituals that create predictability, difficulty improvising or tolerating the unexpected. The tense body is also a way of holding on: if I clench hard enough, I won't fall. What this strategy doesn't solve is that chronic tension, besides being energetically costly, dulls proprioceptive sensitivity and makes the body feel less and less like a resource and more and more like a burden.

Bodily disconnection, or dissociation, is another frequent response, especially when one's history includes experiences of physical threat, pain, or intrusion. When the body has become a dangerous place, the system learns to leave it. Not literally, of course, but in the sense of turning down the intensity of bodily signals. Lowering the volume of physical sensation so it doesn't overwhelm. The result is exactly what María's therapist described: someone who knows they have feet but doesn't feel them as a living part of their experience.

And then there is the search for a base outside oneself. When there is not enough inner ground, it is sought externally: in other people's approval, in measurable success, in possessions, in perpetual motion. All of this can feel like a temporary base, but it is never enough, because the need it tries to meet is a need for inner sensation, not outer information. Like trying to quench your thirst with soda: the taste satisfies for a moment, but the thirst comes back.

2.6. The Vicious Circle and How to Break It

All these strategies generate a vicious circle worth describing clearly, because once named, it becomes recognizable.

The less the lower body is felt, the more the mind needs to compensate with activity. The more mental activity, the less presence remains for direct sensation. The less direct sensation, the more diffuse insecurity. The more diffuse insecurity, the more need for control and thinking. And so on, with the impressive efficiency of self-reinforcing loops.

The loop can go on for years without anyone noticing from outside, because the person keeps performing, keeps delivering, keeps functioning. Only when something makes it visible—the crack that can't be ignored—does the opportunity to intervene appear.

What breaks the loop is not, paradoxically, more intellectual understanding. It is not understanding the mechanism better. That helps as context, but it doesn't change the state of the nervous system. What breaks the loop is the repeated experience of the opposite: descending. Bringing attention downward. Feeling the weight of the body on the ground. Breathing into the abdomen. Perceiving the contact of the feet with the surface that holds them.

Not once. Not as a weekend spiritual exercise. But as an everyday practice woven into the ordinary moments of life: on getting up, before difficult meetings, during the commute, when sitting down to eat, before sleep. Repeated signals that the ground is there. That the body has a base. That the sense of safety doesn't have to be built mentally every time, because it is already available in the physical territory of the lower segment.

This practice has effects that go far beyond what one would expect from something so apparently simple. It changes what neuroception—that unconscious process that is constantly assessing whether the environment is safe—detects, because the sensation of being supported by the ground sends signals to the nervous system that alter its state of activation. It changes muscle tone, because the muscles that hold the body up from below can partly relax when the ground is felt to be reliable. It changes breathing, because when there is a base, the breath can descend into the abdomen

instead of getting stuck in the chest. And, over time, it changes something deeper: one's relationship with one's own body. The body stops being that logistical appendage that ferries the head from meeting to meeting and becomes what it always was: home.

2.7. The Transgenerational Dimension

Before closing this chapter, there is one dimension we cannot ignore, uncomfortable as it may be for those of us who prefer to believe we are the starting point of our own story: the wound of the lower segment often does not begin with us. It is inherited.

Not in a magical or metaphorical sense, but in perfectly understandable ways. Attachment patterns are passed down through generations: a mother who was never adequately held in her own childhood will find it harder to hold her baby—not because she doesn't want to, but because she never learned how that is done with the body. A father who grew up in an environment where emotions were dangerous will implicitly pass that same signal on to his children, even if he never says it out loud.

Epigenetics adds an even more surprising dimension: certain effects of sustained stress can be transmitted biologically, through changes in gene expression that do not alter DNA itself but do alter how it functions. Research with survivors of intergenerational trauma—wars, famines, persecutions—has documented biological markers in children and grandchildren who did not directly live through those experiences—for example, among people prenatally exposed to the Dutch Hunger Winter of 1944–1945 and among the offspring of Holocaust survivors (Heijmans et al., 2008; Yehuda et al., 2016). The field is young, and the mechanisms of transmission in humans are still debated, but the direction of the findings is consistent.

This does not mean being condemned by one's ancestors' history. It means that understanding difficulties with grounding sometimes requires widening the lens beyond personal biography. And that the work of building a base is not only individual work but also, in some sense, a work of repair that reaches beyond oneself.

2.8. The Plasticity That Changes Everything

And now we come to the good news, which in this book always arrives in the end, even if it sometimes takes a while. The human nervous system is plastic. It can change. It can reorganize itself. It can learn new ways of regulating itself, even in adulthood, even after decades of maladaptive functioning. It isn't magic. It's neuroplasticity. And it is probably the best news science has given us in the last fifty years. Think about it: your brain is changing right now, as you read this sentence. None of this is a New Age promise: it is biology.

Working with the lower segment in adulthood means, in a sense, offering ourselves the experience of being held that may not have been sufficiently available at other times. Not to rewrite history—that isn't possible—but to update the nervous system's expectations. So that the organism learns, through repeated experience, that the ground is in fact available. That there is a base. That safety doesn't have to be earned through more performance or built through more thinking. Just feeling oneself, from below, from the base, from those feet that have been waiting their whole life to be noticed.

After all, the deepest psychology is not only in what we think about ourselves. It is in how we feel inside our own body. And when that "how we feel" includes the simple but powerful certainty of being supported, everything else—the mind included—can begin to operate from a radically different place.

2.9. The Map and the Territory: Why Understanding Isn't Enough

There is a sentence therapists hear more often than they would like, usually spoken by very intelligent people in a moment of entirely legitimate frustration: "I understand it perfectly. And still nothing changes."

The sentence matters because it points to a real problem that the exclusively cognitive approach to psychology took far too long to recognize: understanding a problem is not the same as solving it. And in the case of grounding difficulties, the distance between intellectual understanding and experiential change can be considerable.

The reason is that conceptual knowledge and changes in the state of the nervous system happen in different circuits. The prefrontal cortex, where intellectual understanding resides, can know perfectly well that there is no real danger. But the amygdala doesn't read books. The amygdala responds to bodily signals, to repeated experiences, to the perceived temperature of the environment, to muscle tone, to the quality of the breath, to the presence or absence of safety signals coming up from the lower body.

Real change, the kind that alters the system's baseline state, requires experience. Not understanding of experience: direct, repeated, felt experience. And that is where work with the lower segment has a structural advantage over exclusively cognitive approaches: it offers direct access to the level where the change needs to happen.

2.10. Attachment, Trauma, and the First Center

In *The Body Keeps the Score*, Bessel van der Kolk (2014) articulated something body therapists had been observing in practice for decades, and which neuroscience took a long time to validate but finally did: the body does not forget. Experiences that were not processed verbally or cognitively at the time do not disappear. They remain recorded in the body as states, as tension patterns, as organizations of the nervous system that repeat with almost perfect fidelity in response to stimuli the system associates, consciously or unconsciously, with the original situations.

This has an important therapeutic implication: talking about a difficult experience is not always enough to switch off the patterns that experience left in the body. One can analyze one's attachment history in great detail, understand the mechanisms of emotional reactivity, and articulate dysfunctional patterns with surgical precision, and the body may still respond exactly the same way to the same stimuli—because the patterns that generate that response are not verbal patterns but physiological ones, and physiological patterns are changed by physiological experience, not by intellectual understanding.

Here lies the deeper reason why work with the lower segment proves effective in situations where purely verbal work had not reached. Not because

it is magic, nor because the body is more important than the mind. But because it gives direct access to the level—the subcortical nervous system, automatic muscular patterns, visceral regulation—where many of the patterns that generate psychological suffering are encoded.

2.11. Healing from Below

Healing the first center is, by definition, a slow process. Not because humans are slow, but because the systems involved—the autonomic nervous system, neuromuscular patterns, somatic memory—reorganize through repeated experience, not through sudden decisions. The nervous system is an organ of learning, and learning takes time. Especially when what is being learned contradicts implicit expectations that have been in place for decades.

What clinical experience suggests again and again is that change begins with small experiences the system can absorb without overload: the brief but clear sensation of the feet on the ground, a breath that reaches the abdomen and produces not activation but a slight calm, the weight of the pelvis felt more distinctly than usual. Nothing lasting more than a few seconds at first. The system needs to learn that this is safe before it can go deeper.

Over time, those small experiences accumulate. The system gradually develops what attachment theory calls an internal secure base (Bowlby, 1988): a sense of support that no longer depends exclusively on external circumstances but is available from within, from one's own body. Not always, not perfectly, not without effort. But consistently enough to make a real difference to the quality of everyday experience.

That is what María gradually found. Not all at once, not with fanfare, not in one of those transformative sessions that fill self-help books. But gradually, unevenly, with setbacks and advances, with good days and days when the anxiety came back as always—and days when the anxiety came back but she could feel her feet, and that changed something. Slow but real. The way roots grow.

CHAPTER AT A GLANCE

- **Basic safety is built implicitly** in early development through attachment (Bowlby; Winnicott).
 - **The silent wound:** the base can be disturbed without dramatic trauma—chronic unpredictability or emotional absence may suffice.
 - **Common compensations:** hypermentalization, postural rigidity, dissociation, and seeking the base outside oneself.
 - **A self-reinforcing loop** results, which understanding alone does not break; repeated felt experience does.
 - **Patterns can be transmitted across generations**—and they can change through neuroplasticity.
-

CHAPTER 3

Interrelations of the Segment

The body as an integrated system. The silent conversation between below and above.

KEY CONCEPTS fascia · psoas · diaphragm · pelvis · tensegrity · proprioception

One of the great illusions of modern thought is to imagine that the body works like a collection of independent parts. Bones on one side, muscles on the other, organs here, nerves there, and emotions floating somewhere ambiguous that nobody quite knows where to place on the org chart. A sort of biological construction kit in which each element could be studied and treated in isolation, as if the quadriceps knew nothing of the diaphragm and the diaphragm had no opinion about what the jaw was doing.

Classical anatomy has contributed a good deal to this illusion, partly because it needed to simplify in order to teach, and simplification has its pedagogical virtues. But when we observe the body in motion—or, more revealing still, when we observe someone who cannot move freely—that fragmented view falls as short as an umbrella in a hurricane. The human body does not work like a collection of watertight departments. It works like a single conversation carried on in many dialects at once.

And within that conversation, the lower segment—the region from the feet to the pelvis, including the sacrum, the pelvic floor, and the lumbar area—occupies something like the engine room: not the most visible place, nor the one that gets the most visitors, but the one that determines whether ev-

everything else can work fluidly or is forced to keep compensating for a base that doesn't provide enough support.

This chapter is a map of those interrelations. A tour of the conversations the lower segment holds with the rest of the organism, and of the ways its silence or dysfunction sets off a domino effect that climbs up the body with a regularity that no longer surprises any body therapist, but which still proves revealing for most people who discover it for the first time in their own experience.

3.1. The Body as a Tower with Its Own Foundations

Imagine for a moment that your body is an architectural structure. Not a modern steel-and-glass skyscraper, but something more organic: a stone tower built over centuries, with additions, restorations, and adaptations reflecting the conditions of each age. The lower segment would be the foundations and the ground floor: the place where all the weight rests, where pressure coming up from the ground and pressure coming down from above are managed, where the building is anchored to the earth.

If that ground floor is barely inhabited—if attention almost never comes down that far, if muscle tone is uneven, if support feels vague and uncertain—the rest of the structure has to compensate. The upper floors become stiffer, as if trying to give themselves the stability the foundations don't adequately provide. The shoulders rise a little. The neck juts forward, turned into an anxious periscope trying to get a better view of a horizon that, paradoxically, grows blurrier the closer we lean toward it. The jaw clenches. The breath rises and gets trapped in the chest, like a bird that doesn't dare come down to the ground.

The whole building is still standing. The lights work. But there is a background creaking that nobody can quite locate, and a persistent sense that something doesn't quite fit.

This compensation is not a system error. It is a brilliant adaptation. The nervous system, always pragmatic and always putting immediate survival ahead of long-term efficiency, looks for the fastest way to maintain balance when the base doesn't provide it. The problem is that compensating is ex-

pensive. A structure that is constantly compensating uses far more energy than one that simply rests well on its base. And that bill—energetic, muscular, neurological—is paid sooner or later in the form of chronic fatigue, aches that appear for no apparent reason, tensions no massage ever fully resolves, and that vague feeling of always being slightly “above” oneself, slightly on the edge of something that never quite happens.

3.2. The Pelvis: The Great Organizer Nobody Consults

At the center of all these interrelations—literally at the geometric center of the body—is the pelvis. It deserves time and space, because it usually gets far less than it merits, except when it hurts, when some specific function fails, or when someone with inside knowledge points to it as the source of problems nobody had connected with it.

The pelvis is not just a bony structure that supports the trunk. It is a postural organizer, a load distributor, a bridge between the lower and upper body and—this is what classical anatomy books mention least—a territory of emotional expression with a density and richness few other body segments can match.

In purely mechanical terms, the pelvis acts as the central knot of two great chains of movement. From below, it receives the supporting forces that rise from the feet through the legs. Every step generates mechanical impulses that travel up from the heel, through the ankle, along the tibia, across the knee, and into the hip joint, where the pelvis receives them, distributes them, and passes them on to the spine. From above, it receives the weight of the trunk, arms, and head, which comes down the spine and reaches the sacrum. The pelvis is constantly managing these two currents of force, and does so with a biomechanical sophistication any structural engineer would admire.

For this to work well, the pelvis needs to be in what anatomists call a neutral position: neither tipped excessively forward nor rolled back. This neutral position allows the curves of the spine to keep their optimal shape, the hip and pelvic floor muscles to work efficiently, and loads to be distrib-

uted evenly. When the pelvis chronically leaves neutral, everything above and below it has to adapt, and those adaptations leave their mark.

Exaggerated anterior pelvic tilt—the pelvis tipped forward, the pubis dropping, the tailbone lifting—increases the lumbar curve and shortens the hip flexors. The body takes on an arrangement that often conveys readiness for action, alertness, drive, a certain difficulty in fully resting. Not because posture on its own causes a psychological state with simple cause-and-effect mechanics, but because posture and emotional state reinforce each other in a cycle that can run in either direction: changing posture changes state, and changing state changes posture.

Posterior tilt—the pelvis rolled back, the tailbone dropping, the lumbar curve flattened—produces the opposite: a body that seems to withdraw into itself, that shrinks its presence in space, that gives the impression of wanting to become smaller. This organization often appears alongside patterns of collapse, inhibition, or withdrawal. Not always, of course. The body is not an anatomical horoscope, and any honest therapist will admit that the correspondence between posture and psychological state is a statistical tendency, not a universal law. But it is a tendency consistent enough to deserve attention.

3.3. The Psoas: The Muscle That Takes Fear Literally

Among all the actors in this story of interrelations, there is one that deserves a chapter of its own within the chapter. The psoas major is quite possibly the most important muscle most people have never heard of.

It originates on the vertebral bodies and discs of T12 to L4—the transition between the lower thoracic and lumbar regions—runs down through the abdominal cavity, passes in front of the hip joint, and inserts on the lesser trochanter of the femur. This path makes it the only muscle that connects the lumbar spine directly to the femur: the principal muscular bridge between above and below, between the trunk and locomotion. (The piriformis, as we will see in Chapter 6, links the sacrum to the femur, but it neither crosses the abdominal cavity nor reaches the lumbar vertebrae.)

Its mechanical functions are many: it flexes the hip, contributes to the stability of the lumbar spine, takes part in transferring loads between the pelvis and the trunk, and plays a role in walking that is still being studied in detail. But what makes it especially relevant to this chapter is its relationship with the nervous system and with stress responses.

The psoas has a very particular neighborhood. It lies in close proximity to the solar plexus, the kidneys, and the adrenal glands. It runs right alongside the abdominal aorta. It shares fascia with the diaphragm. And it lies alongside the lumbar sympathetic chain. In clinical experience, all of this makes it remarkably responsive to the organism's state of activation—an observation widely shared among body therapists (Koch, 2012), though one that controlled research has only begun to examine.

When the nervous system detects threat and triggers the stress response, the psoas contracts. It is part of the preparation for flight or fight: it shortens the distance between trunk and legs and readies the organism for rapid movement. The problem, which is starting to sound familiar, is that this mechanism was designed for acute, physical threats, not for the chronic low-level stress that characterizes contemporary life.

When the psoas stays chronically contracted for months or years—because the nervous system has not received enough safety signals to let it go—the consequences spread in several directions. Mechanically: shortened hip flexors, increased lumbar curve, compression of the lumbar discs, restricted mobility of the hip and sacrum. Physiologically: compression of neighboring structures, altered circulation in the abdominal and pelvic region, effects on digestive and respiratory function. And perceptually: the sensation so many people describe as “not being able to come all the way down,” as if something were pulling upward from the center of the body and wouldn't let attention and presence fully descend.

Releasing the psoas is among the most transformative—and most delicate—tasks in work with the lower segment. Delicate because it is not just a tight muscle that needs stretching: in many cases it has been the guardian of a survival response for years, and when it lets go it can bring with it emotions that were stored in that tension. Not always. Not dramatically. But of-

ten enough that experienced body therapists know to create a context of safety before working on it.

3.4. Fascia: The Network Missing from the Textbooks

For decades, fascia was the forgotten tissue of anatomy. It was there, of course, but it was considered packaging of little functional interest: the plastic wrap on the anatomical sandwich, to be removed to get to the important parts. Dissection manuals cut it away and set it aside so the muscles and bones could be seen. It was, as we would say today, rendered invisible.

We now know that this anatomical indifference was deeply unfair and deeply costly to our understanding of the body. Fascia is not packaging. It is a network. A continuous, three-dimensional web that wraps, connects, and organizes muscles, viscera, nerves, and bones into an integrated structure with no clearly defined beginning or end. Researchers such as Robert Schleip have shown that it is richly supplied with mechanoreceptors and free nerve endings—so richly that some describe it as one of our largest sensory organs (Schleip, 2003). The precise estimate is still debated; the direction of the research is not. The cell biologist Donald Ingber's concept of tensegrity—structures that hold their shape through a balance of continuous tension and discrete compression—offers a useful model for this kind of integrated architecture (Ingber, 1998).

The work of the manual therapist and anatomy educator Thomas Myers, systematized in his concept of *Anatomy Trains* (Myers, 2001), was decisive in changing this perspective. Myers described several myofascial chains running through the organism, linking regions that classical anatomy described as independent. These chains are not anatomical structures in the traditional sense—they cannot be dissected the way a muscle is dissected—but pathways along which tension and movement are transmitted through the fascial web.

The most relevant one for the lower segment is the Superficial Back Line, which connects the plantar fascia on the sole of the foot with the Achilles tendon, rises through the calf muscles, crosses the back of the knee, climbs the hamstrings, continues along the thoracolumbar fascia, ascends

the paravertebral muscles, passes over the skull, and ends at the supraorbital fascia above the eyes. In other words: tension in the plantar fascia can, through this chain, influence the posture of the whole back and even the position of the head. And vice versa: a chronic restriction in the lumbar paravertebrals can increase tension in the hamstrings and change the mechanics of the foot.

This understanding has enormous clinical implications. A pain someone feels in the neck may have its origin—or part of its origin—in a fascial restriction in the lower back, or even in the feet. A problem with how the foot meets the ground can contribute to a chronic tension that climbs the entire back line. The body, as it insists on reminding us every time we look at it closely, is not a sum of parts. It is a continuous conversation.

3.5. Breathing as a Messenger Between Above and Below

If there is one bodily function that reflects especially clearly the state of the lower segment and its relationship with the rest of the organism, it is breathing. Not because it is more important than other functions, but because it is at once automatic and voluntary, involuntary yet open to conscious influence. It is the only physiological process controlled by the autonomic nervous system that we can also modify with attention. And it is, as a result, one of the most accessible doorways to the nervous system.

When the lower segment is disconnected, when the pelvis is locked and the psoas tight, the breath inevitably tends to rise. The diaphragm, which should descend fully with each inhalation to massage the abdominal organs and create room for the trunk to expand, works at half power. The breath stays in the chest, shallow and fast, sending the organism exactly the signal the nervous system interprets as an alert: more movement above, less below, an elevated breathing rate.

This kind of breathing doesn't only affect the oxygen reaching the tissues. It directly affects the balance of the autonomic nervous system. High, rapid breathing preferentially activates the sympathetic system, keeping the organism in a state of mild but persistent alertness—that all-too-familiar feeling of always being “on the verge of something,” of never quite being

able to let one's guard down. It's like a car engine idling too high: it burns more fuel, generates more heat, and, sooner or later, causes wear that has nothing to do with the actual effort.

Breathing work for the lower segment does not mean learning to breathe "correctly" in a prescriptive sense—there is no perfect breath to imitate. It means allowing the breath to drop. Letting the abdomen expand on the inhalation. Giving the diaphragm room to descend. Letting the lower body receive the signal that deep, lowered breathing sends to the nervous system: you are safe. You can let go. There is a base.

3.6. Emotional Interrelations: Posture as Biography

The interrelations of the lower segment don't stop at mechanics or physiology. They also have—and this is what most surprises people who have only thought of the body in physical terms—an emotional dimension that is impossible to ignore once you start to see it.

There is a line attributed to Ida Rolf, the founder of Rolfing and one of the most lucid pioneers of connective-tissue work, worth quoting even at the risk of turning it into a motivational poster: "Posture is a person's biography." Not in the sense that five seconds of looking at someone is enough to reconstruct their childhood with surgical precision—that's what airport books on body language do, and we should not be confused—but in the deeper sense that repeated emotional experiences leave their mark on the body's organization. The body is the embodied memory of everything we have lived (cf. Rolf, 1977).

A body that learned to be on alert from an early age tends to raise the shoulders slightly, shorten the breath, keep the psoas permanently half-engaged, and organize the pelvis defensively. Not as the result of a conscious decision—nobody decides to adopt that posture—but as the accumulated result of thousands of moments in which the nervous system registered threat and activated the corresponding response. Over time, that response became automatic and turned into the baseline. The alarm posture simply became the posture.

A body that has learned to inhibit itself, to make itself small, not to take up too much space, tends to show the opposite: a sunken trunk, a posteriorly tilted pelvis, the head dropped forward, little breadth of gesture. A way of being in the world that says, without words and without intention, “I’d rather not be too much here.”

And a body that has found enough base and safety tends to show something different: not rigidity or postural showing off, but presence. A pelvis that can move freely in every direction. A spine with its natural curves, unexaggerated. Shoulders that rest. A breath that reaches the abdomen. A head that floats on the neck without having to be held up by muscle. A body that occupies the space that belongs to it without asking anyone’s permission.

This emotional dimension of bodily interrelations has a fundamental therapeutic implication: changing posture is not merely mechanical work. At its deepest layers, it is a revision of one’s embodied history. When someone begins to feel their feet more clearly, when the pelvis recovers some mobility, when the breath begins to drop, it is not only biological changes taking place. Something is happening that we might call—with all the nuance the term requires—an update of state. The body begins to register different information: the ground holds. I don’t have to hold myself up from above. I can come down.

3.7. Rhythm and Everyday Movement: The Work That Never Stops

There is one dimension of bodily interrelations that rarely appears in books on somatic work or in corporate wellness programs, and yet it is probably the most important in terms of real impact on the state of the lower segment: ordinary everyday movement.

Not Sunday-morning yoga. Not Pilates twice a week. Not a forty-five-minute personal training session at the gym. But walking, going up and down stairs, squatting to pick something up off the floor, carrying things, scrubbing, making the bed, sitting on the floor and getting up again. The movements that for centuries were the main activity of human life, and which modernity has been eliminating with an efficiency that is both impressive and impoverishing.

Throughout their evolution, human beings kept the lower segment active and calibrated simply by living. Walking on uneven surfaces—mud, stones, grass, sand—demands constant proprioceptive work from the feet and ankles that keeps the nervous system of the lower segment permanently active and updated. Squatting and standing up activates the glutes, the spinal erectors, and the whole stabilizing system of the sacrum. Carrying objects in different positions trains the capacity to distribute loads. The lower body had work to do, rich and varied sensory information, and that constant work was also the most effective way of keeping the lower segment calibrated with the rest of the organism.

Contemporary sedentary life has changed this radically. We spend hours on flat, regular surfaces—carpet, hardwood floors, smooth concrete sidewalks—that give the feet very little sensory information. We sit in chairs that remove any need to engage the pelvic floor and glutes. We get around in vehicles that remove the locomotor load from the legs. And then, to compensate, we go to the gym to do exercises the body would do naturally if we lived the way we were designed to live.

It isn't that exercise is bad. Exercise is very good. But there is a qualitative difference between movement as an isolated practice and movement as the continuous texture of life. The first activates the lower segment for an hour. The second keeps it active all day. The difference in terms of nervous-system calibration, baseline muscle tone, and the sense of rootedness is considerable.

That is why the most efficient and least glamorous advice for the lower segment is also the most forgotten: walk more. Walk without earphones some days, so the nervous system can process information from the environment without distraction. Walk on uneven ground when you can, because imperfect ground is far more interesting to the proprioceptive system than a perfectly flat floor. Take the stairs. Squat to pick things up off the floor instead of bending from the waist. Sit on the floor now and then. Not because it is therapy in any formal sense. But because the body was designed for it, and modern life has turned it into a noteworthy exception.

3.8. The Body That Remembers Without Words

Let us take Ida Rolf's idea one step further. If posture is biography, it is not a metaphor for it, nor a symbolic representation. It is the real biography, written in tissue, in posture, in muscle tone, in the quality of movement, and in the breathing patterns accumulated over decades of experience.

If you spend enough time working with bodies—and in three decades you work with a great many—you begin to develop a capacity to read them that is as comfortable to have as it is unsettling to explain: posture tells things that words have not yet put into form. The body that has spent decades on low-grade alert has slightly raised shoulders, a tight neck, a short breath, a pelvis in a discreet posterior tilt that communicates, wordlessly, a kind of permanent retraction. The body that has learned to take up a lot of space shows an expansion upward and outward that may be presence or may be compensation, and learning to tell the difference is part of the body therapist's art.

But the most revealing reading doesn't come from looking at the body at rest. It comes from watching how it moves. Or, more precisely, how it doesn't.

3.9. The Missing Movement

Modern human beings have a remarkably impoverished movement repertoire compared with the one we used a hundred years ago, and more impoverished still compared with ten thousand years ago. Flat, uniform surfaces have removed the proprioceptive demands of uneven terrain. Chairs have removed the need to keep the pelvic floor and glutes engaged. Vehicles have removed the movement of getting from place to place. The specialization of work has reduced human gesture to a very small set of repeated movements—typing, driving, looking at screens—in which the lower segment appears, if at all, as an accessory, never as the protagonist.

The result is not just a loss of physical fitness. It is a loss of nervous-system calibration. The sensory receptors in the feet, the hip joints, the pelvic floor muscles, and the proprioceptors of the sacrum and pelvis need contin-

uous, varied information to maintain the calibration that allows the organism to feel in contact with the physical reality of the world. When that information becomes impoverished—when the feet only know flat floors, when the pelvis only knows the seated position, when the legs only know the linear movement of walking on asphalt—the system loses the richness of its body map.

And an organism with an impoverished body map has more trouble finding itself when it goes looking. The feet don't feel right because they haven't received enough information for the nervous system to represent them clearly. The pelvis isn't perceived distinctly because it has spent too long in the same position for the system to have needed to update it. The lower body becomes, in the most literal sense, unknown territory.

3.10. Fascial Networks: The Message That Travels Without Words

One of modern anatomy's most important contributions to understanding the lower segment is our grasp of fascial networks and of how tension is transmitted through them in ways that transcend the segmented, local anatomy still taught in medical schools.

As we saw in Section 3.4, Thomas Myers's work on *Anatomy Trains* showed that the organism has pathways for transmitting tension and movement that connect apparently distant regions: the sole of the foot with the occiput, the deep fascia of the thigh with the diaphragm, the pelvic floor with the base of the skull. These lines are not arbitrary. They follow an evolutionary and functional logic that becomes evident when we watch the integrated movement of the body in activities that require coordination of the whole organism: walking, jumping, swimming, dancing.

What this means for the lower segment is that a restriction in the plantar fascia of the feet is not simply a local problem. It is a message that travels upward along the back line, tightening the hamstrings, increasing the load on the lower back, contributing to stiffness of the sacrum, and, potentially, reaching the base of the skull as tension in the suboccipitals that shows up as chronic tension headache. The person who goes to the doctor with a headache that starts at the nape of the neck might never imagine that part of

the solution could lie in working on the plantar fascia and the support of the feet.

This understanding also reverses the direction of the work: we don't work with the lower segment only to influence it directly. We work with it because changing tension at the base changes the entire network. Like loosening the tightest knot in a fishing net: when that point of greatest tension gives, the whole net redistributes its shape.

3.11. The Dialogue Between Posture and Emotion: Who Calls Whom?

One of the most frequent questions in work at the intersection of psychology and bodywork is: does posture reflect emotional state, or does emotional state reflect posture? The answer—which neuroscientists took a long time to formulate but body therapists observed decades ago—is that the relationship runs both ways with equal ease.

The basic experiment is simple: ask someone to hold a slumped posture for several minutes—back curved, head dropped, shoulders closed—and watch what happens to their mood. In one randomized trial, participants assigned to a slumped posture reported lower self-esteem, more negative mood, and greater fear under stress than those sitting upright (Nair et al., 2015). The nervous system interprets posture as information about the state of the organism and adjusts the emotional state accordingly.

The same works in reverse: ask someone to hold a posture of expansion and openness for several minutes, with the spine lengthened and the shoulders open, and mood generally improves modestly. Some of the more spectacular claims made in this field—the so-called power poses, for example—have not held up well in replication studies, a useful reminder to keep expectations proportionate. This does not mean posture is an antidepressant—some conditions require far more specific interventions—but that posture and emotional state are in continuous conversation, and changing one changes the other.

For the lower segment, this has very concrete implications: working on pelvic alignment, the support of the feet, and the tone of the pelvic floor is

not just physical work. It is work that sends the nervous system signals about the state of the organism—signals interpreted as information about the safety of the environment, which change the tone of the autonomic system.

3.12. When the Body Reports and the Brain Listens

In recent years, the neuroscience of interoception has given us a much more precise understanding of the mechanisms through which the lower body influences mental state. The anterior insula, a brain region tucked deep in the Sylvian fissure, acts, according to the influential model proposed by A. D. Craig (2002, 2009), as the integration center for information coming from the body: heartbeat, muscle tension, body temperature, pressure on the soles of the feet, visceral tone. All this information reaches the insula, is integrated there subjectively, and emerges as what we experience as “how I feel.”

What this means is that feeling your feet clearly is not just an exercise in attention. It is a way of giving the insula rich, up-to-date information about the state of the lower body. And when that information says “there is support, there is ground, there is contact with something solid,” the insula processes it as part of its continuous assessment of the organism’s state—an assessment that contributes to what the nervous system interprets as a sense of safety.

3.13. The Way Back Is Not Upward

The title of this book—*The Brain Doesn’t Float*—can be read in different ways. The most obvious: the brain, contrary to what the way we use it might suggest, does not float in a vacuum. It is embodied. It depends on the body. It cannot work well if the body holding it up is not sufficiently organized, nourished, and present.

But there is another possible reading, subtler and perhaps more important: the contemporary problem is not that the brain floats because the body is in poor shape. It is that we have learned to live as if the brain floated—as if psychological experience took place somewhere separate from the

physical weight of the body, from its sensations, from its contact with the world.

The way back, then, is not upward—toward more thinking—but downward, toward the base. The chapters of Part II will show, system by system, why that descent is anatomically, hormonally, and neurologically well founded.

CHAPTER AT A GLANCE

- **The body is an integrated system:** a weak base produces compensations higher up—shoulders, neck, jaw, and breath.
 - **The pelvis** organizes forces from above and below; its tilt both reflects and reinforces emotional states.
 - **The psoas** links the lumbar spine to the femur and responds readily to threat.
 - **Fascia** transmits tension along myofascial lines, from the soles of the feet to the skull.
 - **Posture and emotion** influence each other in both directions; the insula integrates bodily signals into “how I feel.”
 - **Varied everyday movement** keeps the body map calibrated.
-

PART II

From the Sciences

Anatomy, hormones, nerves, and the brain of grounding

The lower segment is not a metaphor: it is anatomy, chemistry, and
neurology.

This second part travels through the body from the pelvis to the
neurotransmitters that regulate the sense of safety.

Five rigorous but accessible chapters that back up with data what
clinical intuition had been pointing to for decades.

Here you will find the scientific basis of grounding: bones, muscles,
hormones, nerves, brain networks, and the bridge to the yogic
tradition.

CHAPTER 4

Musculoskeletal Anatomy

The foundations of the building. The bones, muscles, and joints that hold up the self.

KEY CONCEPTS pelvis · sacrum · psoas · pelvic floor · sacroiliac joints · glutes

So far we have talked about emotions, early development, anxiety, attachment, energy maps, and all those matters that fascinate the mind because they let it believe it understands everything without breaking a sweat. But the time has come to do something far less glamorous and far more useful: to go down into the body. The real body. The one with bones you can touch, muscles that tense and relax, joints that work well or complain, and fascia that connects everything silently with an efficiency that puts any corporate org chart to shame.

Because Muladhara—the first chakra, the lower segment—is not a vaporous idea floating on a cloud of incense, however good incense looks in photographs. Muladhara is bone, muscle, fascia, joint, weight, and gravity. It is matter. Blessed, concrete, measurable matter and, when you work with it, profoundly transformative.

Seen from the body, safety does not begin with a reassuring idea or a well-chosen mantra. It begins with a structure capable of holding us up. And that structure rests on a set of elements that work together with a discipline we ought to admire more: bones that distribute loads, muscles that organize tone, joints that absorb and transmit force, and connective tissues that silently link everything without asking for any recognition. This chap-

ter is a tribute to that invisible architecture and an invitation to get to know it better, because it is very hard to relate well to something you don't know.

4.1. The Sacrum: The Bone the Romans Called Sacred—and They Weren't Exaggerating

We begin with the least known and most fundamental protagonist of the lower segment: the sacrum. That triangular bone that supports the spine and that most people couldn't spell, let alone locate. Ask someone on the street where their sacrum is and they will look at you as if you had asked for the capital of Uzbekistan. And yet without it we couldn't walk. Literally. Everything would collapse. Its name is a historical clue: the Latin *os sacrum* renders the Greek *hieron osteon*, "sacred bone," a term already used by Galen. Why the ancients called it sacred is still debated—perhaps because of its role in sacrificial offerings, perhaps because it was believed to resist decay—but the intuition that this bone deserved special respect turns out to be anatomically justified.

The sacrum is formed by the fusion of five sacral vertebrae which, during embryonic development and the first years of life, gradually consolidate into a single piece. Its triangular shape is no accident: it acts as a wedge between the two iliac bones of the pelvis, distributing compressive forces and helping to stabilize the whole pelvic region with an efficiency structural engineers would love to have invented. At the top it articulates with the fifth lumbar vertebra at the lumbosacral junction, one of the most heavily loaded points in the entire spine. At the bottom it ends in the coccyx, that small caudal appendage that was once a tail in evolutionary terms and which in modern humans serves mainly to remind us of its existence when we sit down hard on a solid surface.

The sacrum receives the weight of the entire trunk—which is a fair amount of weight, let's admit it—and passes it on to the pelvis and, from there, to the legs and the ground. It is the meeting point of the downward forces of the trunk and the upward forces of the legs. When it works well, almost nobody thinks about it. It is one of those tireless workers who only get attention when they complain. And when the sacrum complains, it does so

with a remarkable talent for complicating activities that once seemed trivial: getting up from a chair, turning over in bed, tying your shoelaces, or simply standing for more than twenty minutes without your lower back starting to remind you it exists.

One movement of the sacrum deserves special mention, because it is little known outside manual therapy and osteopathic circles yet has important implications for work with the lower segment: nutation and counternutation. Nutation is the movement in which the base of the sacrum—its upper part—tips forward and down while the coccyx moves back. Counternutation is the opposite movement. These small shifts of the sacrum occur rhythmically during walking and also with deep breathing. They help absorb impact, distribute loads, and keep the sacroiliac joints working efficiently. When the sacrum loses this rhythmic mobility—through chronic muscle tension, restrictive postural patterns, or physical trauma—the effect spreads up the spine and down into the legs with the unfortunate efficiency of every bodily domino effect.

4.2. The Sacroiliac Joints: Where the Body Negotiates

The sacrum joins the iliac bones through the sacroiliac joints: two ear-shaped joint surfaces—hence the anatomical term auricular surfaces—that fit between the sacrum and each iliac bone. For decades the medical community debated whether these joints actually moved at all. The current consensus is that they do, but very little: their range of motion is measured in millimeters and minimal degrees of rotation, not in the wide arcs of the hip or shoulder.

But small does not mean irrelevant. That minimal but real movement is decisive for absorbing the forces that rise from the legs during walking, adapting to changes in posture, redistributing loads toward the spine, and allowing the pelvic mobility of childbirth in women. When these joints lose their small range of movement—because of chronic muscle tension, prolonged inactivity, or fascial stiffness—compensations appear that can show up as pain in the lower back or buttock, or even radiating down the back of the thigh in a way that is often mistaken for sciatica.

One of the most common and most often misdiagnosed ironies of the lumbar region is this: many people who believe they have “lower back pain” in fact have sacroiliac joint dysfunction. The pain is there, in the lower back, and the conclusion seems obvious. But the source may lie in the joints connecting the sacrum to the pelvis, not in the discs or lumbar vertebrae. The solution, accordingly, may be completely different.

4.3. The Pelvis: The Great Bowl Nobody Writes Stories About

If the sacrum is the keystone of the arch, the pelvis is the great structural bowl of the lower segment. It is formed by two hip bones—one on each side—which join the sacrum at the back through the sacroiliac joints and each other at the front at the pubic symphysis. Each hip bone is in turn the fusion of three bones that are separate in childhood and fuse for good during adolescence: the ilium, the ischium, and the pubis.

The ilium is the upper, flared part of the pelvis, which we can feel as the iliac crests on either side of the waist. When someone asks us to put our hands on our hips, we usually rest them on the iliac crests—though in fact we are considerably lower than where many people think “the waist” is, which tends to sit higher in the popular imagination than in actual anatomy.

The ischium forms the lower, back part of the hip bone. Its ischial tuberosities—the “sitting bones,” as they are colloquially and quite accurately called—are the bony points that bear the body’s weight when we sit properly. Sitting on the sitting bones, with the pelvis in neutral, is the basis of a seated posture that doesn’t overload the lumbar spine. Sitting on the sacrum or coccyx, which is what happens when the pelvis slumps back into posterior tilt in a chair, is a recipe for lower back pain, tight hamstrings, and that squashed-down feeling that comes with long hours of desk work.

The pubis forms the front of the pelvis. The two pubic bones—one on each side—meet at the pubic symphysis, a fibrocartilaginous joint that normally allows very little movement but which during pregnancy can soften considerably under the influence of relaxin, in some cases causing the pelvic girdle pain that accompanies many pregnancies.

The pelvic cavity—the inside of that bony bowl—contains the pelvic organs: bladder, rectum, and, in women, the uterus and ovaries. It is a space of containment and protection, but also one dense with nerves, blood vessels, muscle, and fascia. Many sensations felt in the lower belly—pressure, heaviness, warmth, emptiness, tension—arise from the interplay between those organs, the muscles around them, the nerves that supply them, and the state of the autonomic nervous system at any given moment.

4.4. The Feet: The Base of the Base, or What the Body Touches First

And now we come to the feet. Poor things. The only organ in the body we keep locked in darkness eight months of the year, show off for two months as if it were a work of art, and spend two more apologizing for being “a bit dry.” If feet could complain, they would. Because, let’s face it, we squeeze them into shoes that put looks before function with a consistency that would be admirable if it didn’t have so many orthopedic consequences. And we expect flawless efficiency from them for decades while barely investing in their care or in understanding how they work. It’s like owning a race car and only ever taking it to the village garage... every five years. When it won’t start.

And yet they are the most direct physical base of our rootedness. The permanent point of contact with the ground. The place where gravity anchors us and where the ground’s reaction force propels us. The first receivers of all the sensory information coming from below.

Each foot contains twenty-six bones—more than a quarter of all the bones in the human body are in the two feet together—thirty-three joints, and more than a hundred muscles, tendons, and ligaments. This is not a passive platform that simply bears weight. It is a highly sophisticated structure designed to do apparently contradictory things at the same time: to be stable enough to support the body’s weight and adaptable enough to respond to uneven ground; to absorb impact during the stance phase of each step and propel the body forward at push-off; to inform the nervous system about the ground and carry out the balance adjustments that information calls for.

The arches of the foot are perhaps the body's most underrated biomechanical stroke of genius. The medial longitudinal arch—running from the heel to the base of the big toe and creating that empty space you can see in the footprint of a healthy foot—works like an elastic spring that stores energy during stance and releases it at push-off, significantly reducing the energy cost of walking. The transverse arch and the lateral longitudinal arch complement this function, creating a three-dimensional system for absorbing and distributing loads that no high-tech sneaker has yet surpassed.

When the arches are balanced—neither collapsed nor excessively rigid—walking is efficient, economical, and comfortable. When they collapse, the mechanical load is redistributed in ways that overwork structures not designed for it: the knees rotate inward, the hip loses alignment, the lower back compensates. And when they become too rigid—the high-arched or cavus foot—shock absorption is reduced and impact travels straight up to the knee, hip, and spine.

The sensory receptors of the feet deserve special mention. The sole is densely populated with mechanoreceptors—receptors that detect pressure, vibration, and mechanical deformation—and with proprioceptors that tell the nervous system about the foot's position in space and the qualities of the ground it touches. This information travels up to the central nervous system continuously, contributing to the regulation of balance, to postural adjustment, and—most relevant for this book—to the sense of being supported.

When the feet spend all day enclosed in rigid shoes on uniform surfaces, the amount of sensory information reaching the nervous system drops dramatically. It's like trying to read wearing boxing gloves: technically possible, but with a loss of resolution that has consequences. A nervous system that receives too little information from the feet loses calibration in its assessment of support, and that loss contributes to the sense of floating, of having no base, of never quite having landed, which so often appears in people who lead very sedentary or very indoor lives.

4.5. The Glutes: The Forgotten Nobility of Standing Upright

We now turn to a muscle group that has gained a certain aesthetic popularity in contemporary culture but remains deeply misunderstood in functional terms. The glutes. Together, the gluteus maximus, medius, and minimus form the most powerful musculature in the human body, and their role in grounding is fundamental.

The gluteus maximus is the largest muscle in the human body. Yes, that one. The one we have turned into an aesthetic obsession, the subject of gym selfies, a measure of success in life. And yet most people don't feel their glutes. They can't contract them at will. It's like having the most powerful engine in the car and not knowing where the key is—an oversight that, it must be admitted, keeps a good many therapists in work.

The gluteus medius and minimus play a specific role in stabilizing the pelvis laterally. When we walk or stand on one leg, these muscles stop the body from collapsing toward the side of the unsupported leg. The Trendelenburg test—which physical therapists use to assess the strength of these muscles—consists precisely of standing on one leg and seeing whether the pelvis stays level or drops toward the side of the raised leg. When the gluteus medius is weak, the pelvis drops. And when the pelvis drops with every step, compensations accumulate in the hip, knee, and spine.

What is striking is how many people have underactive glutes despite decades of walking on them. A sedentary lifestyle and long hours in chairs with the trunk leaning forward hold the glutes in a lengthened position and progressively inhibit them. Technically present. Functionally absent. When they stop doing their job, the lower back takes on tasks that aren't its own. And therapists who work with chronic lower back pain know that in a significant share of those cases, strengthening the gluteus medius is as important as working on the spine directly—or more so.

4.6. The Pelvic Floor: The Structure That Deserves More Conversation Than It Gets

Few structures in the human body receive so little attention until they fail as the pelvic floor. The name has an almost geographical clarity, yet the structure itself is rarely discussed outside specialist settings, perhaps because naming that lower territory precisely still causes a certain cultural discomfort that has more to do with the history of our relationship with the body than with anatomy itself.

The pelvic floor is a set of muscles and fasciae that closes off the bottom of the pelvic cavity, forming a kind of three-dimensional hammock that supports the bladder, the rectum, and, in women, the uterus. It is made up of several overlapping muscle layers organized around the openings through which the urethra, the rectum, and, in women, the vagina pass. Its coordinated functioning is essential for urinary and fecal continence, for lumbopelvic stability, for breathing, and for sexual function.

What is rarely mentioned outside specialist settings is its relationship with the state of the nervous system. The pelvic floor responds reflexively to the organism's state of activation: when there is perceived threat, it tends to contract as part of the overall protective response. When there is safety and calm, it tends to release. A person living in chronic alert very often has a hypertonic—overly tight—pelvic floor that contributes to the sense of pelvic rigidity, of being unable to let go, of having the center of the body permanently “held.” And that holding, which began as a form of protection, ends up also standing in the way of deep relaxation, of rest, and of the sense of having a base.

4.7. The Lumbar Spine: Where the Body Carries What It Can't Set Down

The lumbar spine, with its five vertebrae, performs an underrated mediating role. It transmits the weight of the entire trunk to the pelvis and, from there, to the legs and the ground. Each intervertebral disc works like a hydraulic shock absorber designed to absorb compression and redistribute it. We are the only species that carries its whole spine on two points of sup-

port, and that evolutionary detail has a mechanical price the lumbar discs know only too well.

The physiological lumbar curve—the lordosis—allows loads to be distributed more efficiently. When the curve is exaggerated or flattened, pressures concentrate in places not designed for them. Disc herniations, which so often cause sciatic pain, usually occur under repeated loading in positions that don't respect the spine's natural curve.

There is a clinical observation worth stating plainly, because it connects the anatomical and the psychological directly: the lower back tends to accumulate tension when a person is carrying loads they cannot set down—in both the most literal and the most metaphorical sense. People who find it very hard to delegate, who habitually bear more than their share, whose responsibilities pile up with no room for relief, often show chronically tight lumbar paraspinal muscles. A sustained emotional state changes muscle tone, and that altered tone changes mechanics. Body and mind are not two things that interact. They are two languages for the same thing.

Knowing this architecture—the sacrum, the sacroiliac joints, the pelvis, the feet, the glutes, the pelvic floor, the lumbar spine—is not an academic exercise. It is a way of relating to your own body with more intelligence, more respect, and, when needed, more effectiveness. Because it is very hard to take good care of something you don't know. And very hard to truly inhabit it. The lower segment was already there before this chapter named it. It was just waiting for someone to take the time to look at it closely.

4.8. When the Body Speaks Before the Mouth Does

There is a kind of information the body processes before the conscious mind has a chance to intervene. The patient Antonio Damasio (1994) called Elliot—the man with a lesion in the ventromedial prefrontal cortex who had lost the capacity to make adaptive decisions despite keeping his rational intelligence intact—illustrated dramatically something researchers have been measuring in the laboratory for decades: the body takes sides before the mind does.

What Damasio called somatic markers are bodily signals that act as guides to decision-making: a kind of commentary from the organism saying “this doesn’t smell right” or “this looks good” before the prefrontal cortex has finished weighing the variables. These signals don’t come from anywhere abstract. They come from the muscles, the viscera, the heart rate, the quality of the breath. They come, to a great extent, from the lower segment.

4.9. The Sacrum as a Coherence Detector

There is an observation that recurs among therapists who practice craniosacral work—one drawn from palpation rather than from controlled measurement, and therefore to be read as clinical experience rather than established fact: the sacral region responds almost immediately to changes in the state of the autonomic nervous system. Not in the sense that the sacrum “thinks” or “feels” independently, but in the sense that patterns of sacral mobility—the rhythmic micro-movement craniosacral therapists have learned to perceive—change consistently when the organism’s state of activation changes.

When someone shifts from a sympathetic to a parasympathetic state—when the system takes a breath and lowers its guard—the rhythm of sacral movement changes. When someone is highly activated, that rhythm speeds up and becomes shallower and more irregular. If the observation holds, it requires no mystical explanation: the sacrum, with its sacroiliac joints, muscles, and deep fascia, is densely innervated by the autonomic nervous system and responds to its state with the honesty of everything that cannot lie.

4.10. The Architecture of Standing Upright

Human beings are the only mammals that habitually walk with the spine vertical. This evolutionary choice—which freed the hands for manipulation, raised the gaze to the horizon, and made possible a relationship with the world radically different from that of four-legged animals—has a biomechanical cost that the lower segment pays especially heavily.

The lumbar spine, bearing the weight of the entire trunk on just five vertebrae and two points of contact with the ground, works under conditions any structural engineer would consider borderline. The lumbar discs are the ones that degenerate most, herniate most, and most often require intervention. Not because they are badly designed—they are remarkable pieces of biological engineering—but because upright bipedalism subjects them to loads evolution has not yet had time to fully optimize for.

What protects the lumbar discs most is not rest. It is the dynamic activation of the muscles surrounding the spine. The deep core—the transversus abdominis, the lumbar multifidus, the diaphragm, and the pelvic floor—acts as an active corset that distributes loads, protects the discs from excessive pressure, and provides the dynamic stability the spine needs to move without harm. When this system fails—through weakness, inhibition, or compensatory movement patterns—the load falls directly on the passive structures: discs, ligaments, joints. And those structures protest with an eloquence nobody can ignore.

Deep core work, then, is not merely an exercise in abdominal aesthetics. It is a way of protecting the lumbar spine by redistributing load intelligently. And in the lower segment it also has a neurological dimension: the transversus abdominis and the lumbar multifidus are involved in regulating intra-abdominal pressure, which in turn affects the quality of breathing and, through it, the state of the autonomic nervous system.

One last anatomical consideration deserves a space of its own: the body is not a static structure that is put into certain positions and stays there until someone changes them. It is a living, dynamic structure in constant adjustment. Every breath moves ribs, spine, pelvis, and diaphragm. Every step redistributes loads from the feet to the spine. Every change in muscle tone alters the relative position of the bones. The body is always moving, even when it looks still from outside.

This dynamism has an important implication for work with the lower segment: there is no static “correct posture” to be maintained as if the body were a sculpture. There is a dynamic organization that favors efficient movement and the smooth functioning of the nervous system. And that organi-

zation is cultivated not by searching for the perfect position but by practicing attention to movement: noticing how the foot meets the ground with each step, how the pelvis organizes itself as you sit down, how the trunk breathes when the body is at rest.

A body that learns to feel itself is a body that begins to correct itself. The nervous system, when it receives enough good-quality information, knows what adjustments to make. The problem is not that the body doesn't know how to move well. The problem is that it has stopped listening to itself closely enough.

CHAPTER AT A GLANCE

- **The sacrum** is the keystone between spine and pelvis; its small rhythmic movements (nutations and counternutations) absorb load.
 - **The sacroiliac joints** move only millimeters, but their dysfunction is often mistaken for lumbar pain.
 - **The feet:** twenty-six bones, thirty-three joints, and arches that act as both springs and sensors.
 - **Underactive glutes** shift work onto the lower back.
 - **The pelvic floor** tightens with threat and releases with safety.
 - **There is no static “correct posture,”** only dynamic, well-informed organization.
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CHAPTER 5

Glands, Hormones, and Behavior

The chemistry of grounding. The adrenals, cortisol, and the biology of chronic fear.

KEY CONCEPTS cortisol · adrenaline · HPA axis · testosterone · oxytocin · estrogens · adrenal glands

The organism doesn't regulate itself through mechanical structures alone. It also does so through messages. Precise, constant, and highly effective chemical messages that travel through the blood and the nervous system, telling each tissue what to do at each moment: speed up, rest, save energy, stay alert, sleep if you can, wake up now. This chemical communication is as old as multicellular life and so sophisticated that, despite decades of endocrine research, it still holds surprises.

The hormonal system is not a list of substances with fixed functions. It is a conversation in constant evolution, sensitive to context, modifiable by experience, and deeply connected to psychological states. And when we talk about safety, survival, and grounding, two small protagonists deserve special attention: the adrenal glands. They are discreet, weigh barely ten grams each, and sit on top of the kidneys like two little hats nobody would choose for a wedding but which are indispensable for everything else. Their influence on the bodily and psychological experience of the lower segment is enormous, and understanding it is essential to see why chronic stress and uprootedness are not just related metaphors but biologically intertwined experiences.

5.1. The Adrenal Glands: Two Departments, One Mission

Each adrenal gland has two distinct regions with different embryonic origins and complementary functions. At the center is the adrenal medulla, of nervous-tissue origin; surrounding it is the adrenal cortex, of epithelial origin. One responds to immediate danger. The other manages prolonged adaptation. Together they form the organism's oldest and most reliable emergency system.

The adrenal medulla produces catecholamines, mainly adrenaline and noradrenaline. These are the rapid-response hormones, the ones that turn the body into a near-instant survival machine when the brain—more precisely, the amygdala—detects threat. Adrenaline speeds up the heart, dilates the airways to maximize oxygen intake, mobilizes glucose from the liver to give the muscles immediate energy, and redistributes blood flow, favoring the skeletal muscles and brain over the digestive organs. Noradrenaline contributes to vigilance, attentional focus, and raised blood pressure. Together they prepare the organism for the only thing that mattered on the African savanna: moving fast or keeping very still.

The problem that has arisen since then—with direct consequences for the subject of this book—is that this system was designed for acute, physical dangers, not for the chronic low-level stress that characterizes contemporary life. The organism releases the same cascade of adrenaline in response to a four-hundred-pound predator as to a tense meeting, a worrying mortgage, or an email that opens with the dreaded words “we need to talk.” The amygdala is not very good at telling physical threats from symbolic ones, present dangers from anticipated ones, what is happening from what might happen. And the body pays for that lack of discrimination in the currency of accumulated stress.

5.2. Cortisol: The Ally That Doesn't Know When to Leave

The adrenal cortex produces several steroid hormones, but the one that interests us most here is cortisol, the main medium- and long-term stress hormone. If adrenaline is the guard on duty who shouts “alarm” and switches

everything on in seconds, cortisol is the crisis manager who arrives afterward and keeps the response going for minutes, hours, or days, as the situation requires.

Cortisol is, in theory, your friend. It's the companion who wakes you at three in the morning when there's an intruder in the house. It mobilizes energy, modulates the immune response, raises blood sugar, and helps you remember—with uncomfortable clarity—dangerous experiences. Without cortisol, the stress response would be brilliant for its first minute and utterly ineffective from the second minute on. It is an indispensable hormone.

The trouble is that it also wakes you at three in the morning when all there is is an unanswered email. It's like having a guard dog that barks at the mailman, at a falling leaf, at its own shadow. Useful, but exhausting.

The problem—and in endocrinology the problem always has a “but” built in—is that cortisol was designed for bursts, not marathons. When cortisol secretion stays chronically elevated for weeks or months, its adaptive effects become destructive. Immune suppression; altered glucose metabolism, with a higher risk of type 2 diabetes; reduced bone density; impaired hippocampal memory; disturbed sleep; and one effect particularly relevant to grounding: the sustained sense of alertness and diffuse threat that makes it almost impossible to feel the body from below (McEwen, 1998; Sapolsky, 2004).

5.3. The HPA Axis: The Thermostat That Goes Haywire

Cortisol is not secreted chaotically. It is regulated by a feedback axis connecting the hypothalamus, the pituitary, and the adrenal glands: the HPA (hypothalamic–pituitary–adrenal) axis. When the nervous system detects stress, the hypothalamus releases corticotropin-releasing hormone (CRH), which stimulates the pituitary to release ACTH, which in turn stimulates the adrenals to produce cortisol. Once in the bloodstream, cortisol feeds back negatively on the hypothalamus and pituitary to put the brakes on its own production. An elegant regulatory system under normal conditions.

Under chronic stress, this thermostat can go haywire. The receptors in the hypothalamus and pituitary, exposed to cortisol for too long, can be-

come less sensitive to the braking signal. The result is cortisol production that is no longer well regulated: it may be chronically high, or it may become blunted—a pattern of low or flattened cortisol output observed in some people with long-standing stress, burnout, or post-traumatic stress, often accompanied by deep, hard-to-shake fatigue (Fries et al., 2005).

For the lower segment, this dysregulation has direct consequences. An overactive HPA axis keeps the organism in a state of subchronic alertness that makes it very hard to bring attention down to the lower body. The safety signals that should come from feeling the feet, breathing into the abdomen, perceiving the ground—signals that normally calm the autonomic nervous system—are drowned out by the activation of the stress axis. It's like trying to listen to soft music in a room where someone has left the fire alarm ringing in the background: technically the music is there, but you can't hear it.

5.4. Testosterone and Estrogens: Beyond Gender Stereotypes

The lower segment is also the bodily region most closely linked to the sex hormones, in men and women alike. The ovaries and testes—located in or immediately next to the pelvic region—produce estrogens, progesterone, and testosterone in proportions that differ by sex but are present in both.

Testosterone has effects that go well beyond the strictly sexual. It contributes to overall energy, muscle tone, initiative, risk tolerance, and a certain quality of presence and occupation of space which, at healthy levels, shows up as the capacity to act, to sustain effort, and to feel effective in the world. Its decline—in men with hypogonadism or andropause, but also in anyone under severe chronic stress—is associated with fatigue, loss of initiative, reduced muscle tone, and a quality of withdrawal that has direct bodily expression.

Estrogens, predominant in women but also present in men, take part in regulating mood, protecting bone, supporting cardiovascular function, and maintaining the sensitivity of pelvic tissues. Their fluctuations—during the menstrual cycle, pregnancy, breastfeeding, or menopause—have bodily and psychological effects that medicine was slow to take as seriously as they de-

serve, including direct effects on the sense of base, rootedness, and inner stability.

5.5. Prolactin and Oxytocin: The Hormones of Bonding and Rest

Two more hormones deserve mention on this tour of the hormonal landscape of the lower segment, because both directly affect the sense of safety and grounding.

Oxytocin, popularly known as “the love hormone”—a simplification that makes endocrinologists smile the way neurologists smile when people call the entire central nervous system “the brain”—has far more specific effects than the romantic label suggests. It reduces the stress response, facilitates interpersonal trust, improves the perception of positive social signals, and is directly involved in the sense of safety that allows the organism to lower its guard (Heinrichs et al., 2003; Uvnäs-Moberg, 2003).

Prolactin, mainly associated with breastfeeding, also has broader effects related to rest, relaxation, and a reduced stress response. Its nighttime release during deep sleep contributes to repair and to maintaining hormonal balance. Insufficient sleep—one of the great public-health problems of our time—disrupts this release, with consequences that spread across the whole hormonal system.

5.6. What Feeds Stress and What Calms It

The vicious circle linking chronic stress, HPA dysregulation, and difficulty grounding can be interrupted from several angles. None is complicated in concept, though all require consistency in practice.

Repeated safety signals are the most direct mechanism. Feeling the feet. Breathing into the abdomen. Perceiving the weight of the body on the ground. These experiences, repeated often enough, send the HPA axis information that changes its calibration. Slowly but durably, the system learns that the environment is safe enough to lower its guard.

Rhythmic movement—walking, swimming, rocking, dancing—has specific effects on HPA regulation that go beyond the metabolic benefits of ex-

ercise. Rhythm in itself calms the nervous system in ways neuroscientists are still characterizing but which the contemplative traditions of the world have known for millennia: chanting, ritual dance, walking meditation, rhythmic prayer. All of these use the moving body to regulate the nervous system, and all share that element of rhythm and repetition that seems to be the language in which the oldest parts of the nervous system listen.

Good-quality sleep, nutrition that avoids glucose spikes, warm social contact, affectionate physical touch, and time in natural surroundings—especially exposure to real ground, uneven surfaces, plants, soil, water—are other regulators of the HPA axis that contemporary life has drastically reduced, with consequences this book cannot separate from the epidemic of uprootedness it has been documenting since the prologue.

Working with the lower segment from a hormonal perspective does not require pharmacological intervention. Above all, it requires recovering living conditions the organism recognizes as safe. And in that, the body—with its adrenals, its cortisol, its oxytocin, and its feet on the ground—is a formidable ally when given the chance to act as what it is: a system designed, first and foremost, to live.

5.7. The Body That Can't Tell the Boss from the Bear

There is a fundamental mismatch between the design of the human nervous system and the demands of contemporary life that nobody has resolved yet and probably won't in the foreseeable future: the human organism was designed to handle acute, sporadic dangers, not chronic, ubiquitous worries. From the nervous system's point of view, the difference between the two kinds of threat is enormous. From the physiological point of view, the difference in the organism's response to them is almost nil.

When the amygdala fires—at the bear on the savanna or at Monday morning's threatening email—the body responds with the same sequence it used to escape from lions. Only now there are no lions. There are bills. There are deadlines. There are meetings nobody wants to go to. And our poor nervous system reacts as if it were a matter of life and death... because, for it, it is. It doesn't know the difference. It never did.

The problem is that the emergency response isn't designed to last for months. It is designed to last for minutes. And when the stressors are not bears that appear and disappear but mortgages, relationship conflicts, work pressure, and alarming news refreshed every hour, the emergency system can't switch off. It stays on. And an emergency system that is permanently on is exactly that: a permanent emergency.

5.8. Beyond “Adrenal Fatigue”: When the System Loses Its Rhythm

The concept of adrenal fatigue, popularized in integrative medicine, is not recognized as a medical diagnosis, and a systematic review found no consistent evidence that the adrenal glands “run out” under stress (Cadegiani & Kater, 2016). The popular term nevertheless points to something that does have a solid basis: the HPA axis can become dysregulated in ways that have real, measurable physiological consequences. The most common outcome after a prolonged period of overactivation is not necessarily dramatic adrenal exhaustion, but a loss of the axis's normal rhythm.

Normally, cortisol follows a precise circadian rhythm: it rises significantly in the first hours of the morning—the cortisol awakening response, which helps us wake up and get going—and falls over the course of the day, reaching its lowest point at night to allow restorative sleep.

When the HPA axis is dysregulated by chronic stress, this rhythm becomes distorted: morning cortisol may be insufficient—which explains the difficulty getting started in the morning that so many people experience—while evening cortisol stays too high, which explains trouble falling or staying asleep.

This desynchronization of the cortisol rhythm has effects that go far beyond sleep. It affects glucose metabolism, immune function, mood, concentration, and emotional regulation. A person with a dysregulated HPA axis is not simply tired: the hormonal coordination system that regulates almost everything else has been thrown off.

The sex hormones and our basic vitality are also compromised. When chronic stress keeps the HPA axis overactive, sex hormone production can suffer. A popular explanation holds that cholesterol—the common precur-

sor of all steroid hormones—is “stolen” from sex-hormone production to make cortisol. The actual physiology is more complex and better documented: the stress axis actively suppresses the reproductive axis, as corticotropin-releasing hormone and glucocorticoids inhibit the hypothalamic release of gonadotropin-releasing hormone (Whirledge & Cidlowski, 2010). The underlying logic, however, is the same: in a perceived emergency, the priority is immediate survival, not reproduction.

Under chronic stress, the production of testosterone, estrogens, and progesterone can decline. The result can be a range of symptoms often treated in isolation—fatigue, reduced libido, menstrual irregularities, loss of muscle tone, mood swings—without connecting them to the underlying hormonal system or, more fundamentally, to the state of the stress axis that is coordinating it all.

The connection between the state of the lower segment and sex-hormone health is, in this sense, direct: working with the first center—reducing chronic sympathetic tone, improving sleep quality, restoring the sense of safety—is not only psychological work. It is also, indirectly, endocrine work.

CHAPTER AT A GLANCE

- **Adrenal medulla:** adrenaline and noradrenaline for immediate response. **Adrenal cortex:** cortisol for sustained adaptation.
 - **The HPA axis** regulates cortisol through negative feedback; chronic stress can disrupt it.
 - **Dysregulation** appears as an altered daily cortisol rhythm, poor sleep, fatigue, and suppressed reproductive hormones.
 - **“Adrenal fatigue”** is not a recognized diagnosis, but HPA dysregulation is real and measurable.
 - **Oxytocin and safe social contact** dampen the stress response.
 - **Regulators:** repeated safety signals, rhythmic movement, sleep, touch, and time in nature.
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CHAPTER 6

The Sacral Plexus: The Wiring of the Base

The autonomic nervous system, Polyvagal Theory, and the neural anatomy of grounding.

KEY CONCEPTS sacral plexus · sciatic nerve · vagus nerve · Polyvagal Theory · neuroception · piriformis

We have now covered a good part of the territory of the lower segment: its bony and muscular architecture, its interrelations with the rest of the organism, the hormonal landscape that regulates it, and the psychology that inhabits or abandons it. But something fundamental is still missing: the wiring. Because impeccably designed bones, committed muscles, and hard-working hormones would be of little use without a network capable of coordinating everything, transmitting information in real time, and continuously adjusting the system to the changing circumstances of each moment.

That wiring is organized, to a great extent, by the nervous system. And in the specific territory of the lower segment, one of the most important structures is the sacral plexus: a nerve distribution hub deep in the pelvis on which depend the movement of the legs, pelvic and perineal sensation, sphincter control, essential aspects of sexual function, and—less obviously but very decisively—the quality of our relationship with the ground.

6.1. The Autonomic Nervous System: Accelerator and Brake, or the Mechanics of Safety

To understand the sacral plexus and its role in grounding, we'll start a little higher up: with the autonomic nervous system. I promise it's more interesting than it sounds.

The autonomic nervous system has two great branches that work in a complementary and, in health, balanced way. The sympathetic nervous system prepares the body for action: it speeds up the heart, dilates the pupils, inhibits digestion, increases muscle tone, directs blood toward the skeletal muscles, and raises the overall state of vigilance. It is the accelerator pedal. The parasympathetic nervous system does exactly the opposite: it promotes rest, activates digestion, lowers the heart rate, reduces peripheral muscle tone, and creates the conditions for repair and recovery. It is the brake pedal.

Health doesn't mean always braking or always accelerating, but being able to switch fluidly between the two according to the real demands of the situation. The problem with chronic stress is precisely that the accelerator gets stuck: the sympathetic system stays switched on longer than the real situation requires, while the parasympathetic can't assert itself effectively enough to compensate. The result is that diffuse activation, that low-intensity alert tone, so many people describe as "always being on edge, even when nothing's happening."

A body in accelerator mode cannot ground itself. It cannot come down. It cannot feel the ground with the quality and depth needed for that sensation to calm the nervous system.

6.2. The Vagus Nerve and Polyvagal Theory

Before getting into the anatomy of the sacral plexus—don't worry, I'll explain it in ordinary words—let's stop at the scientific discovery that has most transformed our understanding of the autonomic nervous system in recent decades: Porges's Polyvagal Theory.

Beginning in the mid-1990s, Porges (1995, 2011) proposed that the autonomic nervous system does not simply operate as the classic two-pedal

model suggested. He described three hierarchical, evolutionarily distinct circuits that come into play in a set order depending on the level of perceived threat.

The most evolutionarily recent circuit—the ventral vagal system—controls the muscles of the face, head, and neck and takes part in social engagement behaviors: facial expression, tone of voice, the capacity to listen and to make eye contact. When this circuit is active, the organism feels safe, can connect with others, can explore the environment with curiosity, and can rest. It is the state in which the lower segment can be fully felt.

The second circuit—the sympathetic system—switches on in the face of perceived threat and prepares for mobilization: fight or flight. In this state, the body is placed in the service of active survival. Attention narrows, the senses sharpen in some dimensions and shut down in others, and the lower segment loses its place in the face of the urgency of the emergency response.

The third circuit—the dorsal vagal system, the oldest in evolutionary terms—switches on in the face of threats the system perceives as insurmountable. It is the circuit of collapse, of immobilization, of the “playing dead” mammals resort to when fight and flight are not options. In humans it shows up as dissociation, numbness, a sense of unreality, a collapse of tone, and a profound disconnection from the lower body.

What makes Polyvagal Theory especially relevant to work with the lower segment is the idea of neuroception: the unconscious, involuntary process by which the nervous system constantly assesses whether the environment is safe, dangerous, or life-threatening. Neuroception is not a conscious decision. It happens before the mind can form any opinion about the situation. And it relies on cues from the environment—the faces of people nearby, tones of voice, movement in space—but also, and this is crucial, on cues from inside the body itself.

The feeling of the feet on the ground. The pelvis resting on the chair. The breath reaching the abdomen. The weight of the body resting on the surface that supports it. All of these are interoceptive signals the nervous system continuously assesses as part of neuroception. When they are clearly present, they help activate the ventral vagal system and the sense of safety.

When they are absent—when the feet aren't felt, when the pelvis is locked, when the breath doesn't drop—the system lacks those safety cues and tends to stay more activated than necessary.

6.3. The Anatomy of the Sacral Plexus

The sacral plexus proper is a nerve network formed by the anterior rami of the spinal nerves L4 to S4—that is, from the lower lumbar spine and the first four sacral segments. It lies deep in the pelvis, resting on the piriformis muscle and surrounded by a remarkably crowded anatomical neighborhood: muscles, bones, ligaments, organs, and fasciae living in such close proximity that the region is as strategic as it is sensitive to disturbances of any kind.

The term plexus—from the Latin for “braided” or “interwoven”—already gives an important clue to its organization. It is not a single, clean, solitary nerve. It is a network in which nerve roots cross, regroup, and form combinations that then give rise to different peripheral nerves. This arrangement builds in redundancy and flexibility: if one root is irritated or compressed, the others can partly compensate. The network is more robust than a single cable.

Several nerves of great functional importance emerge from the sacral plexus. The sciatic nerve is the best known—and the most cursed when it gets irritated. It is formed mainly by fibers from L4 to S3 and is the longest nerve in the human body: it leaves the pelvis through the greater sciatic notch, passes below or through the piriformis muscle depending on individual anatomy, runs down the back of the thigh, divides behind the knee into the tibial and common peroneal nerves, and continues to the feet with branches that supply the muscles and skin of the entire leg and foot.

The pudendal nerve supplies the perineum, the external anal sphincter, the external urethral sphincter, and the external genitals. Its dysfunction—through compression, trauma, chronic inflammation, or pelvic floor hyper-tonicity—can show up as perineal pain, sexual dysfunction, continence problems, or a vague sense of “inner tension” in the pelvic region that many people cannot attribute to any specific cause.

6.4. Piriformis Syndrome: When the Muscle Is Mistaken for the Culprit

The piriformis muscle plays a special role in the functional anatomy of the sacral plexus. It originates on the front surface of the sacrum and inserts on the greater trochanter of the femur. Its main function is external rotation of the hip. But its importance in this chapter comes from its anatomical relationship with the sciatic nerve and its sensitivity to the state of the nervous system.

The piriformis and the sciatic nerve are very close neighbors. In most people, the sciatic passes beneath the piriformis. In roughly one person in six (Smoll, 2010), some or all of its fibers pass through the muscle. In both cases, when the piriformis contracts chronically—through stress-reactive muscle tension, prolonged sitting, or compensatory movement patterns—it can compress or irritate the sciatic nerve, producing pain that radiates down the back of the thigh and leg and is often mistaken for a lumbar disc herniation. This is what is called piriformis syndrome.

What makes the piriformis especially interesting for grounding is its sensitivity to the state of the autonomic nervous system. It is one of the muscles that respond directly to sympathetic activation: when there is perceived threat, it contracts as part of the preparation for flight. A person who has spent years in a state of chronic diffuse alertness very often has a tight piriformis. And a tight piriformis doesn't only irritate the sciatic: it reduces the mobility of the sacroiliac joint, restricts pelvic movement, and contributes to that feeling of stiffness and blockage at the base of the body that so many patients describe.

6.5. The Vagus Nerve and the Sacral Plexus: The Dance Between Head and Base

The connection between the vagus nerve and the sacral plexus is perhaps the least known but most fascinating aspect of the neurophysiology of grounding. The vagus nerve—the tenth cranial nerve, which descends from the brainstem, passes through the neck, supplies the heart and lungs, continues

to the gut, and reaches the abdominal organs—is the main conductor of the parasympathetic system in the upper half of the body.

In the pelvic region, the parasympathetic system is represented by the pelvic splanchnic nerves, which emerge from sacral segments S2 to S4. These sacral parasympathetic nerves supply the pelvic organs—bladder, rectum, genitals—and play a role complementary to the vagus in regulating the autonomic nervous system.

Some clinicians speak of a “safety response” distributed between the cranial vagus and the pelvic sacral nerves, as the two ends of an axis that, when well regulated, creates the conditions for rest and connection. This image—a safety axis running from the head to the base of the sacrum, with the vagus at one end and the sacral nerves at the other—is anatomically suggestive and clinically useful, though it should be understood as a working model rather than an established finding. It suggests that grounding is not just a matter of paying attention to the feet, but has a neurophysiological basis in which the sacrum and the sacral plexus act as generators of safety signals that travel up to the central nervous system.

When the sacral plexus works well—when it is not compressed, when the muscles around it have balanced tone, when the pelvis is mobile enough not to squeeze the neural structures—information flows freely between the lower body and the brain. Signals of support, plantar pressure, pelvic mobility, and pelvic floor tone travel up clearly and contribute to the sense of having a base. When there are restrictions in that system—from a chronically tight piriformis, a locked sacroiliac joint, or a hypertonic pelvic floor—the signals are distorted or reduced, and the sense of base deteriorates.

Working with the sacral plexus doesn't mean manipulating the nervous system directly, which is beyond the reach of everyday practice. It means creating the conditions—postural, muscular, respiratory, movement-based—in which the nervous system of the base can function more freely. It means reducing piriformis tension. Mobilizing the sacrum. Activating the parasympathetic system through slow, deep breathing. Feeling the feet. Bringing attention down.

All of which, once again, begins with something apparently simple but profoundly real: coming back to the body. To the lower body. To the body that holds us up.

6.6. The Vagus Nerve as an Ally of Grounding

The vagus nerve has something the body's other nerves don't: it can be influenced voluntarily, at least in part. Most nerves of the autonomic system are entirely beyond conscious control: we cannot decide to lower our heart rate, to stop our adrenal glands secreting cortisol, or to speed up our digestion. But the vagus, in the portion that supplies the larynx, pharynx, and lungs, responds to something we can modulate: breath and voice.

Slow, prolonged exhalation activates the parasympathetic system via the vagus nerve (Zaccaro et al., 2018). This is the mechanism by which practices as varied as yogic pranayama, coherent breathing in biofeedback, choral singing, humming, chanted prayer, compassion meditation, and simply singing in the shower regulate the nervous system: all of them involve extended exhalation, which stimulates the vagus, which activates the parasympathetic system, which lowers the organism's overall level of activation.

For work with the lower segment, this has very concrete implications. Breathing into the abdomen with an exhalation slightly longer than the inhalation—a ratio of 4–6 seconds in to 6–8 seconds out is enough for most people—activates the vagus directly. The effects of a single breath are modest, but consistent and cumulative with regular practice.

6.7. The Enteric Nervous System: The Second Brain That No Longer Surprises Anyone

The gut contains more neurons than the spinal cord. When people hear this for the first time, most react the same way: a mixture of disbelief and “well, that explains a lot.” And yes, it explains a lot. The enteric nervous system—some 200 to 600 million neurons distributed along the gastrointestinal tract (Furness, 2006), a network Michael Gershon (1998) famously called “the second brain”—is not just a local control system for digestion. It can

process information, coordinate complex reflexes on its own, and communicate in both directions with the brain via the vagus nerve.

This means that what happens in the gut influences mental and emotional state, and what happens in mental and emotional state influences the gut. Not poetically, but through concrete nervous, endocrine, and immune pathways. Irritable bowel syndrome, the nausea of anxiety, “butterflies in the stomach” in emotionally significant situations: all of these are expressions of that two-way communication.

For the lower segment, the enteric nervous system is one more player in the web of interrelations that makes up the first center. Gut health—including the composition of the microbiome, the permeability of the intestinal lining, and the tone of the enteric nervous system—is part of the ecology of the first center, though it is rarely mentioned in books on somatic work.

6.8. Walking Barefoot: Information from the Ground

There is a practice that contemplative traditions have recommended for millennia, that pediatricians encourage for children’s motor development, and that modern life has turned into an exotic rarity: walking barefoot. Not on flat artificial surfaces, but on soil, grass, sand, stone. On real ground.

The reason is neither nostalgic nor romantic. It is physiological. The feet, with thousands of mechanoreceptors in each sole, are among the most densely innervated sensory organs of the body. When those endings are allowed to touch uneven surfaces, varied textures, and changing temperatures, the nervous system receives a quantity and quality of information that is simply not available through rubber soles and uniform floors. That information has effects that go beyond the foot: it improves proprioception along the whole lower chain, activates the intrinsic foot muscles that stay idle in rigid shoes, stimulates the peripheral nervous system, and contributes to the nervous-system calibration that allows us to feel the ground as real support rather than as an abstract surface.

CHAPTER AT A GLANCE

- **Autonomic nervous system:** sympathetic (accelerator) and parasympathetic (brake).
 - **Polyvagal Theory:** ventral vagal (safety and connection), sympathetic (fight or flight), dorsal vagal (collapse).
 - **Neuroception** assesses safety unconsciously, drawing on cues from the body itself.
 - **Sacral plexus (L4–S4):** sciatic, gluteal, and pudendal nerves; the pelvic splanchnic nerves (S2–S4) carry sacral parasympathetic fibers.
 - **A tense piriformis** can irritate the sciatic nerve.
 - **Slow exhalation** engages vagal pathways; barefoot contact enriches sensory input from below.
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The Neuropsychology of Grounding

The brain that listens to the body. Amygdala, hippocampus, and insula on home ground.

KEY CONCEPTS amygdala · hippocampus · insula · prefrontal cortex · interoception · neuroplasticity

So far we have talked a great deal about feet, pelvis, ground, roots, plexuses, and adrenal glands. If you have come this far while kindly continuing to think for yourself, you may be asking a perfectly legitimate question: all this about the body is very well, but what exactly about the brain? After all, it seems to call the shots in almost everything we do. It is the one that interprets what happens, decides whether something is dangerous or harmless, generates the narratives we take for autobiography, and sets off the alarm when the phone buzzes with a message that begins “we need to talk”—a phrase that rarely announces anything pleasant.

The short answer is yes, the brain matters enormously. The long answer—and a far more interesting one—is that the brain does not work like an absolute monarch locked in its skull, sending orders to the rest of the organism like someone governing a remote colony from the capital. It works more like a biological parliament, sometimes brilliant, sometimes contradictory, where the body has as much right to speak as the prefrontal cortex. Signals don’t only travel from the top down. They rise from the feet, the viscera, the muscles, the breath, the skin. The brain listens. And what it hears from the lower body largely determines what kind of world it perceives and how it responds to it.

7.1. The Amygdala: The Threat Detector That Prefers False Alarms

One of the leading actors in this story is the amygdala, a subcortical structure deep in the temporal lobe whose name comes from the Greek for almond—which is reassuring, because it reminds us that something very small can have a very intense character.

The amygdala works as an automatic threat detector: it continuously assesses stimuli from the environment—sounds, images, smells, movements at the edge of vision, changes in the facial expressions of people nearby—as well as internal stimuli—bodily sensations, variations in heart rate, changes in breathing—searching for signs that might indicate danger.

The amygdala's logic is elegant in its brutality. It prefers a false alarm to an ignored threat. In the evolutionary environment in which our nervous system developed, this preference made perfect sense: mistaking a branch for a snake was a nuisance. Mistaking a snake for a branch was terminal. The asymmetry of consequences justified a bias toward over-detection. The result is a brain structure that fires before asking for more information and that, when in doubt, always chooses the most dangerous scenario.

The problem that has arisen since then is that the same mechanism still operates today with the same urgency and the same lack of discrimination. The amygdala is not very good at telling a four-hundred-pound predator from a tense meeting with the boss. It doesn't distinguish a present threat from one anticipated in the mind. It doesn't distinguish a real danger from an image of danger on the eleven o'clock news. The body responds in much the same way in all these cases, with varying intensity but in the same direction: sympathetic activation, rising cortisol, increased muscle tone, reduced perception of the lower body.

The good news is that the amygdala is not immutable. It can learn. Not in the sense that it starts reflecting philosophically on its responses, but in the sense that its activation thresholds can be modified through repeated experience. When the organism has enough experiences of safety—physical, relational, interoceptive—the amygdala gradually recalibrates its threshold. It learns that this particular environment, in this particular body, at this particular moment, doesn't call for the level of alarm it used to. And that recal-

bration affects every system the amygdala regulates, including the muscle tone of the lower segment.

7.2. The Hippocampus: Where Past and Present Learn to Tell Themselves Apart

Next to the amygdala, in the same depths of the temporal lobe, sits the hippocampus, a structure shaped like a seahorse—hence its Greek name—which performs entirely different functions. The hippocampus organizes episodic memory and puts fear responses in context. When the nervous system detects threat and the amygdala fires, the hippocampus supplies the context: is this the same kind of threat as before? Am I in a genuinely dangerous situation, or in one that resembles a dangerous situation I lived through long ago?

The hippocampus is, in a way, the one charged with telling the amygdala: “Wait. We’ve been through this. Remember how it ended.”

In people with unprocessed trauma or a history of chronic early insecurity, the hippocampus’s capacity to contextualize may be compromised. Neuroimaging research, beginning with the work of J. Douglas Bremner and colleagues, has documented that sustained traumatic stress is associated with reduced hippocampal volume and impaired hippocampal function (Bremner et al., 1995; van der Kolk, 2014), with direct consequences for the ability to tell past from present. The nervous system of someone whose hippocampus doesn’t contextualize well may respond to present situations as if they were the past situations that left their mark. Today’s tense meeting triggers the response that organized a difficult family relationship twenty years ago. The body doesn’t distinguish time.

This temporal disconnection is one reason somatic work is especially valuable in some therapeutic contexts. The body’s memories of trauma are not narrative memories that can be edited by changing the story. They are state memories—memories of how the body felt in certain circumstances—and they are only updated through new bodily experiences that give the nervous system different information. Understanding what happened is not enough. You have to feel it differently.

7.3. The Prefrontal Cortex: The Director with Limits

Above the amygdala and hippocampus, in the frontal lobes, lies the prefrontal cortex: the most highly evolved region of the brain, the one that makes possible abstract thought, long-term planning, complex decision-making, inhibition of impulsive responses, and voluntary emotional regulation. It is, to put it very simply, the “adult in the room” of the nervous system.

The prefrontal cortex can modulate the amygdala’s response. When it works well, and when activation is not too intense, it can send inhibitory signals that reduce amygdala reactivity and allow a more nuanced, contextual response. This is what happens when someone feels a surge of anger and, instead of acting on it immediately, is able to pause, assess the situation, and consciously decide how to respond. The prefrontal cortex working at full capacity.

But the prefrontal cortex has a well-documented Achilles’ heel: it is the first casualty of severe stress. When nervous-system activation crosses a certain threshold—when cortisol is very high, when the amygdala is very active, when the body is in a real or perceived emergency—the prefrontal cortex loses effectiveness. Its regulatory capacity deteriorates. Thinking becomes more rigid, perspective narrows, creativity drops, and the tendency toward reactive responses increases. This is why people do things under severe stress that, once calm again, they can’t believe they did.

For grounding work, this has an important implication: we cannot think our way into grounding when we are badly dysregulated. The prefrontal cortex we need in order to reflect, to decide to practice, to apply the resources we have learned, is the first to fail just when we need it most. That is why work with the lower segment cannot be only cognitive. It has to be bodily first. It has to be available even when the mind can’t cooperate, through bodily gestures that require no reasoning: feeling the feet, lowering the breath, perceiving the weight of the body. Things the older brain can do even when the prefrontal cortex has decided to take a break.

7.4. The Insula: Where the Body Listens to Itself

There is a brain region that rarely appears in popular neuroscience writing but which is absolutely central to the subject of this book: the insula, or insular cortex. It lies deep in the Sylvian fissure, hidden between the frontal and temporal lobes, as if anatomy had decided to keep its especially important function in an especially protected place.

The insula is the center of interoception—the process by which the nervous system perceives the internal state of the body: heart rate, body temperature, hunger, thirst, pain, muscle tension, breathing, visceral sensations. It is, in the terms developed by A. D. (Bud) Craig and Antonio Damasio, the place where the brain “listens” to the body (Craig, 2009; Damasio, 1999). And it is also where those bodily sensations are transformed into what we subjectively call “feelings”: not emotions in the sense of behavioral responses, but the conscious sense of being in a particular internal state.

People with a high capacity for interoception—those whose insula integrates bodily information well—tend to have better emotional regulation, a greater capacity for adaptive decision-making, and greater overall psychological well-being (Khalsa et al., 2018). This is no coincidence. The capacity to feel one’s body from within—to detect accurately when the heart speeds up, when the breath shortens, when the stomach tightens—provides valuable information about the state of the system that the mind alone cannot generate.

And here is the direct bridge to work with the lower segment: feeling the feet, perceiving the weight of the pelvis on the chair, noticing contact with the ground are exercises in interoception. Every time you practice bringing attention to the lower body, you are training the insula. You are developing the capacity to receive information from your own body with greater precision and greater confidence. You are rebuilding, cell by cell and synapse by synapse, the capacity to inhabit yourself.

7.5. The Reward System and Grounding: Why Working on the Base Becomes a Pleasure

There is one aspect of the neuropsychology of grounding that is rarely mentioned and yet is decisive for understanding why some people find bodywork deeply satisfying while others experience it as an obligation: the reward system.

The nucleus accumbens and the dopaminergic pathways of the mesolimbic system are the neural substrates of motivation, pleasure, and the sense of meaning. When we do something the nervous system judges beneficial for survival or well-being, this system releases dopamine, generating a “this is worth it” signal that makes us more likely to repeat the behavior. It is the biological basis of reinforcement learning.

Grounding activates the reward system. Not always dramatically or intensely, but consistently when practiced regularly enough. Feeling the feet clearly, perceiving the weight of the body resting on the ground, experiencing the breath dropping into the abdomen: when these experiences happen in a context of sufficient safety, they generate well-being signals the system records and remembers. Over time, the body starts to “ask” for those experiences. Practice becomes easier not because discipline has increased, but because the nervous system has learned that it feels good.

This is the neurobiological basis of something many somatic practitioners describe in more intuitive terms: once someone has experienced, even briefly, the genuine sensation of being grounded, the body wants to go back. Not through abstract willpower but through memory. The body remembers.

7.6. Neuroplasticity: The Foundation of Clinical Hope

Everything described so far—the hypersensitive amygdala, hippocampal impairment, reduced prefrontal effectiveness, impoverished interoception—could sound like a list of life sentences. An inventory of what stress and uprootedness do to the brain, with no way out.

But contemporary neuroscience has shown convincingly that the adult brain retains a far greater capacity for structural and functional reorganization than was believed until a few decades ago. Neuroplasticity—the nervous system’s capacity to modify its connections in response to experience—is not exclusive to childhood. It persists throughout life, though with different characteristics and rhythms depending on age.

Aerobic exercise has been shown to increase hippocampal volume in older adults (Erickson et al., 2011). Reductions in perceived stress after mindfulness training have been correlated with decreases in amygdala gray-matter density (Hölzel et al., 2010). A hypersensitive amygdala, in other words, can be recalibrated with repeated experiences of safety. Impoverished interoception can develop with systematic practice. The prefrontal cortex can regain regulatory effectiveness when the level of chronic activation drops enough for it to do its work.

None of this is quick or linear. The nervous system doesn’t reorganize itself in a weekend workshop or after reading a book, however well written. It reorganizes through the repeated, sustained, felt experience of being different. Through thousands of moments in which attention comes down, the feet are felt, the breath descends, and the nervous system receives the information that this, here and now, is safe enough.

That, in the end, is what grounding means neurophysiologically. It is not a poetic metaphor. It is a description of what happens when the nervous system learns, cell by cell, that there is ground.

7.7. The Memory of Fear and How It Changes

One of the questions people most often ask when they begin working with the first center is: “How long does it take to change?” The question is understandable—nobody wants to embark on a process of indefinite length—and the honest answer is: it depends. But something more useful can be said.

Research on fear extinction—the process by which conditioned fear responses weaken through repeated exposure to the stimulus without negative consequences—suggests that new learning does not erase old learning

(Bouton, 2004; Milad & Quirk, 2012). Instead, new learning is created that competes with the old. The nervous system doesn't forget: it learns that, in this context, with this person, at this level of activation, the stimulus that used to predict danger no longer does. The old memory is still there, latent. But the new one carries more weight in the present context.

This has several implications. First: change is possible at any age, because the nervous system remains plastic throughout life. Second: change requires repeated experience, not intellectual understanding. Third: context matters a great deal—the experiences of safety that modify the amygdala need to happen in an environment safe enough for the system to absorb them without setting off the defense response again.

That is why work with the lower segment must always begin at a manageable level of activation. It is not about diving into difficult emotions to “process” them. It is about working within what Daniel Siegel (1999) called the window of tolerance: gradually approaching its edge, feeling that the body can handle it, returning to the comfort zone, and, from there, exploring a little further. This is what Peter Levine—the creator of Somatic Experiencing—calls titration (Levine, 1997, 2010): approaching difficult material in doses the system can absorb without being overwhelmed.

7.8. The Prefrontal Cortex as Ally, Not Director

There is a tendency in cognitive psychology to treat the prefrontal cortex as the region that should always be in charge—regulating, inhibiting, controlling. The implicit ideal is a perfectly rational human being who assesses situations objectively and decides on optimal responses. A human being, in short, who doesn't exist.

The prefrontal cortex is a valuable ally. But it is an ally, not a director. Its optimal function is not to suppress the responses of the limbic system but to integrate them: to receive information from the amygdala, put it in context with the help of the hippocampus, weigh it against goals and values, and facilitate a response that integrates both the emotional signal and the cognitive context.

When the prefrontal cortex tries to ignore or suppress the information coming from the body and the limbic system instead of integrating it, the result is not more control but more dysregulation. The nervous system doesn't accept being ignored: if silenced, it turns up the volume. That is why people who have spent years "controlling" their emotions with cognitive strategies often experience crises of disproportionate intensity: the system has been building up pressure that mental control could not contain indefinitely.

The alternative offered by work with the lower segment is not surrendering to emotions but learning to receive them from a more stable place. When there is a base—when the feet are felt, when the breath reaches the abdomen, when the nervous system has enough safety cues—emotions can arise and be processed without overwhelming. The lower segment doesn't suppress emotions. It holds them.

CHAPTER AT A GLANCE

- **Amygdala:** a threat detector biased toward false alarms—and one that can be recalibrated.
 - **Hippocampus:** places fear in context; chronic stress can impair it.
 - **Prefrontal cortex:** regulates, but is the first to falter under high stress; grounding must therefore be bodily first.
 - **Insula:** the hub of interoception; attention to the feet trains it.
 - **Fear extinction** creates new learning that competes with the old; titration and the window of tolerance set the pace.
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CHAPTER 8

The Muladhara Chakra

The root. The yogic tradition, the history of the chakras, and the psychophysical correlates of the first center.

KEY CONCEPTS Muladhara · Svadhisthana · earth · LAM · Kundalini · Brahma · Dakini

After several chapters devoted to bones, nerve plexuses, hormones, amygdalae, and neuroplasticity as the foundation of clinical hope, the time has come to pause and change language without changing seriousness. So far we have described the lower segment through anatomy, physiology, and contemporary psychology. Now it is time to look at the same territory from another register: the symbolic and the contemplative.

Not to abandon rigor—which would be a bad idea at this point in the book—nor to surrender to a mystical fog where everything vibrates a great deal but nobody explains anything. Quite the opposite: the aim is to widen the map. Because however useful it is to know what role cortisol plays or how the amygdala takes part in assessing threat, nobody goes through everyday life saying, “My neuroception is a bit off today and my HPA axis got up wanting to be the center of attention.” Life presents itself as sensations, meanings, intuitions, blocks, reliefs, images, and experiences that need a language capable of capturing them without reducing them. And at that level, the chakra system—and Muladhara in particular—still has a power and usefulness that deserve to be taken seriously.

8.1. The Chakra System: A History with Nuances

The chakra system is one of the best-known and most misunderstood maps of the yogic tradition. Its roots lie in ancient India. The word *cakra* —“wheel”—appears early in Indian literature, but the systematic doctrine of bodily centers was elaborated far more fully in the tantric traditions from the latter half of the first millennium onward, culminating in texts such as the sixteenth-century *Ṣaṭ-cakra-nirūpaṇa*, the treatise Woodroffe would later translate. The seven-chakra scheme that circulates today as canonical in Western popular culture—with its colors, elements, seed syllables, and psychological correspondences—spread in the West mainly through the work of Sir John Woodroffe, a British scholar and lawyer based in Bengal, who in 1919, under the pseudonym Arthur Avalon, published *The Serpent Power* (Woodroffe, 1919), a translation of and commentary on classical tantric texts about the chakra system.

This historical detail is not academic pedantry. It is a vaccine against dogmatism. The seven-chakra system as we know it today was not revealed on some sacred mountain in its final form. It is, rather, the result of centuries of elaboration, debate, synthesis, and reinterpretation that continues to the present. Different yogic and tantric schools describe different numbers of chakras—four, five, six, seven, ten, twelve—and different psychological and spiritual correspondences for each. The seven-chakra version with its rainbow colors is, in a sense, the most exportable and accessible one, but neither the only one nor necessarily the most elaborate.

This context does not diminish the system's value. What it does is put it in its proper place: as a symbolic and functional map of human experience, developed over centuries by people who observed body and mind with the same seriousness with which modern scientists observe the nervous system. Different methods, different vocabularies, territories that overlap with surprising frequency.

8.2. Muladhara: The Symbolic Anatomy of the First Chakra

Muladhara—from the Sanskrit *mula*, root, and *adhara*, base or support—is the first chakra in the classical seven-chakra system. It is located at the base of the spine, in the region of the perineum, between the anus and the genitals. It is the lowest chakra in the physical body and the first in the hierarchy of energy centers, which in the yogic tradition means it is the foundation on which all the others are built. Not in the sense of being the most important—the tradition sets up no such hierarchy of value—but in the sense that without a well-established base, the development of the higher centers remains suspended, like a pyramid without a base or a tree without roots.

Muladhara's symbolism is rich and coherent. It is depicted as a red lotus with four petals—red being the color the yogic tradition associates with vitality, blood, the energy of the earth, and the survival instinct. Each petal bears a Sanskrit seed syllable (*vaṁ, śaṁ, ṣaṁ, saṁ*), and some commentaries associate the four petals with four forms of bliss. At the center of the lotus sits a yellow square, the symbol of the earth element.

The earth element is perhaps Muladhara's most revealing correspondence. Earth as stability, as weight, as density, as what holds without moving. Earth as what remains when everything else changes. Work with Muladhara is always, in some sense, work on the capacity to be still without being rigid, to let oneself be held without needing to control everything, to trust that there is something beneath one's feet.

Inside the yellow square is a downward-pointing triangle—toward the earth, toward the root—and above the triangle sits the bija mantra, the seed syllable of this chakra: LAM. This sound, pronounced during meditation or chanting, is said to resonate at the vibrational frequency of this center and to contribute to its balance. From a more physiological perspective, voicing LAM with resonance in the lower chest and abdomen activates the deep respiratory muscles, sets the abdominal organs vibrating, and—as research on vocal contemplative practices has documented—can activate the parasympathetic nervous system via the vagus nerve (Bernardi et al., 2001). Science and tradition pointing in the same direction, in different dialects.

8.3. The Associated Deities: Brahma, Dakini, and the Kundalini Serpent

The Hindu tradition associates Muladhara with two deities who, read symbolically, bring considerable psychological richness.

Brahma, the creator of the universe in the Hindu trinity, represents the principle of manifestation, the force that turns the potential into the actual, that gives form to the formless. His association with the first chakra suggests that creativity—in the broadest sense of the word, including the capacity to bring projects, relationships, and ways of life into being—requires a solid base. There is no lasting creation without foundation.

The goddess Dakini, who in some versions of tantra is Muladhara's guardian, represents the consciousness of the earth and the instinctive. She is the wisdom that resides in the lowest part of the body, in the most archaic layers of the nervous system, in the instincts that precede language and that know, with a certainty that needs no argument, when the ground is safe and when it isn't. Her association with this chakra is a reminder that the intelligence of the lower body is not a primitive intelligence to be surpassed through cognitive refinement. It is a different intelligence, equally valid, which complements analytical reason and in some contexts surpasses it.

Above all these symbolic elements rests the best-known image of the tantric tradition in relation to the chakra system: the serpent Kundalini, coiled three and a half times around the *svayambhu linga* at the base of Muladhara, asleep, waiting. On the tantric map, Kundalini is the unawakened vital energy which, when activated and rising up the spine through the chakras, produces the experience of enlightenment or expanded consciousness.

From a psychological perspective—without reducing spiritual experience to psychology or denying its transcendent dimension—the sleeping Kundalini can be read as the individual's unrealized vital potential. The energy that is there, at the base, waiting for conditions safe enough and a channel open enough to rise. Yogic practice, in this frame, is not a technique for producing mystical experiences but a systematic way of creating the condi-

tions—bodily, psychological, ethical, relational—in which that potential can unfold progressively and sustainably.

8.4. Convergences Between Science and Tradition

One of the most intellectually satisfying experiences offered by integrative work between the contemplative traditions and contemporary somatic psychology is discovering how often they describe the same territory in completely different vocabularies.

What developmental psychology calls a “secure base”—the state of implicit trust in the availability of the environment that Bowlby (1988) identified as the foundation of healthy psychological development—resonates with what the tradition calls a balanced Muladhara: that sense of being supported, of being able to explore without fear of losing the ground, of existence having a foundation.

What Porges describes as the activation of the ventral vagal system—the state of perceived safety that allows social connection, rest, and exploration—corresponds functionally to what the tradition describes as an active, balanced Muladhara: that quality of calm presence, of being here without the need for constant vigilance.

What van der Kolk documents about the body’s inability to complete interrupted defense responses, which remain stuck as chronic muscle tension and nervous-system dysregulation, has its parallel in the concept of a blocked Muladhara: energy that doesn’t flow, that stagnates at the base, unable to rise or descend freely.

It is not that science “confirms” the chakra system, nor that the chakra system “explains” neuroscience. It is that two traditions of systematic observation of the human being, with very different methods and in completely different cultural contexts, have reached convergences too frequent to be coincidental.

8.5. Muladhara Imbalances: Excess and Deficiency

The tradition works with the idea of a chakra's balance as a point between two poles of imbalance: deficiency and excess. This polarity is clinically very useful because it names precisely two kinds of presentation that often appear in the consulting room and are sometimes confused.

Muladhara deficiency shows up as difficulty feeling the lower body, a sense of floating or unreality, lack of contact with material reality, difficulty sustaining long-term projects, inability to commit, a feeling of having no place in the world. People with this pattern tend to live “upstairs”: in their heads, in plans, in ideas, in abstract spirituality, or in fantasy. Their feet, quite literally, are not clearly felt. The image that emerges is of a plant without roots: it can grow, it can even flower for a while, but in the first storm it loses its footing.

Muladhara excess presents completely differently: rigidity, difficulty changing, excessive attachment to material security, a tendency to accumulate, resistance to the new, territoriality, a conservatism bordering on paralysis. People with this pattern are so far “down” they can't move. Their roots are so thick they have become an obstacle rather than a resource. The image is of a tree whose roots go so deep they have begun to invade the foundations of nearby buildings: what was once protection has become restriction.

The aim of work with Muladhara is not to eliminate these tendencies—which will always find natural expression in each person's temperament and history—but to move toward a point of greater flexibility between them. A lower segment that can be felt when safety is needed but doesn't become an anchor when it's time to move.

8.6. Practical Work with Muladhara: From Theory to the Body

Working with Muladhara does not mean visualizing a red lotus at the base of the spine for twenty minutes a day, though that has its place within a broader meditative practice. It consists, fundamentally, of the same things this book has been describing from different angles.

Feeling the feet. Bringing attention to the soles, the heels, the contact with the ground. Noticing how the body's weight falls downward and how the earth responds. This simple practice, repeated several times a day for a minute, is the most basic and most effective work with Muladhara there is.

Working with the earth. Direct contact with natural ground—grass, soil, sand, stone—stimulates the sensory receptors of the feet and provides interoceptive information that modern culture has drastically impoverished. It is not a New Age practice: it simply gives the nervous system back the sensory information it was designed for.

Postural practices. The yoga postures associated with Muladhara—Tadasana (mountain), Virabhadrasana (warrior), Malasana (squat)—share an emphasis on support, rootedness, and connection with the ground. The point is not to achieve perfect postures but to experience, in one's own body, the quality of being supported.

The LAM mantra. Voicing this syllable with the intention of making the sound resonate in the pelvic area combines the physiological effects of deep breathing with the neurological effects of sound, and with the element of intention and attention that contemplative practice adds as its distinguishing dimension.

And finally, working with the earth in the broadest sense: commitment to the concrete, the material, everyday responsibilities, projects that require patience and sustained presence. Muladhara isn't worked on only on the meditation cushion. It is also worked on in the kitchen, in the garden, in the difficult conversations one doesn't avoid, in the responsibilities one takes on without fleeing into abstraction.

8.7. The Tree, the Mountain, and the Cave: Images the Body Understands

Over the centuries, the contemplative traditions that work with Muladhara have developed a repertoire of images whose practical effectiveness transcends their symbolic dimension. They are not just metaphors illustrating a concept: they are images the nervous system can use as anchors, that activate

specific sensory representations, and that, when held with attention long enough, produce measurable physiological effects.

The tree is the richest of these images. When someone visualizes themselves as a tree—with roots descending from the feet into the earth, a trunk that can move in the wind without losing its anchorage, branches spreading up and out from a solid base—several things are happening at once. Attention drops to the feet, activating interoceptive perception of the lower body. The image of roots organizes the intention of connecting with the ground in a form the nervous system can process more easily than an abstract instruction. And the image of the trunk moving without losing its anchorage offers the system a model of flexibility that requires no rigidity, of movement that requires no loss of ground.

The mountain is another central image. A mountain is stability without rigidity: solid, present, immovable in essence, yet able to bear on itself the movement of trees, water, animals, and wind. When someone visualizes themselves as a mountain during seated meditation—rocky masses beneath the buttocks, a solid sacrum, the pelvis supported, the trunk rising from that base—the nervous system receives an image of stability that can counteract, at least for a moment, the sense of floating and precariousness.

The cave represents another quality of Muladhara: the protected interior, the womb of the earth, the place nothing can enter without permission. The practice of imagining a warm, protected inner space in the pelvic region—not as a dissociative exercise but as a way of developing familiarity with that territory of the body—is the reverse of dissociation: instead of leaving the body, one enters it. One inhabits it.

The most productive approach is to treat science and tradition as two languages describing the same territory. Each can say things the other cannot. And whoever learns both can find their way through the territory with more resources than someone who knows only one.

CHAPTER AT A GLANCE

- **Chakras are symbolic maps**, not anatomical organs; the seven-chakra model reached the West largely through Woodroffe (1919).
 - **Muladhara**: base of the spine, earth element, four-petaled red lotus, seed syllable LAM.
 - **Symbols**: Brahma (manifestation), Dakini (instinctive wisdom), Kundalini (latent potential).
 - **Convergences**: the secure base (Bowlby), ventral vagal safety (Porges), stored defensive responses (van der Kolk).
 - **Imbalance** may take the form of deficiency (floating) or excess (rigidity).
 - **Working images**: the tree, the mountain, and the cave.
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PART III

The Practical Tools

Breath, posture, movement, and attention

Theory alone doesn't ground anyone. Practice is needed.

This third part offers four concrete paths back to the body: breath, posture, movement, and mindful attention.

Each chapter includes techniques with clear instructions and precautions for different kinds of bodies.

Here you will find the tools: pranayama, asanas, Atmandance, and somatic mindfulness, explained so you can start applying them from day one.

Pranayama: The Breath That Holds Us Up

Breath as a regulator of the nervous system. Techniques for the lower segment.

KEY CONCEPTS diaphragmatic breathing · Kapalabhati · Nadi Shodhana · Bhramari · Ujjayi · vagus nerve · extended exhalation

If the body is the temple and the mind is the priest, the breath is the usher who opens the doors. Or, if you prefer a less religious metaphor: if the body is the house and the mind is the obsessive tenant who reorganizes the drawers at three in the morning, the breath is the thermostat. The thermostat doesn't argue with the tenant. It simply changes the temperature. And suddenly the tenant stops reorganizing drawers and sits down. That is what breath does.

By this point in the book we have covered a good deal of ground. We have drawn a fairly complete map of the lower segment. But there comes a moment when the map, however detailed, is no longer enough. Because understanding is not the same as inhabiting. You can know a great deal about nervous-system regulation and still breathe as if you were fleeing the tax collector in the middle of the night.

The breath is the fastest, humblest, and most democratic way into the body. It requires no special equipment, no silent retreat, and no particularly convincing robe. It requires something simpler and far less glamorous: attention. No monthly subscription, no notifications at seven in the morning.

9.1. We Already Know How to Breathe; Breathing Consciously Is Something Else

The brainstem handles breathing with admirable competence. While we sleep, while we think, while we argue with someone over text: it breathes. Without complaining. Without asking for a raise. Without vacations. If the brainstem had a union, it would have gone on strike long ago. But it doesn't. So it carries on. More loyal than a dog. Quieter than a monk.

But breathing to stay alive is one thing; breathing to regulate inner experience is quite another. That is where pranayama begins.

The word comes from Sanskrit. *Prana* is usually translated as breath or vital energy, and *ayama* as expansion, regulation, or mastery. What matters is this: pranayama is not about turning the breath into a heroic act or competing with yourself to see how long you can hold it. It is about consciously modulating the rhythm, depth, and quality of the breath to influence your physiological and psychological state.

What is fascinating is that here the yogic tradition and modern neurophysiology shake hands rather elegantly. What the tradition described as the regulation of prana we can now understand, at least in part, as modulation of the autonomic nervous system through the breath, the vagus nerve, heart rate, and the balance between activation and rest.

9.2. Diaphragmatic Breathing: More Than Ventilation

The first fundamental idea is diaphragmatic breathing. It sounds obvious, but it isn't. A great many people, especially under stress, anxiety, or mental overactivity, breathe predominantly into the chest. That is, they move the chest, shoulders, and neck more than the abdomen and diaphragm. The result is a high, fast, shallow breath—very useful if you are escaping a fire and not recommended if you are just answering emails.

The diaphragm is a dome-shaped muscle separating the chest from the abdomen. When it contracts, it descends and allows the lungs to expand. But it doesn't just ventilate. It also massages the viscera, modulates internal pressures, takes part in posture, influences lumbar stability, and, decisively,

sends signals to the nervous system. It also has a close relationship with the psoas: a stiff diaphragm usually coexists with a tight psoas. It is no coincidence. It is an anatomical dialogue.

That is why recovering diaphragmatic breathing is not just “breathing better.” It is offering the body a different experience of itself. It is telling it: there is no need to live on permanent alert.

9.3. The Exhalation: The Key to the Brake

If we had to sum up in a single sentence much of how breathing regulates the nervous system, it would be this: a longer exhalation tends to favor a calmer body.

Inhalation tends to be associated with activation. A prolonged exhalation, on the other hand, facilitates the parasympathetic response. During exhalation, the vagus nerve takes part in modulating heart rate and the signal that the organism can let go is reinforced.

9.4. Pranayama Techniques for the Lower Segment

Grounding breath (diaphragmatic, with a long exhalation). The simplest and probably the most important technique. You can practice it sitting with your feet well planted or lying on your back with your knees bent. Place one hand on your chest and the other on your abdomen: the hand on the chest should move little; the one on the abdomen should rise as you inhale and fall as you exhale. Breathe in through the nose for a count of four and out slowly for a count of six. The numbers are not what matters. What matters is that the out-breath is longer than the in-breath. When this practice is sustained for five or ten minutes, the breath stops getting trapped in the chest and begins to descend toward the abdomen and pelvis. The experience of the breath starts to cooperate with the sense of having a base.

Ujjayi (ocean breath). Breathe through the nose while gently narrowing the glottis, so that a soft, continuous sound appears. The key is for the sound to be subtle, not forceful, and to serve as a sensory anchor. It helps focus the mind, gives a sense of continuity, and can induce an experience of

greater inner stability. The slight vibration in the throat also has an interesting effect on the vagus nerve.

Bhramari (bee breath). Especially useful when the system is overloaded, scattered, or anxious. Breathe in through the nose and, as you breathe out, let a sustained hum escape with your mouth closed. It can be done with the ears gently closed by the thumbs to intensify the inner resonance. The effect is often surprisingly fast: the sound vibration calms, reorganizes attention, and stimulates the parasympathetic system.

Nadi Shodhana (alternate-nostril breathing). Performed by alternating the inflow and outflow of air between the nostrils, using the fingers to close one and then the other. Its effect is organizing: it can improve heart rate variability, reduce perceived stress, and improve attentional balance. Put simply, it brings a little order where there was too much inner traffic.

Kapalabhati (breath of fire). This belongs to another family of practices: more active, more energizing, and more stimulating. It consists of a series of short, fast, forceful exhalations driven by abdominal contractions, while the inhalation happens passively. Its effect is cleansing, activating, and toning. But precisely for that reason, it is not the ideal technique for an already overactivated body. It is also generally contraindicated during pregnancy and in cases of uncontrolled hypertension, epilepsy, or recent abdominal surgery. For the lower segment, it can be useful in a gentle, well-dosed form, especially in the morning.

9.5. Common Mistakes

The first is forcing. Many people treat the breath as if they were at a summer boot camp: deeper, faster, more... intense. Breathing is not a competitive sport. There are no medals. There is no panel of judges. If you feel exhausted after breathing, you are doing something wrong. And yes, I mean it: people come to my office exhausted from “breathing properly.”

The second mistake is competing. Some people turn even pranayama into a race: “I do twenty minutes of alternate-nostril breathing.” “Well, I do thirty.” Congratulations. You are both breathing. It’s like competing over

who whitens their teeth more times a day. If you're breathing, you've already won. The rest is detail.

The third mistake is practicing without presence. Technique without attention may improve ventilation somewhat, yes, but it loses half its power. If the mind is entirely elsewhere, the experience becomes superficial. It's not serious. You just come back. Always come back.

9.6. Practice Woven into Everyday Life

What is truly transformative usually happens when the practice enters everyday life. Three breaths on waking before you look at your phone. A long exhalation before opening your email. Breathing into the abdomen while you wait for the light to change. A short Bhramari before bed if your head won't be quiet. Feeling the support of your feet while lengthening the out-breath in the middle of a tense situation. Small gestures, yes, but cumulative. And this is exactly how the nervous system learns: through minimal repetitions that, over time, reorganize the whole.

9.7. Breath as Emergency Intervention

Everything described so far can be practiced in relatively comfortable conditions: at home, in a quiet space, without hurry. But the breath has a virtue no other practice can match: it is always available. You don't need a mat, a free hour, or for anyone to see you. When anxiety rises without asking permission, when a meeting starts getting out of hand, or when you receive news that throws you off balance, the fastest way back runs through the breath. Not because breathing solves everything, but because it can change the physiology of the moment enough for the mind to function again.

The simplest protocol I know for moments of acute activation is this: breathe out. That's all. Before doing anything technical, before counting, before closing your eyes, let go of the air inside you. One long, complete exhalation. The body will inhale on its own afterward. You don't have to decide to.

From there, if there is room, you can add structure. Breathe in for four counts. Breathe out for eight. Do it three times. That is enough to shift the autonomic nervous system in a more parasympathetic direction. It isn't magic. It is basic neurobiology: a long exhalation activates the vagus nerve and eases off the sympathetic accelerator.

Every time you have practiced pranayama in calm conditions, you have built a circuit. When you use it in difficult conditions, the brain recognizes the pattern and responds faster. Practice in calm is not a luxury. It is the training that makes access possible at the moment it matters most.

CHAPTER AT A GLANCE

- **The breath** is the one autonomic function we can also govern voluntarily.
 - **Diaphragmatic breathing with a longer exhalation** favors parasympathetic activity.
 - **Techniques:** grounding breath, Ujjayi, Bhramari, Nadi Shodhana, and—with caution—Kapalabhati.
 - **Avoid** forcing, competing, and practicing without presence.
 - **In acute activation:** exhale first; then breathe in for four counts and out for eight, three times.
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Asanas: The Geometry of Stability

Posture as information for the brain. Asanas for grounding.

KEY CONCEPTS Tadasana · Malasana · Virabhadrasana · Vrksasana · Supta Baddha
Konasana · Balasana · alignment

We now come to movement. If the breath was the software that modulates the system, movement is the hardware: the geometry with which the body organizes space. And this is where a lot of people get nervous. “Do I have to do yoga?” No. You have to move. With intention. With attention. With the body you have now—not the one you had at twenty, nor the one you’d like to have. This one. The one you have now. Which is the only one that works.

Posture is information. It is not an aesthetic detail or the fixation of fussy teachers who enjoy correcting other people’s shoulders. It is a constant source of signals for the brain. When the body adopts a stable, open, well-supported shape, the nervous system receives a fairly clear message: there is a base, there is ground, there’s no need to live as if you were about to face a pop quiz. When the opposite happens, the message changes: better not trust things too much.

That is why asanas are not gymnastics with exotic names or a flexibility contest for social media. Properly understood, asanas are a way of using bodily geometry to produce function, regulation, and presence. Patanjali defined the matter with surgical precision in the *Yoga Sutras* (II.46): *sthira sukham asanam*—posture should be steady and comfortable (Bryant, 2009). Firmness and ease. Neither collapse nor rigidity. A well-organized body is not a tense body; it is an available body.

10.1. Tadasana: The Mountain

The first great teacher of this chapter is Tadasana, mountain pose. It may seem ironic to devote so much attention to something as apparently simple as standing. And yet that is exactly the trap. Almost nobody really knows how to stand. We fake it quite convincingly, which is not the same thing. The body does the bare minimum and the mind assumes that's enough.

Practicing it means truly standing up. Place your feet together or hip-width apart. Weight is distributed between the heels, the base of the big toe, and the base of the little toe. It helps to sway a little forward and back, then side to side, to find the midpoint. That point where the body stops "falling" somewhere and actually begins to hold itself up.

The toes spread gently. The arches of the feet engage without rigidity. The knees don't lock. The pelvis looks for a living neutral. The deep abdomen engages discreetly. The chest opens without theatrics. The shoulders drop. The back of the neck lengthens. The crown of the head rises.

When the body finds this alignment, many people immediately notice an inner change. More presence, more clarity, more sense of being gathered without being shut in. Postural organization changes the information the brain receives about the state of the organism. The mountain is not just a shape. It is an experience of axis.

10.2. Malasana: Reclaiming the Ground

In contrast to that serene verticality comes Malasana, the deep squat, which reminds us of a somewhat uncomfortable truth: modernity has made it easier for a large part of the population to handle a spreadsheet than to squat down without negotiating with their ankles.

For millennia, the human species rested, worked, ate, and eliminated in this position. Malasana restores a profoundly human function. The feet are placed wider than the hips, the toes point slightly out, and the body descends. Ideally, the heels stay on the floor. If they don't, a folded blanket or a prop can be placed under them. What matters is not achieving an ideal image, but creating a stable posture in which the pelvis can approach the

ground, the spine can stay long, and the breath can descend toward the base.

There is nothing ornamental about Malasana. It is a deeply functional posture. It restores mobility, activates the legs and glutes, frees the pelvic area, and reminds the body of something it seemed to have forgotten: that going down isn't always losing; sometimes it is returning to the base.

10.3. The Warriors: Rootedness and Direction

Virabhadrasana I introduces a different quality. Grounding is not only rest and receptivity; it also needs power. But not a hysterical or overacted power. Rather a calm, directed, well-supported strength. What makes this posture interesting is not its heroic look but the relationship between rootedness and direction. The back foot pushes into the earth. The back leg supports. The front leg receives the weight and organizes the forward projection. The body thus learns a crucial lesson: moving forward without losing its base.

Virabhadrasana II changes the quality of the gesture. Here frontal verticality gives way to lateral expansion. It is a posture that teaches something fundamental in work with the first center: taking up space without apologizing. For many people that isn't as easy as it sounds. Some bodies have learned to fold in, to minimize themselves, not to be a bother. Warrior II proposes the opposite: breadth with support.

Utkatasana (the invisible chair) has the charm of practices that seem simple until you try them. It consists of sitting in a chair that doesn't exist. Sounds easy, right? Try it. Forty seconds. The legs burn, the glutes wake up, and you wonder why something so absurd is so effective. It strengthens the quadriceps, glutes, and ankles, stabilizes the knee, and generates internal heat that gets the circulation going. It's like doing squats, but with a better relationship.

10.4. Vrksasana: The Tree and Balance

Vrksasana introduces the dimension of balance. And balance has a very interesting therapeutic virtue: it doesn't lie. You can lie to yourself quite suc-

cessfully about philosophical, emotional, or work matters. But when you try to stand on one leg, the body gives a pretty honest answer.

What matters in Vrksasana is not perfection but recovery. Every small wobble and every return to center teaches the nervous system something. It also teaches you not to brag about your balance in front of demanding audiences. Balance is not stillness. It is the capacity to reorganize. And that applies as much to posture as to life, which also frequently shifts the ground under us without making an appointment.

10.5. Restorative Postures: Letting Go with Dignity

Supta Baddha Konasana. Lying on your back with the soles of the feet together and the knees falling out to the sides, the body enters an experience of passive opening and support. The beauty of this posture is that gravity works in the system's favor. The body stops actively holding itself up and begins to allow. Sometimes that is the most regulating instruction of all: there's nothing you need to do right now.

Balasana, child's pose, also belongs to this family of homecomings. From kneeling, the body folds forward, the forehead rests on the floor or a prop, and the back begins to breathe differently. There is something very primal and deeply regulating in this way of gathering in. It is not collapse, but a safe withdrawal.

Setu Bandhasana, bridge pose, takes us in another direction. Lying on your back with knees bent and feet close to the buttocks, the body lifts the pelvis by pressing the feet into the floor. The feet push, the legs work, the glutes engage, and the chest opens. This posture strengthens the back of the body and gives a very concrete experience of how support is born from pushing against the ground.

10.6. Principles That Run Through All Practice

Quality matters more than quantity: twenty postures without presence usually transform less than two or three truly inhabited. The breath is the most honest alarm signal; when it becomes clumsy or forced, something in the

posture has gone too far. And grounding is always active: it is not about letting yourself drop but about feeling the contact with the ground and letting everything else organize itself from there. Alignment, finally, is not dogma but dialogue.

A brief, consistent practice can change far more than you imagine. Fifteen or twenty minutes a day, if well inhabited, are worth far more than sporadic weekend heroics. The nervous system learns through repetition, not epic feats.

In the end, that is what asanas offer to work with Muladhara: a pedagogy of support. They teach the body that it can trust gravity a little more, that it doesn't always need to organize itself defensively, and that firmness is not the same as hardness.

10.7. When the Body Can't Get There: Adaptations and Alternatives

One of the most frequent objections to asana practice is also one of the most legitimate: "My body can't do that." Stiffness, chronic pain, joint limitations, or simply lack of previous practice can make some postures seem out of reach. It is worth making clear from the start that the asanas described in this chapter are not goals to be reached. They are invitations to an experience, and the experience can be adjusted without losing its essence.

Malasana with the heels raised on a folded blanket is Malasana. Vrksasana with a hand on the wall is Vrksasana. A Warrior with the back knee on the floor instead of the leg extended is still a Warrior. The question that guides adaptation is not "Are you doing the posture correctly?" but "Are you receiving the information the posture wants to give you?" If the answer is yes, the posture is well done.

For people with chronic lower back pain, hip-opening postures should be introduced with particular care, always respecting the natural curve of the lumbar spine. Malasana can aggravate certain lumbar conditions if done without proper preparation. Setu Bandhasana, on the other hand, is usually very well tolerated and can even relieve lumbar tension when done with at-

tention to the push of the feet and the engagement of the glutes. When in doubt, always consult a professional before persisting.

For some people the most suitable work is not formal asana but conscious functional movement. Getting up from a chair while paying attention to the weight in your feet. Squatting to pick something up off the floor with the hips back and the spine long. Climbing stairs while feeling the push of the glutes. Walking barefoot at home for ten minutes. None of these activities is yoga in the formal sense, but all of them are ways of educating the first center through patterns of conscious movement. And for someone who has never practiced and who recoils at the sight of a mat, they may be the most honest and sustainable entry point there is.

CHAPTER AT A GLANCE

- **Posture is information** for the nervous system: *sthira sukham asanam*—steady and comfortable.
 - **Tadasana** (axis), **Malasana** (descent), the **Warriors** (rootedness with direction), **Vrksasana** (balance as recovery), **Utkatasana** (strength).
 - **Restorative postures:** Supta Baddha Konasana, Balasana, and Setu Bandhasana.
 - **Quality over quantity;** the breath is the alarm signal; alignment is a dialogue.
 - **Every posture can be adapted;** conscious functional movement is a valid point of entry.
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CHAPTER 11

Atmandance: When I Dance, My Mind Stops

Movement as the language of the nervous system. The dance that heals and connects.

KEY CONCEPTS Atmandance · somatic movement · bodily expression · rhythm · co-regulation · interoception

After breath and asanas, we enter different territory. Freer, more intuitive, and, in a sense, older than any formalized technique. If asanas are the conscious architecture of the body, Atmandance is its mother tongue.

Here we are not looking for the impeccable posture, the correct gesture, or alignment worthy of a framed anatomical plate. We are looking for something harder and truer: movement that is born from within.

That changes a great deal. In structured work like asana, the body learns to organize itself. In Atmandance, it also learns to express itself. And that is not an artistic luxury or a poetic concession. It is a clinical, psychological, and bodily necessity. Because human beings were not designed only to hold shapes. They were also designed to turn them into life.

11.1. What Atmandance Is

Atmandance: A Neurointegrative Movement System is a psychocorporal method of harmonious expression that integrates conscious movement, breath, meditation, and group interaction into a structured process of self-regulation and human development.

Its name sums up its essence. *Atman*, a Sanskrit term associated with the deep core of being, and *Dance*, understood not only as dancing but as vital movement, emotional expression, and bodily awareness. The “D” also evokes direction, dynamics, and development. The result is a bridge between body psychology, Eastern contemplative traditions, and contemporary knowledge about neuroregulation and bodily awareness.

Atmandance starts from a simple but profound idea: the body doesn’t only express what we live through; it also organizes the way we feel, think, and relate. Every sustained muscle tension, every breathing block, and every postural pattern is part of an emotional biography inscribed in the organism. The right movement can help loosen those patterns and open new possibilities of experience.

The system works mainly with the seven bodily and energetic segments described by various body-psychology schools and Eastern traditions. Sessions combine two major dimensions: a dynamic, expressive part and a part devoted to meditative and respiratory integration.

In the first half of the session, the first five segments are worked through specific corrective and expressive movements, accompanied by music designed to facilitate particular emotional and bodily experiences. These movements are done in a group—preferably in a circle—fostering synchrony, emotional resonance, and a sense of belonging. There are moments guided by the instructor and spaces for free movement where each participant can explore their own inner expression without formal constraints.

The sixth segment is approached from a more observational and meditative dimension: the first five segments are “danced” without corrective intent, developing self-observation, presence, and witness awareness. The seventh segment integrates the work of rootedness and opening in particular, with emphasis on the first and fourth segments, accompanied by specific breathing and meditative practices.

The second half of each session is oriented toward inner integration through asanas, pranayamas, and meditation. Here movement becomes quieter and more conscious, fostering nervous-system regulation, bodily perception, and emotional integration.

Finally, the experience culminates in a space for sharing and talk therapy, where the group can put into words what was lived during the practice. This moment is not an incidental add-on but an essential part of the method: the body mobilizes experiences that words help to integrate. As Freud observed more than a century ago, what cannot be remembered and put into words tends to be repeated in action (Freud, 1914/1958). And the body, when nobody listens to it, has the awkward habit of insisting.

Atmandance does not aim to train dancers or impose a bodily aesthetic. Its goal is to foster processes of awareness, psychocorporal integration, and human development through a universal language: movement lived from within.

11.2. The Dancing Body Remembers

In therapy there are moments when language goes as far as it can and then just stands there, with the look of someone who has done all it could. The person understands their story, recognizes their patterns, can explain with admirable lucidity why they react the way they do. And yet the body remains just as contracted, just as alert, just as absent, or just as loaded. That is where movement starts doing work that explanation alone cannot complete.

Sometimes all it takes is a clear, deep, repetitive rhythm. A drum, firm percussion, a sound base with enough pulse. Then the body, which had been behaving like a model citizen for a while, begins to remember that it is also an animal. An oscillation appears, a discharge, a tremor, an expansion of the chest, a spontaneous bending of the knees. The body remembers. Not because the music is magical, but because the nervous system holds memories at many levels, and not all of them take the form of a story.

11.3. Principles of Corrective-Expressive Movement for the First Center

Work with the first center within Atmandance rests on a few fundamental principles.

First principle: the relationship with verticality and gravity. Movement starts from an organized spine and a clear connection with the ground, but it is not seeking a military uprightness—rather, a living verticality. The image of bamboo helps a great deal: firm, yes, but flexible; rooted, yes, but sensitive. Gravity stops being an enemy to defeat and becomes an ally to cooperate with.

Second principle: understanding the ground not as scenery but as a partner in conversation. Every step is a dialogue with the earth. When the ground enters experience as something alive, the sense of having a base stops being abstract.

Third principle: inhabiting weight. Many people with wounds in the first center move as if they didn't want to disturb reality. Overly soft steps, minimal contact, a tendency to pass through without leaving a trace. Attendance invites another possibility: inhabiting body weight as presence, not as burden. The difference is not a small one.

Fourth principle: rhythm. Deep rhythm, especially when carried by percussion, has a very direct effect on the nervous system. The body recognizes the pulse before it understands it. That is why certain kinds of music make work with the first center so much easier: they help reorganize movement from a primary, almost archaic base.

Fifth principle: expression without judgment. In expressive movement there are no right or wrong gestures in the conventional sense. There are more authentic movements and more forced ones. More alive or more defensive. The critical voice may show up, of course. It usually does. There's no need to argue with it. It's enough not to let it steer the practice.

11.4. The Grounding Dance: A Guided Practice

The grounding dance can be done alone or in a group and usually lasts about thirty-five minutes. The sequence is not presented as choreography but as an experiential progression.

Preparation: a place where you can move without obstacles, ideally barefoot, with music that accompanies without intruding. Begin standing, in a simple, attentive posture. Let the breath quiet down. Let attention

drop to the feet. Before moving, it helps to feel where you are going to move from.

Phase one: waking up the feet. The movement is small, almost microscopic if need be. Gentle swaying, shifts of pressure, exploring heels, balls of the feet, arches, toes. The point is not yet to travel, but to inhabit once more that part of the body so many adults use a lot and feel very little.

Phase two: descent. Little by little, the movement can take the body downward. Bending the knees, exploring the squat, sitting, placing the hands, passing through the floor. Descending is not falling; it is recovering intimacy with gravity. In that closeness to the ground, another kind of information appears: the back making contact, the weight of the pelvis, the support of the hands.

Phase three: ascent. Then comes the rise. And it's best done slowly. Going from the floor to sitting, from sitting to all fours, from there to kneeling, and then to standing lets you notice how the relationship with gravity changes at each stage. The body doesn't rise like someone leaving one posture for the next. It rises like someone rebuilding an axis.

Phase four: free dance. Once standing again, the dance proper can begin. Swaying, turning, stepping, changing direction. But with one fundamental condition: don't lose the relationship with the ground. If the movement becomes disorganized or the mind starts needing too much control again, simply return to the feet. That return, in fact, is the central practice.

Phase five: stillness. Finally, the movement settles until it comes to rest in simple stillness. Standing, breathing, feeling what has changed. Sometimes the change is clear: more weight, more warmth, more presence, more calm. Sometimes it is subtle. That's enough. The body doesn't always translate quickly.

11.5. The Music of the First Center

In Atmandance, music is not decoration. It is a tool of intervention.

For the first center, music with a strong rhythmic base, a clear presence of percussion, and a quality we might call earthy tends to work especially well. African drums, tribal rhythms, ritual percussion, some world-music

fusions, and, in some cases, electronic music with a deep, steady bass can be excellent allies. Not because there is a secret frequency that instantly awakens the sleeping root while the universe silently applauds, but because rhythm has a real capacity to organize movement, modulate the nervous system, and return the body to very primary patterns of coordination.

Movement is our oldest language. Long before narrative or the word appeared, gesture was already organizing experience. And it is still probably one of the most sincere. When we move with enough freedom, without excessive judgment, and supported by the base, something very deep begins to fall into order.

Atmandance gives back to work with Muladhara something more structured techniques sometimes forget: that grounding also has rhythm, weight, gesture, and expression.

11.6. Group Work: The Shared Base

Atmandance can be practiced alone, and many of its benefits are fully available in an individual format. But there is a dimension of group work that deserves special attention, especially where the first center is concerned.

Human beings regulate themselves in groups. For most of our evolutionary history, collective rhythmic movement was one of the main technologies of social and emotional regulation in the human communities that used it. Dance rituals, ceremonies, communal singing were not just entertainment. They were maintenance for the group's nervous system.

When several people move together to the same rhythm, a phenomenon of neurophysiological synchronization occurs that goes beyond motor coordination. Heart and breathing rhythms tend to converge (Müller & Lindenberg, 2011). Each participant's nervous system begins to regulate itself in relation to the others'. Co-regulation, which in individual therapy happens between patient and therapist, happens in the group in a multilateral, amplified way.

In group Atmandance sessions focused on the first center, this becomes especially visible when the group passes through the phase of contact with the floor. Several people lying down or squatting, moving slowly over a

shared rhythmic base, create a field of presence that holds each individual more than any of them could hold themselves alone at that moment. The base becomes, literally and metaphorically, shared.

For people with histories of severe disconnection or relational trauma, a well-facilitated group can offer something individual therapy sometimes cannot provide as efficiently: the experience that the presence of others can be a resource rather than a threat. Moving close to other bodies without needing anything from them, feeling their weight, their breath, their rhythm, and discovering that it helps rather than frightens, is in itself a therapeutic lesson of the first order.

CHAPTER AT A GLANCE

- **Atmandance** integrates corrective-expressive movement, breath, meditation, and group process across seven bodily segments.
 - **Session structure:** a dynamic, expressive first half; an integrative second half (asanas, pranayama, meditation); a closing verbal sharing.
 - **Principles for the first center:** living verticality, the ground as partner, inhabited weight, rhythm, and expression without judgment.
 - **The grounding dance:** feet, descent, ascent, free dance, stillness.
 - **Moving together** fosters co-regulation and a shared base.
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Mindfulness and Meditation

Full presence from the ground up. The body as the anchor of attention.

KEY CONCEPTS mindfulness · body scan · foot meditation · LAM mantra · somatic presence · neuroplasticity

This is a delicate point. After crossing anatomy, psychodynamics, neurobiology, breath, asanas, and expressive movement, a word appears that has met a strange fate in our time: meditation.

Depending on whom you ask, the word evokes a variable mix of Tibetan monasteries, phone apps with little bells, and apparently serene people telling you to “just observe your thoughts” while you are trying not to argue with your bank, your ex, or your own head at three in the morning.

And yet, behind all the knockoff spirituality, meditation remains a remarkably precise tool. Not because it turns us into ethereal beings or lifts us above the human condition, but for something far more useful: it teaches us to be here without running away from ourselves.

For the first center, this is crucial. Because Muladhara doesn’t need a spirituality that separates us from the body. It needs a practice that brings us back to it. Not a meditation for “getting out of the mind,” as if the mind were an enemy, but a meditation that integrates mind, body, and presence. An attention that doesn’t float. An awareness that doesn’t dissociate. A silence that isn’t escape but rootedness.

12.1. Meditating Is Not Disappearing

There is a widespread and rather unfortunate idea that meditating means emptying the mind or reaching a state of flawless peace. This idea has done more harm than it seems, because thousands of people sit down to practice expecting not to think, not to feel discomfort, not to get restless... and when they discover they are still human beings, they conclude that meditation “isn’t for them.”

But meditating isn’t about stopping thought. Meditation, in its most honest form, is about changing our relationship with what happens inside us. Thoughts keep appearing. So do emotions. What changes is that we stop automatically identifying with every inner wave. We stop being swept along by every impulse.

Now—and this is decisive—that space of observation must not become a new form of exile. Some people, especially those with histories of trauma, chronic anxiety, or dissociation, can use meditation as an elegant refinement of disconnection. They are far from the body. They have traded chaos for a polished absence.

That is why, in work with the first center, we need embodied meditation. A practice in which attention descends into the body.

12.2. The Danger of Meditating “Upward”

Not all meditation favors grounding. Some practices, especially when done without enough preparation or good bodily support, can intensify the sensation of floating. In people with a fragile base they can reinforce exactly what we want to change: the tendency to vacate the body (Treleaven, 2018).

A nervous system that becomes disorganized easily doesn’t usually benefit, at first, from overly abstract or expansive practices. It first needs to learn to tolerate the bodily present. To feel without being overwhelmed. To stay without escaping.

That is why this chapter proposes a meditation that goes down. That descends. That simplifies. Not a meditation for fleeing the earth, but for making peace with it.

12.3. Practice 1: Foot Meditation

This is probably the simplest and most powerful practice for the first center.

Sit comfortably. Close your eyes if that helps. Take three calm breaths. Now bring all your attention to your feet. Don't imagine. Don't analyze. Don't judge whether you're doing it right. Just observe. What do you feel in your right foot? Temperature? Pressure? Pulsing? An area that's more alive? An area that's more asleep? Stay there for a minute or two. Then move to your left foot. And finally feel both feet at once.

If thoughts appear, that's normal. If you get distracted, that's normal. If you feel almost nothing, that's normal too. The practice consists of coming back. That in itself is meditating.

12.4. Practice 2: Descending Body Scan

The body scan is a cornerstone of Mindfulness-Based Stress Reduction (Kabat-Zinn, 1990). For the first center, it helps to give special prominence to the lower half of the body. Begin by feeling the breath. Then bring attention to the pelvis. Then to the thighs, knees, calves, ankles, feet. In each area, stay for a few breaths.

The point of this practice is not to relax the body by force. It is to map presence. To reclaim inner territory. Many people live as if their body ended at the diaphragm. This scan helps rebuild the map.

12.5. Practice 3: Breath Anchored at the Base

Sit with your feet on the floor. One hand on your lower abdomen. Breathe in gently through the nose, feeling the air descend toward the belly. Breathe out for longer than you breathed in, as if the body were releasing weight downward. In for four counts. Out for six. Continue for seven to ten minutes.

This practice is especially useful for people prone to diffuse anxiety, racing thoughts, or a sense of unreality.

12.6. Practice 4: Walking Mindfulness

Seated meditation is valuable. But the first center benefits a great deal from attention in motion.

Find a quiet space. Walk more slowly than usual. Not theatrically slowly, just a little slower than normal. Pay attention to the phases of each step: lifting, moving forward, placing, shifting weight. Feel the heel, the sole, the toes, the other leg receiving the weight.

This practice trains the capacity to keep attention on the base while the body moves. It is, in a way, the contemplative version of dynamic grounding.

12.7. Mountain Meditation

This practice was popularized in clinical settings by Jon Kabat-Zinn (1994). Sit or stand. Feel your base. Now imagine you are a mountain. Not a rigid or grandiose mountain to feed the spiritual ego, but a stable presence. The base anchored in the earth. The body rising. Clouds, storms, wind, snow, and sun can pass over you. Everything changes. You remain. Not because you harden, but because you are deeply supported.

This meditation works especially well for people with a chronic-alert profile because it offers stability without total immobility. The image of the mountain allows a presence that doesn't feel trapped: you can be a mountain and keep breathing, keep feeling the ground, keep being a living body. The metaphor doesn't ask you to freeze. It asks you to settle.

12.8. LAM Mantra Meditation

Sit comfortably. Breathe in through the nose. As you breathe out, slowly pronounce: LAM. Let the final M resonate softly. Feel the vibration in your chest, your throat, your head, and, with a little practice, in your pelvis as well. You can repeat it for five or ten minutes. The vibration helps steady attention, naturally lengthens the exhalation, and stimulates the vagus nerve.

12.9. A Twenty-Minute Practice for the First Center

- **Minutes 1–3:** Posture and natural breathing. Feel the feet and pelvis.
- **Minutes 4–7:** Breathing with a longer exhalation (in 4, out 6).
- **Minutes 8–12:** Sustained attention on both feet.
- **Minutes 13–16:** Descending scan (pelvis, legs, feet).
- **Minutes 17–19:** LAM mantra, or simple silence with attention at the base.
- **Minute 20:** Integration. Ask yourself inwardly: “What do I notice now?”

If I had to sum up this chapter in a single idea, it would be this: in the first center, meditating means learning to come back. Back to weight. To the breath. To the pelvis. To the feet. To the present.

There’s no need to do it perfectly. There’s no need to feel a lot. There’s no need to become enlightened. What’s needed is practice. Enlightenment, if it comes, can come. Meanwhile, the feet. Because the base is not built with grand heroic gestures but with thousands of small returns.

12.10. Why Sitting Still Is Sometimes the Hardest Thing

There is a paradox in mindfulness practice aimed at the first center: the people who most need grounding are often precisely the ones who find it hardest to stay still. A nervous system on chronic alert can interpret stillness as danger. For that system, becoming still carries a connotation of vulnerability that constant activity helps it avoid.

Recognizing this matters because it prevents a common mistake: pushing someone to sit and meditate when their nervous system experiences it as a threat doesn’t speed up grounding. It often complicates it. The path may need to be different.

For some people, practice has to begin in movement. Walking with attention, doing housework with presence, mindful gardening, cooking while paying attention to smells and textures. Attentive movement, especially when it includes contact with the ground, activates the same interoceptive

circuits as formal meditation, with the added advantage that the nervous system doesn't register it as a threat.

Over time, once the system has accumulated enough experience that being present is not dangerous, stillness can be introduced gradually. Five minutes at first. Then ten. With eyes open if necessary. Standing if sitting is too activating. Without demanding results within any set time frame, because the nervous system learns at its own pace, which doesn't always match ours.

A final observation about the practice setting: meditation aimed at the first center works best in an environment that physically reinforces the experience it seeks. Firm ground underfoot. A comfortable temperature. Natural light if possible. No screens. Not because screens are bad in themselves, but because the nervous system, in order to learn to rest, needs experiences that don't demand vigilance. The environment can do much of the work before formal practice even begins.

CHAPTER AT A GLANCE

- **Meditation is not emptying the mind**, but changing one's relationship with experience.
 - **"Upward" or abstract practices** can deepen dissociation in a fragile system; embodied practice descends.
 - **Practices:** foot meditation, descending body scan, breath anchored at the base, walking mindfulness, mountain meditation, and the LAM mantra.
 - **A twenty-minute sequence** integrates them.
 - **For a nervous system on alert**, stillness may need to follow movement rather than precede it.
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PART IV

The Action Plan

Clinical cases and the eight-week program

What theory promises, clinical cases confirm or qualify.

This fourth part closes the journey with eight in-depth stories and the eight-week program that brings everything you have learned into your daily routine.

Here the book stops being a map and becomes an itinerary.

Here you will find the proof: how work with the lower segment changes real lives, and how you can start applying it in yours on Monday morning.

CHAPTER 13

Eight Clinical Cases

Eight conversations between the body and life. Names and details changed to protect confidentiality.

KEY CONCEPTS lower segment · grounding · somatic trauma · nervous-system regulation · body therapy · rootedness

Over thirty years of clinical practice, I have accompanied hundreds of people on the journey back to their bodies. No two stories are the same, but all share a common substrate: a nervous system that at some point learned it was not safe to rest, and that has been carrying out that instruction with absolute fidelity for decades.

The cases that follow are composites—blends of several clinical experiences—with names and details changed to protect confidentiality. But their essence is entirely real.

13.1. Silvia, or the Body That Decided Enough Was Enough

Silvia was fifty-one when she first came to see me. She came because her family doctor, worn out by seeing her with pains that had no organic explanation, had written a single sentence on the referral: “She needs to talk to someone who understands the body.” Silvia had had a lumbar muscle spasm for seventeen years that no treatment had resolved for good. Physical therapy, osteopathy, yoga, acupuncture. Everything worked for a few weeks and then it came back, like a tenant you can’t evict.

The first session revealed something the X-rays and reports couldn't show. When I asked her to stand up and describe how she felt her body at that moment, Silvia did something that immediately caught my attention: she stood up straight with a rigidity that wasn't posture—it was alertness. Like someone waiting to be hit who doesn't want to be caught off guard. Her feet touched the floor but didn't feel it. Her breath stopped every time she began an emotionally meaningful sentence.

Silvia had grown up in a house where her father's mood set the tone of every day. There was no physical abuse. There was something quieter and more pervasive: unpredictability. You never knew which father you would find when you got home from school. Silvia's nervous system had developed the only possible response to that chronic uncertainty: constant monitoring. Scan the environment before relaxing. Prepare for whatever might come. That preparation had settled into the body as architecture.

For the first few months we worked exclusively with the most basic thing: Silvia's relationship with the ground. Exercises in conscious contact with the feet. Breath that reached the pelvis. Positions in which the body had real support, not just symbolic support. Silvia did them with the meticulousness of an accountant, recording in a notebook what she felt, when, and for how long. That record was already, in itself, part of the change: the body was starting to become a source of information rather than just a source of alarm.

The turning point came in the tenth session. Silvia arrived that day quieter than usual. When I asked what was going on, she said: "I think I woke up today without the knot." She hadn't gone looking for it. It had happened. That spontaneity was the sign we were waiting for: the nervous system had found, on its own, a moment of regulation without conscious effort.

Eighteen months after we began, the lumbar spasm still appeared at moments of intense stress. But Silvia knew how to read it. "When I notice it coming back, it's because I've forgotten my feet," she told me in one of our last sessions. She had gone from having an incomprehensible symptom to

having a bodily message she could decipher. That is not cure in the classical sense. It is something more useful: dialogue.

13.2. Damián, or the Man Who Lived in His Head

Damián came to me at forty-six with the same complaint I have heard more times than I could count: “I don’t feel anything.” He didn’t say it sadly, but with the discomfort of someone stating a technical fact. He was a philosopher by profession, a university lecturer, someone whose intellectual life was extraordinarily rich and whose bodily life was, by his own admission, practically nonexistent. He slept little. He ate in order to function. Physical pleasure was almost foreign to him.

The initial assessment confirmed what his account suggested. Damián’s interoception—the capacity to perceive the body’s own signals—was minimal. Not because he was insensitive, but because for decades he had trained his mind to process reality, and the body had been left as a dispensable appendage. When I asked him to close his eyes and describe what he noticed physically at that moment, he was silent for almost a minute. Then he said: “I think I have hands.” He said it with humor, but it was also an accurate description.

Damián’s story was not one of trauma in the dramatic sense. It was the story of a gifted child in an environment that valued abstract intelligence above everything else. The body had not been punished; it had simply been ignored. And what isn’t named, what doesn’t receive sustained attention, ceases to exist as inhabited territory. Damián had built an enormously capable identity in the cognitive dimension and an almost nonexistent one in the somatic dimension.

With Damián the approach had to be completely different. The body-awareness practices that work with other patients would have felt too strange to someone who lived mainly through language. We started with something that came naturally to him: talking. But talking about physical sensations instead of ideas. What did he notice in his chest when he lectured? What happened in his abdomen when he read something that unset-

tled him? Language as a doorway into the body, instead of the body as a doorway into language.

Progress with Damián was slow and fascinating. At six months, he reported something he himself described as disconcerting: he had begun to feel hungry. Not the cognitive signal of “it’s been five hours, I should eat,” but real hunger—visceral, bodily. For a forty-six-year-old man, recognizing hunger for the first time was an event. For me, it was the sign that his nervous system had begun to reconnect territories that had been disconnected for decades.

Two years after we began, Damián published a philosophical article on the embodiment of thought that drew, without naming it directly, on his own therapeutic experience. I asked him if he felt comfortable with that. He said yes—that it was the first time he had written something that came from the body and not just from books. That was exactly it.

13.3. Natalia, or Learning to Step Without Permission

Natalia was thirty-one, with a life that on the surface seemed to work perfectly. A stable job, a solid relationship, friends, good health. What didn’t work was harder to name: a chronic feeling that her presence in the world was provisional, as if someone might withdraw her permission to be there at any moment. She put it like this: “I feel like I’m taking up borrowed space.”

In the first session I noticed something that gave me a sense of where the work would go. Natalia always sat on the edge of the chair. Not at the back, where the body has full support, but on the edge, like someone getting ready to stand up at any moment. When I pointed it out, she was genuinely surprised: she had never noticed. She had spent thirty-one years on the edge of the chair.

Natalia was the second of three sisters. The eldest had for years absorbed the family’s expectations; the youngest had absorbed its worries. Natalia had taken the space that was left: the one who caused no trouble, who didn’t need attention, who could wait. That apparently comfortable position had generated a deep bodily belief: being at the center, being seen, fully taking up space was something that belonged to others.

The work was, in many ways, work on permission. Not in the abstract sense of self-help books, but in the bodily, concrete sense: can this body sit fully in the chair? Can this body take up its full height? Can this voice come out whole, without the filter of caution? Each of these questions was answered in the body before it was answered in the mind.

One exercise that proved particularly meaningful was something as simple as walking consciously. Natalia tended to walk close to the walls in hallways, leaving room for anyone coming the other way, even when nobody was there. We practiced walking down the middle. It sounds trivial. It isn't. The body learns to take up space long before the mind can reason its way there.

Three years later, Natalia came to a closing session she herself had requested. She said she wanted to tell me something: she had gotten a promotion she had been putting off for a long time because she didn't think she was ready. "I asked for it," she said. "I went in to ask for it and I sat all the way back in the chair." It was a professional achievement. It was also, and above all, proof that, at thirty-four, she was no longer living on the edge of the chair.

13.4. Félix, or the Body That Kept the Score

Félix was sixty-three and came, as he put it himself, "because my grandson asked me why I never hug him, and I didn't know what to say." It was the first time in his life he had sought psychological help of any kind. He had built a mid-sized company, supported his family through hard times, and survived a separation and several crises. He had never needed anyone—or so he thought.

Félix presented what the clinical literature calls alexithymia: a marked difficulty in identifying and describing one's own emotions (Sifneos, 1973). It wasn't that he didn't feel emotions; it was that emotions had never found an outlet and had ended up encapsulated in the body as unprocessed information. The visible result: pronounced rigidity in the chest, an almost total inability to make spontaneous physical contact, and a sensation he described as "having something stuck right here," pointing to his sternum.

Félix's history included what clinicians sometimes call strict normative masculinity: he had grown up in an environment where men didn't cry, didn't complain, didn't need. Emotions were women's territory. The body was a work tool, not a place of experience. These rules had never been spelled out; they were simply the air one breathed in his family, his neighborhood, his generation.

With Félix I had to be especially careful with language. The word "emotion" triggered immediate resistance. We started by talking about "information." The body gathers information, processes information, stores information that sometimes never reaches the conscious circuits. That was a language Félix could handle. And it was also, at bottom, an entirely accurate description of what we were doing.

The first advances were modest and deeply meaningful. Félix began to notice when his jaw was clenched. Then when he was holding his breath. Then, months later, when he felt a pressure in his chest that wasn't physical. The body map he was gradually building represented the first time in sixty-three years that he had paid sustained attention to the signals of his own interior.

The moment Félix would remember as decisive came in the twenty-seventh session, sixteen months after we began. We were doing a diaphragmatic breathing exercise when he stopped, closed his eyes for a moment, and when he opened them his eyes were moist. He didn't cry, exactly. But he came close. "I don't know what this is," he said. I told him he didn't need to know what it was. It was enough that his body was letting it go.

Six months later he sent me a short message. It said his grandson no longer asked why he didn't hug him. There was no need.

13.5. Clara, or the Executive Who Couldn't Feel the Ground

Clara was thirty-four and managed the most important accounts at a multinational. Seen from outside, her life was a resounding success: rapid promotions, a team reporting to her, a salary most people would envy. Seen from inside, it was constant vertigo. She came to my office with a sentence she herself considered the problem: "I have anxiety attacks on planes." She flew

every week, and every takeoff had become a small agony of palpitations, cold sweat, and the irrational certainty that something terrible was about to happen.

But two or three sessions were enough for the plane to stop being the main character. The plane was simply where the anxiety became visible. The real pattern showed up in any situation in which Clara couldn't control her surroundings: a meeting that went off script, a delay, a conversation that took an unexpected turn. The plane, that sealed capsule in which you hand control to a pilot you can't see, was simply the purest version of her underlying fear.

In the first bodily exploration, something happened that I had seen many times and that surprised her deeply: her feet didn't appear on her body map. When I asked her to bring her attention to her soles, her heels, the contact with the ground, she fell silent. "I know I have feet," she said at last, "but I don't know what I'm supposed to notice in them." Her entire experience of herself lived from the collarbones up. The rest of the body was logistical: what carried her from one meeting to the next.

Work with Clara was, to a large extent, work of descent. We started with something deliberately humble, almost ridiculous for someone used to solving complex problems: three times a day, for one minute, notice the contact of the heels with the floor. No goal, no measurement, nothing to optimize. For someone who lived by performance indicators, the absence of a goal was itself an intervention. There was nothing to achieve. There was only feeling.

The change didn't arrive as a revelation, but as an almost absent-minded discovery. Two months in, in the middle of an especially tense meeting, Clara realized she was pressing her feet into the floor. She hadn't decided to. Her body had done it on its own, looking for the base her head couldn't provide. That night she wrote to me: "Today my feet held me up without my asking them to." The nervous system had begun to regulate itself from below. Slowly, Clara learned that safety isn't negotiated from the head with arguments and contingency plans; it is felt from the base, in the silent contact of a foot with the ground.

13.6. Marc, or the Architect Who Couldn't Sleep

Marc was forty-two and enjoyed considerable prestige as an architect. He designed buildings others admired, spaces thought through down to the last detail so that people would feel good inside them. The irony, which wasn't lost on him, was that he himself didn't feel good anywhere, least of all inside his own body. He slept four or five broken hours a night, and had for years. He had tried medication, sleep hygiene, herbal teas, apps. Nothing held its effect for more than a few weeks.

What struck me most about Marc in the first sessions was his extraordinary capacity to analyze his own emotions. He could describe what he felt with enviable precision: he named nuances, distinguished anguish from unease, irritation from resentment. He was, quite literally, an expert on his inner life. And yet all that knowledge lived in his head. His body, by contrast, was a complete stranger. When in the first session I asked him to close his eyes and bring his attention to his pelvis, he opened them after ten seconds with a sentence I had heard hundreds of times: "There's nothing there."

Marc's story had the shape of so many others. A brilliant, successful, admired father who was emotionally absent—present in achievements and missing in everything else. An anxious mother who, without meaning to, had passed on a permanent vigilance: the world was a place where you had to stay alert, where something could go wrong at any moment. Marc had inherited that vigilance and turned it into a professional virtue. The problem is that vigilance doesn't switch off when you go to bed. Marc's insomnia was not an isolated disorder; it was the nighttime continuation of a state of alert that had gone forty years without rest.

So we didn't work on sleep directly. Working on sleep would have given Marc one more problem to solve, and solving problems was precisely what kept his system switched on. Instead, we worked on the base. Slow diaphragmatic breathing before bed, with attention placed not on falling asleep but on feeling the weight of the body sinking into the mattress. The long exhalation, the one that applies the parasympathetic brake, repeated without hurry. I asked him to give up the goal of sleeping and stay only with the sensation of letting go.

The first night he slept six hours straight, Marc sent me a message the next morning. It said, tersely: “I dreamed of earth.” He didn’t write it as poetry, and I didn’t read it as such. It was a literal description of what his nervous system was finally doing after forty years: putting down roots. The man who designed foundations for others had begun, slowly, to build his own.

13.7. Laia, or the Teenager Trapped in Anxiety

Laia was seventeen and, on paper, did everything you are supposed to do to be well. She meditated every morning with an app that tracked her streaks and gave her badges. She went to the gym. She ate healthily, slept her hours, avoided screens before bed. And still she carried a diffuse anxiety she didn’t understand and couldn’t locate anywhere in particular. “I have no reason to feel bad,” she told me in the first session, with a mix of guilt and bewilderment that is, sadly, very common in her generation.

Her parents meant well, and that was precisely part of the problem. Concerned about their daughter’s well-being, they had surrounded her with tools: mindfulness courses, self-care routines, lists of healthy habits. Without realizing it, they had turned well-being into one more task, a subject with its own homework and its own demands. Laia’s meditation was not a space of rest; it was another thing to do well, another way of not being still. The gym was not joyful movement; it was control. Her whole well-being practice had been contaminated by the same anxiety it was meant to relieve.

In the clinic we see this more and more: young people who have learned the vocabulary of well-being before ever having the experience of calm. They know how to talk about anxiety, boundaries, and self-care, but they have never felt the body truly at rest. Hyperactivity disguises itself as mindfulness. Vigilance disguises itself as responsibility. And the body, beneath it all, stays tense, waiting for an order to rest that never comes because even rest has become a task.

The first time I asked Laia to do nothing—simply to notice the weight of her body on the chair, with no technique, no special way of breathing,

no counting—she was silent for a long while. Then, somewhat warily, she asked: “This isn’t doing anything, is it?” I told her yes, that was exactly it: doing nothing, just feeling. I saw in her face something like relief and something like fear, because for someone who has defined herself by activity, stillness can be dizzying.

Work with Laia consisted largely of taking tools away rather than adding them. She learned to tell hyperactivity disguised as calm from real presence. She learned that feeling the body isn’t a technique you execute, but a permission you give yourself. Her anxiety didn’t disappear completely—it rarely does, and it doesn’t need to—but it stopped functioning as a constant alarm. There began to be moments, longer and longer ones, in which she simply was. And that, for a teenager of her time, was almost revolutionary.

13.8. Joan, or the Man Who Felt Nothing

Joan was fifty-five and a business owner. In his whole life he had never complained about anything he considered important. He didn’t come on his own initiative: it was his wife who persuaded him, after a complete medical checkup found absolutely nothing and yet he kept feeling, in his own words, “strange.” He couldn’t describe the discomfort any better. There was no pain, no locatable symptom, nothing a test could detect. Only a dull sense that something didn’t fit, that he lived at some distance from his own life.

In the first session, when I asked him to close his eyes and bring his attention to his body, he opened them after ten seconds with an expression of genuine bewilderment. “There’s nothing there,” he said. It is the sentence I have heard most often in my career, and it moves me every time, because it is almost never true. It isn’t that there’s nothing there. It’s that the person stopped looking so long ago that they have mistaken silence for absence. Joan’s body had been speaking for decades in a language he had stopped listening to.

Joan had built his entire identity on action and problem-solving. He was good, very good, at it: see an obstacle, analyze it, solve it, move on to the

next. The world was a succession of problems, and he was an efficient machine for solving them. In that scheme, the body didn't appear anywhere, because the body is not a problem to be solved. The body is something to be inhabited, and inhabiting was not a category Joan worked with. His organism had served him faithfully for fifty-five years as a tool, and he had never treated it as anything else.

Work with Joan was deliberately slow, because any attempt to speed things up would have reactivated his problem-solving mode. We started with what was most concrete and least threatening: walking consciously, noticing the support of each foot. Then feeling his glutes as he climbed the stairs at his office, something he did dozens of times a day without ever noticing. I didn't ask him to feel emotions, because the word would have scared him off. I asked him to feel mechanics: weight, contact, movement. Things a practical man could accept.

The change came through an unexpected door. One day, about six months in, Joan said, almost in passing: "I noticed my shoulders were up around my ears in a meeting—and I let them drop." For anyone else it would have been a trivial remark, not worth mentioning. For Joan it was enormous. For the first time in years, an internal signal had reached his awareness before it turned into pain or exhaustion. The body had recovered a voice, however faint. And after that first signal came others: hunger before the clock announced lunch, tiredness, a vague unease before difficult meetings. It wasn't the end of a process, but the beginning of finally inhabiting his own life.

13.9. What the Lower Segment Teaches Through These Eight Stories

Silvia, Damián, Natalia, Félix, Clara, Marc, Laia, and Joan have little in common when seen from outside. A woman with unexplained chronic pain, a philosopher disconnected from his body, a professional occupying borrowed space, an older man who never learned to feel, a successful but empty executive, a sleepless architect, an overcontrolled teenager, a businessman who feels nothing. Eight lives, eight stories, eight completely different

ways of arriving at the same place: a nervous system that at some point learned it was not safe to rest.

What these stories share is not the symptom but the substrate. In every case, the most essential work was not making the troublesome thing disappear, but restoring a relationship of trust between the nervous system and the ground. Literally and metaphorically: the sense that there is something underneath that holds, that the body can lower its guard without that meaning danger.

Thirty years of clinical work have taught me that this restoration is possible. Not always quickly. Not always in a straight line. Not always completely. But possible. And that the path does not run through the head—through understanding, analyzing, and solving—but through the body: through feet that feel the ground, an abdomen that breathes, a chest that allows itself to open one millimeter more than yesterday.

The lower segment is not a metaphor. It is the place from which everything else becomes manageable.

CHAPTER AT A GLANCE

- **Eight composite cases:** chronic pain, a disembodied intellect, borrowed space, alexithymia, performance anxiety, insomnia, adolescent overcontrol, and numbness.
 - **Different symptoms, one substrate:** a nervous system that learned it was not safe to rest.
 - **Interventions are adapted** to each person's language and tolerance.
 - **Change shows up as small signals:** waking without the knot, feeling hunger, sitting back in the chair, feet that hold without being asked.
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Practical Exercises and Everyday Life

Grounding as a habit. Practices for bringing the lower segment into every day.

KEY CONCEPTS daily practice · somatic habit · S.A.B.A. · micro-practice · 8-week program · signs of progress

Imagine for a moment that you are an architect called in to inspect an old building. You wouldn't arrive, look at the façade for five seconds, and announce, with the authority of a TV pundit, "all fine" or "this needs knocking down." You would look at the foundations, listen to the creaks, observe the small cracks, see how the weight is distributed, and try to understand what is really happening in the structure.

Assessing the lower segment calls for exactly that attitude: less instant drama and more informed curiosity. The point is not to diagnose quickly or to label yourself with phrases like "I'm badly grounded" or "my first chakra is blocked." Assessment has a far more useful and much less melodramatic purpose: to understand how you are inhabiting your own body.

14.1. The Four Pillars of Assessment

Grounding doesn't depend on a single factor. It is expressed in four interrelated dimensions: physical, emotional, relational, and existential. They don't work separately. A physical tension can generate emotional anxiety. A relational difficulty can produce a bodily sense of insecurity. An existential crisis can show up as shallow breathing. It all happens in the same organism.

Physical dimension: the body as a map. Stand with your feet hip-width apart and your arms relaxed. Don't judge or correct. Just observe. Are your feet really planted, or do they seem to float? How is your weight distributed? Are your toes relaxed or gripping the floor? Then observe your knees, pelvis, lower back, shoulders, head. One very revealing dynamic exercise: stand up, close your eyes, and stay still for about thirty seconds. A sufficiently grounded system sways a little, like a tree. It doesn't stay rigid like a lamppost, nor wobble like a makeshift antenna. The breath counts too: if the air stays in the chest, if there are tense pauses, if the exhalation comes out forced, the body is telling you something about its degree of inner safety.

Emotional dimension: the inner landscape. Ask yourself what comes up in you when you think of the word safety. Use a subjective scale from one to ten to rate your current sense of grounding. What matters is not so much the number as the conversation it opens. If you give yourself a four, the useful question is not why you aren't a spiritual nine, but what keeps you from being a two. Which resources are already there.

Relational dimension: connection as regulator. Ask yourself whom you feel calmest with. Notice whether it's easy for you to ask for help. The way a relationship makes you feel in your body matters a great deal. There are people whose presence softens your breathing. Others automatically make your shoulders rise.

Existential dimension: the sense of belonging. Do I have a place in this world? Feeling grounded involves experiencing some degree of belonging. To a community, a place, a practice, nature, a meaningful task.

14.2. The Minimum Viable Practice

The golden rule here is the minimum viable practice. Most people don't give up body practices because they don't work, but because they start as if preparing for a high-intensity monastic retreat with goals of cosmic transformation. Then fatigue arrives, real life arrives, the old habit of not looking after oneself arrives, and the whole thing falls apart.

It's better to start small. Ridiculously small, if need be. Small things sustained transform far more than grand things abandoned.

The S.A.B.A. method is the smallest unit of practice. Four steps that fit into thirty seconds to a minute:

1. **Stop:** halt automatic movement for a moment.
2. **Attend:** bring your attention to the body, specifically to the area of the first center (feet, legs, pelvis).
3. **Breathe:** two or three diaphragmatic breaths with attention on the abdomen.
4. **Anchor:** notice the contact with the ground or the surface that holds you up.

It can be done standing in an elevator, sitting in a meeting, before a difficult conversation, or while waiting for your computer to finish starting up.

14.3. An Eight-Week Program

Eight weeks is the realistic minimum for the nervous system to begin registering the new pattern as part of the everyday landscape.

- **Week 1 — Contact with the feet.** Three S.A.B.A. practices a day (on getting up, at midday, before bed). No ambition beyond remembering that the lower body exists.
- **Week 2 — Add diaphragmatic breathing.** Keep the three S.A.B.A. practices. Add five minutes of diaphragmatic breathing in the morning (in 4, out 6).
- **Week 3 — Awareness of the pelvis and pelvic floor.** Keep everything so far. Add: when you sit down, become aware of your sitting bones. Several times a day, take three breaths while feeling the base of the pelvis.
- **Week 4 — Glute activation and Tadasana.** Add: standing, gently contract the glutes five times, then walk while feeling them. Practice Tadasana (mountain pose) for three minutes a day.

- **Week 5 — Functional movement.** Walk ten minutes a day without earphones, paying attention to the phases of each step. Climb stairs feeling the push of the glutes and feet.
- **Week 6 — Foot meditation and body scan.** Add five minutes a day of foot meditation. One day a week, a descending body scan (ten minutes).
- **Week 7 — Integration with basic asanas.** Three times a week, 10 minutes: Tadasana, Malasana (with support if needed), Vrksasana (at the wall), Balasana. Keep up the diaphragmatic breathing.
- **Week 8 — Maintenance and personalization.** Design your own daily routine of ten to fifteen minutes combining the practices that have worked best. The goal: for grounding to become a background, not one more task.

14.4. Basic Exercises for Every Day

- **Waking up the feet (30 seconds):** sitting or standing, bring your attention to the sole of one foot, then the other, then both.
- **Glute activation (1 minute):** standing, gently contract the glutes for a few seconds and release. Repeat several times, then walk while keeping a light activation.
- **Psoas release (2 minutes):** lying on your back with your knees bent, breathe into the lower abdomen, letting the tension drain into the floor.
- **Pelvic floor awareness (1 minute):** sitting with the spine available, bring your attention to the base of the pelvis and imagine that as you inhale this area widens slightly, and as you exhale it gathers in a little.
- **Breathing to the root:** inhale while bringing attention to the lower abdomen, and exhale imagining energy descending through the legs and feet into the earth.
- **Emergency S.A.B.A. grounding pause:** feet press into the floor, three deep breaths, acknowledge “I am here,” anchor your attention in a bodily sensation.

14.5. Signs of Progress and When to Seek Help

Progress in grounding is usually gradual—not very cinematic—and in this area it pays to distrust spectacular expectations. Grounding doesn't resemble the transformations promised by self-help books or the before-and-afters of social media. It is a slow, cumulative, and, above all, discreet process. It changes the texture of the days more than their plot. That is why many people who are truly making progress believe for a while that nothing is happening—simply because they were expecting fireworks, and what they get is a light that keeps growing without their noticing the exact moment the room stopped being dark.

Some early signs deserve recognition, because they are easy to miss. The first is noticing sooner when you disconnect: catching, almost in real time, the instant attention rises into the head and the body is left behind. That awareness of the moment of escape is already, in itself, a return. The second is that the breath drops more easily, that the air reaches the abdomen without having to be forced. The third is somewhat less reactivity: situations that used to throw you completely start to produce a briefer jolt, from which you recover sooner. None of these signs is heroic. All of them are real.

There are also deeper signs that come later, sometimes after months of practice. The sense of occupying your body rather than dragging it around. The capacity to stay still without the stillness producing anguish. The appearance of a background calm that doesn't depend on things going well but is there, underneath, holding. And perhaps the most valuable: a growing confidence that, whatever happens, there is ground. Not the certainty that nothing bad will happen, but the humbler and more useful certainty that if it does, there will be a base from which to respond.

But individual grounding has limits, and recognizing them is part of the maturity of the process. There are times when working alone is not enough, and insisting on solving everything through one's own practice can become a new form of the old disconnection. If, when you bring attention to your body, overly intense emotions appear, overwhelming traumatic memories surface, anxiety clearly interferes with daily life, or bodily pain is significant

and persistent, the right step is not to persist alone but to seek professional support. A body therapist, a psychologist trained in trauma, a doctor when needed.

Asking for help is neither defeat nor relapse. It is, in fact, one of the purest forms of grounding there is: recognizing one's own need, tolerating the vulnerability of showing it, and allowing another nervous system to help regulate one's own. Co-regulation—the process by which another person's calm presence lowers our activation—is, at bottom, the first grounding we ever knew, the one we received in someone's arms before we had words. Returning to it as adults is not going backward. It is remembering that the base, from the very beginning, was always relational too.

Grounding is not a fixed state conquered once and guaranteed forever. It is a relationship with one's own body and with the world that is cultivated, lost, and recovered, again and again, throughout life.

CHAPTER AT A GLANCE

- **Four pillars of assessment:** physical, emotional, relational, and existential.
 - **The minimum viable practice:** S.A.B.A.—Stop, Attend, Breathe, Anchor (thirty to ninety seconds).
 - **An eight-week program** adds one practice per week.
 - **Daily exercises:** waking the feet, glute activation, psoas release, pelvic-floor awareness, breathing to the root.
 - **Early signs of progress are subtle;** seeking professional help is itself a form of grounding.
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EPILOGUE

The Ground That Was Already There

Thirty years of feet. What working with the body teaches about life.

After thirty years of clinical work, there are things that no longer surprise me. People's capacity to change, for example. I keep seeing it, again and again, and I keep finding it extraordinary. What does still surprise me—though I should know by now—is where that change begins. Not in the head. Not in the great revelation. Not in the dramatic moment. It begins, almost always, somewhere far humbler: in the feet. In contact with the ground. In a breath that suddenly drops a little deeper. Nothing cinematic. Nothing worthy of a soundtrack. Just... body. At last.

And yet, if I am honest, it should no longer surprise me that the movement begins from below. From the feet. From contact with the ground. From a pelvis that regains its weight. From a breath that finally reaches the abdomen. Not because that is all the work—it isn't, it never is—but because that is the possible beginning. The point at which the nervous system can receive the first signal that something is changing.

I have seen people who arrived convinced their anxiety was a character flaw learn, with patience and practice, to feel themselves standing on their own feet. I have seen people who had spent decades in the same relationship pattern—looking outside for what they never found, because they were looking in the wrong place—begin to build, from below, an inner base that didn't depend on anyone's approval. I have seen people who couldn't remember the last time they slept well recover their sleep not with medication but with changes in the way they inhabited their own bodies.

None of this was quick. None of it was linear. None of it was final in the sense that the problem disappeared forever. But all of it was real.

The Promise This Book Doesn't Make

There is something this book does not promise, and it should be said plainly before we close it: it does not promise that feeling your feet three times a day will solve all your problems. It does not promise that work with the lower segment will eliminate anxiety, cure chronic pain, repair difficult relationships, or turn your life into something completely different from what it was.

What work with the first center can offer is something more modest but more solid: a base. A base from which to keep working. A ground that wasn't there before and that now, however imperfect and incomplete, is present enough to make the difference between floating and being here.

And from that base, other things become more possible. Not certain, not guaranteed, not automatic. More possible. Because when there is ground, you can build on it. And when there isn't, everything built above is just acrobatics, however brilliant it looks from a distance.

The Lower Segment Waits

The lower segment isn't going anywhere. It has been there all your life, waiting with the silent patience of roots under the earth in winter. It doesn't demand or reproach. It simply is. And every time someone remembers it—every time attention comes down, the feet are felt, the pelvis regains its weight, the breath descends—the system responds with the faithfulness of something that was always available and never left.

You only had to remember where to look. Beneath your feet, as always. In the ground that doesn't fail, even though for a long time we ignored it with the same unawareness with which we ignore the air we breathe.

Feel them again now. One more time. Like the first time, but with everything you now know.

THE EPILOGUE AT A GLANCE

- In clinical experience, change most often begins from below.
 - This book promises no cure, only a base from which further work becomes possible.
 - The lower segment has always been available; the task is to remember it.
-

LETTER FROM MARCOS

One Year Later

A little over a year ago I received an email that, in a way, would close the circle of this book. It was from Marcos, an engineer, forty-two years old. It began like this:

"I don't really know what's wrong with me. Everything's fine... but I'm not fine."

An apparently simple sentence which, if you stop to listen to it closely, contains a fairly deep mystery. Because often the problem is not that life is objectively bad. The problem is that the body has lost its sense of ground.

For months we worked on exactly what this book proposes: small practices of bodily presence, breathing, observation, relationship with the body and with the base. Nothing spectacular. Nothing dramatic. Just practice.

And as so often happens in real processes, the change didn't arrive with heavenly trumpets. It came slowly, without an appointment and without appearing on any social network. Almost without a sound.

Dear reader:

When Lluís asked me to write these lines, I sat staring at the screen for a while, not quite knowing where to start. Not because I have nothing to tell. But because what I have to tell is so simple it almost seems like too little.

There was no sudden enlightenment. There was no dramatic moment when everything changed at once. If you're expecting that kind of story, I'm sorry to disappoint you.

What happened was something much more discreet. There were mornings when, before getting out of bed, I remembered to feel my feet for a few seconds. There were afternoons at the office when, while my computer took

forever to start up, I caught myself breathing into my abdomen instead of compulsively checking my email. There were nights when, before going to sleep, I went back over the day looking for a moment of presence.

Sometimes I found one. Sometimes I didn't. There were whole days when I completely forgot the practice. And others when I did it grudgingly, just to tick the box, thinking something like: well, since I'm here, let's feel the feet for ten seconds and then get on with my life. Nothing epic. Nothing worthy of a motivational movie. And yet, at some moment I couldn't point to, something began to change. It wasn't a spectacular change. It was more a change of texture.

Anxiety, for example. It didn't disappear. I'm still someone who worries, who gets stressed when deadlines are impossible, who sometimes wakes up at three in the morning with his head churning out not very useful ideas. But it's no longer that constant background noise that used to be there all the time, like a badly tuned radio you can't switch off. Now the anxiety appears. I notice it in my body. And often I can breathe toward that sensation and watch it change.

Another unexpected change happened at work. I used to experience every project as a constant test of whether I was competent enough. Every mistake was a threat. Now I still try to do my job well, but there is more room. More breath. More body. My colleagues say my solutions have something different about them. I couldn't explain it technically, but I think it has to do with this: now there's someone behind the calculations. Not just a mind. A body too.

Another curious change happened walking down the street. I used to wear earphones all the time. Podcasts, music, anything to fill the silence. It was an elegant way of not being fully in the world. Now, sometimes, I leave the earphones at home. I walk. I feel the ground. I look at the trees. I listen to the sounds of the city. It seems like nothing, but it is a small personal revolution.

A few days ago a friend asked me something curious: "So, do you feel grounded now?"

I thought about it for a while. And in the end I gave an answer that still seems to me the best one I can give:

Not all the time. But I know how to come back.

And that, honestly, is enough. It's not about being always present, always calm, always connected. It's about knowing there is a way back. That when I get lost in my head, when anxiety hijacks me, when I start floating too much, I can do something very simple: feel my feet. Breathe into my pelvis. Stop for a few seconds. And come back. The way back is always there. You just have to remember it exists.

With affection,

Marcos. Barcelona, 2026.

A Final Reflection

Marcos's story illustrates something that, in my clinical and teaching experience, happens again and again. Grounding is not a permanent state. It is not a spiritual medal. It is not a certificate of enlightenment. It is a relational skill with oneself. A capacity that is cultivated, lost, recovered, and deepened over time.

The most stable people I have known are not the ones who never come undone. They are the ones who know how to come back. To the body, to the feet, to the moment.

The lower segment doesn't promise a life without fear, uncertainty, or moments of loss. It promises something far more realistic: that even when everything is moving, there can be an inner place to lean on. A place that doesn't depend on external circumstances. A place that doesn't disappear when life's conditions change.

That place is not an idea or a conviction. It is something felt. And like every bodily experience, it is cultivated through practice, patience, and a sense of humor—because human life, observed closely enough, has deeply tragic moments... and others that are rather absurd.

If ever, in the middle of everyday chaos, you remember to notice your feet or take a breath before reacting... then the work is already under way. The way back is open. The ground was always there.

This is the first volume of The Embodied Mind collection, devoted to the first center: the base. The second volume, The Intelligent Pelvis, explores the territory of the pelvis, the second center, where the base learned in these chapters becomes desire, flow, and the art of not fighting with life. But that will be another book. For now, the feet are enough.

Glossary

The book's key concepts in alphabetical order.

Anxiety. A state of nervous-system activation characterized by anticipation of threat, bodily restlessness, and mental worry. From the perspective of the lower segment, anxiety is usually associated with a sense of lacking a base or bodily safety.

Attachment. A relational pattern formed in childhood that influences how we bond with others throughout life. Attachment styles directly influence the sense of basic safety.

Autonomic nervous system. The system that regulates automatic bodily functions such as breathing, digestion, and the stress response. It is divided into the sympathetic system (accelerator, activation) and the parasympathetic system (brake, rest, and recovery).

Bodily base. The subjective sense of physical and emotional stability that arises when the body feels supported and organized. It depends on muscle tone, breathing, posture, and nervous-system regulation.

Chakra. A concept from the yogic tradition describing centers of energy and psychophysical organization in the body. It can be understood as a psychocorporeal model describing functional zones with specific patterns of tension, emotion, and vital organization.

Co-regulation. The process by which two nervous systems regulate each other through interaction. The calm presence of another person can lower physiological activation and increase the sense of safety.

Eight-week program. A progressive two-month program for becoming familiar with work on the lower segment. Each week adds a new practice (contact

with the feet, breathing, pelvic awareness, glute activation, functional movement, meditation, basic asanas) until a personalized routine is built.

Emotional regulation. The nervous system's capacity to modulate the intensity and duration of emotions. It depends largely on interoception and on the state of the autonomic nervous system.

Grounding. The sense of connection with the body, the earth, and the present moment. It involves physical stability, emotional regulation, and a clear perception of the support the ground provides. It is the central aim of work with the lower segment.

Interoception. The nervous system's capacity to perceive the organism's internal signals, such as breathing, heartbeat, or visceral tension. It is directly related to emotional regulation. The insula is its brain center.

Micro-practice. A brief practice, lasting from a few seconds to a few minutes, designed to fit into daily life without requiring extra time or special preparation. In work with the lower segment, the most common micro-practice is the S.A.B.A. pause, which takes thirty seconds and can be done anywhere.

Muladhara. The first chakra in the yogic tradition. It is located at the base of the spine and is associated with the element earth, survival, basic safety, and rootedness.

Neuroception. The unconscious process by which the nervous system constantly assesses whether the environment is safe, dangerous, or life-threatening, modulating responses of connection, defense, or collapse. A concept developed by Stephen Porges (2011).

Neuroplasticity. The brain's capacity to reorganize itself and form new connections throughout life in response to repeated experience. It is possibly the best news in contemporary neuroscience.

Pelvic floor. The set of muscles and fasciae that closes off the bottom of the pelvic cavity, forming a hammock that supports the bladder, rectum, and uterus. It responds to the nervous system's state of activation: it tightens with threat and releases with safety.

Polyvagal Theory. A neurophysiological model proposed by Stephen Porges describing how the nervous system hierarchically regulates responses of safety, defense, and social engagement through three circuits: ventral vagal (safety),

sympathetic (fight/flight), and dorsal vagal (collapse/dissociation). Some of its evolutionary and anatomical claims remain debated.

Psoas. A deep muscle connecting the lumbar spine to the femur. It takes part in hip flexion and is involved in stress and defense responses. It is the only muscle that connects the lumbar spine directly to the femur.

S.A.B.A. A four-step protocol for a grounding micro-practice: Stop automatic movement, Attend to the body, Breathe diaphragmatically, and Anchor attention in contact with the ground. It takes between thirty and ninety seconds. It is the smallest unit of practice in the eight-week program.

Sacral plexus. A nerve network deep in the pelvis, formed by the spinal nerves L4 to S4. Nerves such as the sciatic and pudendal emerge from it. It is the neural wiring of the lower segment.

Trauma. An experience that overwhelms the nervous system's capacity to process a threatening event. It can leave lasting marks on the body and on emotional regulation, even decades later.

Uprootedness. Loss of contact with the bodily base. A sense of floating, of being present but not quite, of never fully inhabiting one's own life. It has no diagnostic code, but it is one of the most common presenting concerns in clinical practice.

Vagus nerve. The main nerve of the parasympathetic nervous system. It regulates functions such as breathing, heart rate, and the response of calm and recovery. It is activated by prolonged exhalation.

Window of tolerance. The range of physiological activation within which a person can function flexibly and adaptively, without tipping into hyperactivation (sympathetic dominance) or collapse (dorsal vagal). The term was coined by Daniel J. Siegel (1999).

Recommended Reading

An annotated selection of the books that have most shaped this work, in order of relevance.

van der Kolk, B. (2014). *The Body Keeps the Score*. Viking. The book that changed the conversation about trauma. Van der Kolk has spent decades showing that the body not only remembers but goes on living what the mind would rather have forgotten. Essential and, at times, uncomfortable reading.

Porges, S. W. (2011). *The Polyvagal Theory*. Norton. Polyvagal Theory redraws the map of the autonomic nervous system. Porges explains precisely why safety is not an idea but a bodily and social experience. A technical book, but one that changes how you understand people in the consulting room.

Damasio, A. R. (1994). *Descartes' Error*. Putnam. With real patients and neuroscientific data, Damasio showed that separating reason from emotion is a philosophical fiction. Emotion and body are constitutive of thought, not its enemies. A classic that remains relevant.

Levine, P. A. (2010). *In an Unspoken Voice*. North Atlantic Books. Levine observed that animals in the wild shake off danger with their bodies and move on. Humans don't. From there he built Somatic Experiencing, a therapeutic approach that gives the body back its leading role in recovery from trauma.

Lowen, A. (1975). *Bioenergetics*. Coward, McCann & Geoghegan. Lowen built on Reich and then went further. His work is sometimes uncomfortable because it forces you to look at character in the face, without euphemisms. For anyone working with body and psyche, reading Lowen is still an honest way of looking in the mirror.

Kabat-Zinn, J. (1990). *Full Catastrophe Living*. Delacorte. The founding text of mindfulness in a clinical setting. Kabat-Zinn took ancient contemplative

tools, removed their religious wrapping, and put them to work with real patients. Decades later, it remains the reference.

Siegel, D. J. (2007). *The Mindful Brain*. Norton. Siegel connects mindfulness, attachment theory, and interpersonal neurobiology in a way that is useful both clinically and for personal practice.

Myers, T. W. (2001). *Anatomy Trains*. Churchill Livingstone. Myers changed the way many body therapists understand movement and posture. The myofascial lines he describes are not metaphor; they are real anatomy.

Koch, L. (2012). *The Psoas Book*. Guinea Pig Publications. A small, dense book about an enormous and misunderstood muscle. Koch treats the psoas not as a mere hip flexor but as the muscle that links our emotional history to our posture.

Sapolsky, R. M. (2004). *Why Zebras Don't Get Ulcers*. Holt. Sapolsky explains the biology of chronic stress with admirable clarity and humor.

Feldenkrais, M. (1972). *Awareness Through Movement*. Harper & Row. Feldenkrais understood that moving well is a matter not of strength but of awareness. His lessons are deceptively simple.

Iyengar, B. K. S. (1966). *Light on Yoga*. Allen & Unwin. The teacher who made yoga a precise discipline without taking away its soul. Anatomical rigor and attention to detail.

Ogden, P., Minton, K., & Pain, C. (2006). *Trauma and the Body*. Norton. A rigorous clinical manual on the sensorimotor approach to trauma. It describes how to intervene through the body within the therapy session.

Maturana, H. R., & Varela, F. J. (1987). *The Tree of Knowledge*. Shambhala. A fundamental work for understanding the biology of knowledge and the organization of living beings. Maturana and Varela show that knowing is effective action in the world, and that emotion and reason are not separate. Essential for anyone wanting to explore the biological foundations of human experience.

Varela, F. J. (2000). *El fenómeno de la vida* [The phenomenon of life]. Dolmen. (In Spanish.) Varela explores the relationship between biology, phenomenology, and cognitive science. His work on enaction—the idea that the mind emerges from the organism's interaction with its environment—resonates deeply with the approach of this book.

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You will also find audio recordings, chapter infographics, and the practice materials that accompany each volume.